Robotics & State Estimation (RiSE) Tutorials

Lecture Notes

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Robotics & State Estimation Tutorials
https://github.com/yangyulin/rise-tutorial

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Lecture Notes - RiSE-2026
Last Updated - July 5, 2026

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1 Lecture 1: Foundations — Probability, Linear Algebra, and Linear Systems

1.1 Probability

1.1.1 Definition

For an event A, the probability $P(A)$ is

$$P(A) = \frac{N_A}{N}, \tag{1}$$

where N_A is the number of occurrences of A and N is the number of all possible outcomes.

1.1.2 Conditional probability

The probability of event A given that event B is true is

$$P(A \mid B) = \frac{P(A,B)}{P(B)}, \qquad P(B) \neq 0, \tag{2}$$

which yields the product rule

$$\begin{aligned} P(A,B) &= P(A \mid B)\, P(B) && (3)\\ &= P(B \mid A)\, P(A). && (4) \end{aligned}$$

1.1.3 Total probability theorem

If $\{A_1, \ldots, A_n\}$ is a partition (mutually exclusive, $P(A_i, A_j) = 0$ for $i \neq j$) and B is an arbitrary event, then

$$\begin{aligned} P(B) &= P(B \mid A_1)P(A_1) + \cdots + P(B \mid A_n)P(A_n) && (5)\\ &= \sum_i P(B \mid A_i)\, P(A_i). && (6) \end{aligned}$$

1.1.4 Bayes' theorem

$$\begin{aligned} P(A_i \mid B) &= \frac{P(B \mid A_i)\, P(A_i)}{P(B)} && (7)\\ &= \frac{P(B \mid A_i)\, P(A_i)}{\sum_i P(B \mid A_i)\, P(A_i)}. && (8) \end{aligned}$$

1.1.5 Independence

With the inclusion–exclusion relation $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, two events A and B are independent when

$$\begin{aligned} P(A,B) &= P(A \mid B)\, P(B) && (9)\\ &= P(B \mid A)\, P(A) && (10)\\ &= P(A)\, P(B) && (11)\\ &= P(B)\, P(A). && (12) \end{aligned}$$

1.2 Random Variables

A random variable defines a function that quantifies an experiment. For a sample space $S = \{s_1, \dots, s_n\}$, a random variable is a map $\mathrm{x} : S \to \mathbb{R}$ with $\mathrm{x}(s_i) = x_i$.

1.2.1 Distribution

The (cumulative) distribution of a random variable x is

$$F_{\mathrm{x}}(x) = P(\mathrm{x} \leq x), \tag{13}$$

i.e. the probability that x takes a value lower than or equal to x.

1.2.2 Probability density function (pdf)

$$\begin{aligned} p(x) &= \lim_{\varepsilon \to 0} \frac{F_{\mathrm{x}}(x) - F_{\mathrm{x}}(x - \varepsilon)}{\varepsilon} && (14)\\ &= \frac{\mathrm{d}F_{\mathrm{x}}(x)}{\mathrm{d}x} && (15)\\ &= \frac{\mathrm{d}P(\mathrm{x} \leq x)}{\mathrm{d}x}, && (16) \end{aligned}$$

hence

$$\begin{aligned} P(\mathrm{x} \leq x) &= \int_{-\infty}^{x} p(\xi)\,\mathrm{d}\xi, && (17)\\ P(\mathrm{x} \leq \infty) &= \int_{-\infty}^{+\infty} p(\xi)\,\mathrm{d}\xi = 1, && (18) \end{aligned}$$

and for an interval

$$P(x_1 \leq \mathrm{x} < x_2) = \int_{x_1}^{x_2} p(\xi)\,\mathrm{d}\xi. \tag{19}$$

1.2.3 Example distributions

- Gaussian: $\mathrm{x} \sim \mathcal{N}(m, \sigma) \iff p(x) = \dfrac{1}{\sqrt{2\pi}\,\sigma}\, e^{-\frac{(x-m)^2}{2\sigma^2}}$.
- Uniform: $p(x) = \begin{cases} 0 & x < x_1 \\ \dfrac{1}{x_2 - x_1} & x_1 \leq x \leq x_2 \\ 0 & x > x_2 \end{cases}$

1.2.4 Conditional distribution and density

$$\begin{aligned} P(\mathrm{x} \leq x \mid A) &= \frac{P(\mathrm{x} \leq x,\ A)}{P(A)}, && (20)\\ p(x \mid A) &= \frac{\mathrm{d}P(\mathrm{x} \leq x \mid A)}{\mathrm{d}x}. && (21) \end{aligned}$$

1.2.5 Total probability theorem for pdfs

For a mutually exclusive partition $\{A_1, \ldots, A_n\}$,

$$p(x) = p(x \mid A_1)P(A_1) + \cdots + p(x \mid A_n)P(A_n). \tag{22}$$

1.2.6 Continuous Bayes' theorem

Using the infinitesimal relations $P(\mathrm{x} = x) = p(x)\,\delta x$ and $P(\mathrm{x} = x \mid A) = p(x \mid A)\,\delta x$,

$$\begin{aligned}
P(A \mid \mathrm{x} = x) &= \frac{P(A, \mathrm{x} = x)}{P(\mathrm{x} = x)} && (23)\\
&= \frac{P(\mathrm{x} = x \mid A)\,P(A)}{P(\mathrm{x} = x)} && (24)\\
&= \frac{p(x \mid A)}{p(x)}\,P(A), && (25)
\end{aligned}$$

so that $P(A \mid \mathrm{x} = x)\,p(x) = p(x \mid A)\,P(A)$, which gives the continuous total-probability and Bayes' relations

$$\begin{aligned}
P(A) &= \int_{-\infty}^{+\infty} P(A \mid \mathrm{x} = x)\,p(x)\,\mathrm{d}x, && (26)\\
p(x \mid A) &= \frac{P(A \mid \mathrm{x} = x)}{P(A)}\,p(x) && (27)\\
&= \frac{P(A \mid \mathrm{x} = x)\,p(x)}{\int_{-\infty}^{+\infty} P(A \mid \mathrm{x} = x)\,p(x)\,\mathrm{d}x}. && (28)
\end{aligned}$$

1.3 Expectation, Variance, and Random Vectors

1.3.1 Mean

The expected value (mean) of a continuous random variable, and its conditional version, are

$$\begin{aligned}
m_x &= \mathbb{E}(\mathrm{x}) && (29)\\
&= \int_{-\infty}^{+\infty} x\,p(x)\,\mathrm{d}x, && (30)\\
m_{x|A} &= \mathbb{E}(\mathrm{x} \mid A) && (31)\\
&= \int_{-\infty}^{+\infty} x\,p(x \mid A)\,\mathrm{d}x. && (32)
\end{aligned}$$

1.3.2 Variance

$$\begin{aligned}
\sigma_x^2 &= \mathbb{E}\big((\mathrm{x} - m_x)^2\big) && (33)\\
&= \int_{-\infty}^{+\infty} (x - m_x)^2\,p(x)\,\mathrm{d}x && (34)\\
&= \mathbb{E}(\mathrm{x}^2) - \big(\mathbb{E}(\mathrm{x})\big)^2, && (35)
\end{aligned}$$

with the conditional variance

$$\sigma_{x|A}^2 = \mathbb{E}\big\{(\mathrm{x} - m_{x|A})^2 \mid A\big\} \tag{36}$$

$$= \int_{-\infty}^{+\infty} (x - m_{x|A})^2 \, p(x \mid A) \, \mathrm{d}x. \tag{37}$$

1.3.3 A set of random variables

The joint pdf of $x_1, \dots, x_n$ satisfies

$$\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} p(x_1, \dots, x_n) \, \mathrm{d}x_1 \cdots \mathrm{d}x_n = 1. \tag{38}$$

If x_i, x_j are independent, $p(x_i, x_j) = p(x_i)\, p(x_j)$. Expectation is linear: for $y = \alpha x_i + \beta x_j$,

$$\mathbb{E}(y) = \alpha \, \mathbb{E}(x_i) + \beta \, \mathbb{E}(x_j). \tag{39}$$

1.3.4 Covariance

$$\sigma_{ij}^2 = \mathbb{E}\big((x_i - m_i)(x_j - m_j)\big) \tag{40}$$

$$= \mathbb{E}(x_i x_j) - \mathbb{E}(x_i) \, \mathbb{E}(x_j). \tag{41}$$

When $\sigma_{ij} = 0$ we have $\mathbb{E}(x_i x_j) = \mathbb{E}(x_i) \, \mathbb{E}(x_j)$ and x_i, x_j are *uncorrelated.*

Discussion questions.

1. If x_i, x_j are independent, then $p(x_i, x_j) = p(x_i)p(x_j) \Rightarrow \mathbb{E}(x_i x_j) = \mathbb{E}(x_i)\mathbb{E}(x_j)$ — so independence implies uncorrelated.
2. If x_i, x_j are uncorrelated, are they necessarily independent?
3. If x_i, x_j are jointly Gaussian, is independence equivalent to being uncorrelated?

1.3.5 Random vector

Let $\mathbf{x} = [x_1, \dots, x_n]^\top$, where all elements are random variables, with mean $\mathbb{E}(\mathbf{x}) \triangleq \mathbf{m}_x$ and covariance

$$\mathbf{P} = \mathbb{E}\big((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top\big) \tag{42}$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x}) \, \mathbb{E}(\mathbf{x}^\top), \tag{43}$$

with entries

$$P_{ii} = \mathbb{E}\big((x_i - m_i)^2\big) \tag{44}$$

$$= \int_{-\infty}^{+\infty} (x_i - m_i)^2 p(x_i) \, \mathrm{d}x_i, \tag{45}$$

$$P_{ij} = \mathbb{E}\big((x_i - m_i)(x_j - m_j)\big). \tag{46}$$

Since $P_{ij} = P_{ji}$, the covariance $\mathbf{P}$ is a symmetric matrix; it is also positive (semi-)definite.

1.3.6 Gaussian vector pdf

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{n/2}\sqrt{|\mathbf{P}|}} \exp\Big(-\tfrac{1}{2}(\mathbf{x}-\mathbf{m}_x)^\top \mathbf{P}^{-1}(\mathbf{x}-\mathbf{m}_x)\Big), \tag{47}$$

where $|\mathbf{P}|$ is the determinant of $\mathbf{P}$, $\mathbf{m}_x = \mathbb{E}(\mathbf{x})$, and $\mathbf{P} = \mathbb{E}\big((\mathbf{x}-\mathbf{m}_x)(\mathbf{x}-\mathbf{m}_x)^\top\big)$.

Homework. Let $\mathbf{x} = [x_1, \dots, x_n]^\top$ with $p(\mathbf{x}) = \mathcal{N}(\mathbf{m}_x, \mathbf{P}_{xx})$. Define

$$\begin{aligned} q &= (\mathbf{x}-\mathbf{m}_x)^\top \mathbf{P}_{xx}^{-1}(\mathbf{x}-\mathbf{m}_x), \\ \mathbf{y} &= \mathbf{P}_{xx}^{-1/2}(\mathbf{x}-\mathbf{m}_x), \\ q &= \mathbf{y}^\top\mathbf{y} = \sum_i y_i^2. \end{aligned}$$

What are the mean and covariance of q?

1.4 Maximum Likelihood and Maximum a Posteriori Estimation

For a measurement model with additive noise $z_i = h_i(\mathbf{x}) + n_i$, stacked as $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$, two estimators are

- **Maximum likelihood estimation (MLE):** $\max\limits_{\mathbf{x}} p(\mathbf{z} \mid \mathbf{x})$.
- **Maximum a posteriori (MAP):** $\max\limits_{\mathbf{x}} p(\mathbf{x} \mid \mathbf{z})$, with $p(\mathbf{x} \mid \mathbf{z}) = \dfrac{p(\mathbf{z} \mid \mathbf{x})\, p(\mathbf{x})}{p(\mathbf{z})}$.

For zero-mean Gaussian noise $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{P})$,

$$\max_{\mathbf{x}} p(\mathbf{z} \mid \mathbf{x}) = \max_{\mathbf{x}} \frac{1}{(2\pi)^{n/2}\sqrt{|\mathbf{P}|}} \exp\Big(-(\mathbf{z}-\mathbf{h}(\mathbf{x}))^\top \mathbf{P}^{-1}(\mathbf{z}-\mathbf{h}(\mathbf{x}))\Big), \tag{48}$$

which is equivalent to the weighted least-squares problem

$$\min_{\mathbf{x}}\ C = (\mathbf{z}-\mathbf{h}(\mathbf{x}))^\top \mathbf{P}^{-1}(\mathbf{z}-\mathbf{h}(\mathbf{x})). \tag{49}$$

Linearizing about $\hat{\mathbf{x}}$ with $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n} = \mathbf{h}(\hat{\mathbf{x}}) + \mathbf{H}(\mathbf{x}-\hat{\mathbf{x}}) + \mathbf{n}$, and defining the residual $\tilde{\mathbf{z}} = \mathbf{z} - \mathbf{h}(\hat{\mathbf{x}})$ and error $\tilde{\mathbf{x}} = \mathbf{x} - \hat{\mathbf{x}}$,

$$\min_{\tilde{\mathbf{x}}}\ (\tilde{\mathbf{z}} - \mathbf{H}\tilde{\mathbf{x}})^\top \mathbf{P}^{-1}(\tilde{\mathbf{z}} - \mathbf{H}\tilde{\mathbf{x}}) = \min_{\tilde{\mathbf{x}}} \big(\tilde{\mathbf{z}}^\top \mathbf{P}^{-1}\tilde{\mathbf{z}} - 2\,\tilde{\mathbf{z}}^\top\mathbf{P}^{-1}\mathbf{H}\tilde{\mathbf{x}} + \tilde{\mathbf{x}}^\top\mathbf{H}^\top\mathbf{P}^{-1}\mathbf{H}\tilde{\mathbf{x}}\big). \tag{50}$$

Setting the gradient to zero gives the normal equations $\mathbf{H}^\top\mathbf{P}^{-1}\mathbf{H}\,\tilde{\mathbf{x}} = \mathbf{H}^\top\mathbf{P}^{-1}\tilde{\mathbf{z}}$, hence

$$\boxed{\begin{aligned} \mathbf{x}^{\oplus} &= \hat{\mathbf{x}} + \tilde{\mathbf{x}} && (51) \\ &= \hat{\mathbf{x}} + (\mathbf{H}^\top\mathbf{P}^{-1}\mathbf{H})^{-1}\mathbf{H}^\top\mathbf{P}^{-1}\tilde{\mathbf{z}}. && (52) \end{aligned}}$$

This is the classical weighted least-squares solution.

Questions.

1. What is the covariance of the new estimate $\mathbf{x}^{\oplus}$?
2. For a prior $\mathbf{x}_p = \mathbf{x} + \mathbf{n}_p$ together with $z_i = h_i(\mathbf{x}) + n_i$, $i = 1, \dots, n$, what is the MAP solution $\max_{\mathbf{x}} p(\mathbf{x} \mid \mathbf{z})$?

1.4.1 Optimization

Nonlinear least-squares problems are solved iteratively, e.g. Gauss–Newton (GN) and Levenberg–Marquardt (LM).

1.5 Linear Systems

A linear (continuous-time) state-space system is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \tag{53}$$
$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}, \tag{54}$$

where $\mathbf{x}$ is the state, $\mathbf{u}$ the input, and $\mathbf{y}$ the output. Its discrete-time counterpart is

$$\mathbf{x}_{k+1} = \mathbf{A}'\mathbf{x}_k + \mathbf{B}'\mathbf{u}_k, \tag{55}$$
$$\mathbf{y}_{k+1} = \mathbf{C}'\mathbf{x}_{k+1} + \mathbf{D}'\mathbf{u}_{k+1}. \tag{56}$$

If $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ are constant, the system is *linear time-invariant* (LTI); if they depend on time, it is *linear time-variant* (LTV).

1.5.1 Homogeneous solution

Consider $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ with $\mathbf{x}(t_0) = \mathbf{x}_0$ and $\mathbf{y} = \mathbf{C}\mathbf{x}$. From $\dot{\mathbf{x}} - \mathbf{A}\mathbf{x} = \mathbf{0}$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(e^{-\mathbf{A}t}\mathbf{x}\big) = e^{-\mathbf{A}t}\dot{\mathbf{x}} + e^{-\mathbf{A}t}(-\mathbf{A})\mathbf{x} \tag{57}$$
$$= e^{-\mathbf{A}t}\big(\dot{\mathbf{x}} - \mathbf{A}\mathbf{x}\big) \tag{58}$$
$$= \mathbf{0}. \tag{59}$$

Integrating from 0 to t gives $e^{-\mathbf{A}t}\mathbf{x}(t) - \mathbf{x}(t_0) = \mathbf{0}$, so

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}_0 \tag{60}$$
$$= \mathbf{\Phi}(t,0)\,\mathbf{x}_0, \tag{61}$$
$$\mathbf{\Phi}(t,0) = e^{\mathbf{A}t}, \tag{62}$$

where $\mathbf{\Phi}(t,0)$ is the *state transition matrix*, and $\mathbf{y} = \mathbf{C}\,\mathbf{\Phi}(t,0)\,\mathbf{x}_0$.

1.5.2 Properties of the state transition matrix

$$\mathbf{\Phi}(t_2,t_0) = \mathbf{\Phi}(t_2,t_1)\,\mathbf{\Phi}(t_1,t_0), \tag{63}$$
$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{\Phi}(t,0) = \mathbf{A}\,\mathbf{\Phi}(t,0). \tag{64}$$

The second follows from $\dot{\mathbf{x}} = \dot{\mathbf{\Phi}}(t,0)\mathbf{x}_0 = \mathbf{A}\mathbf{x} = \mathbf{A}\,\mathbf{\Phi}(t,0)\mathbf{x}_0$.

1.5.3 Output sequence and observability

Sampling the homogeneous response at $t_0, t_1, \ldots, t_n$ with $\mathbf{x}_k = \mathbf{\Phi}(k,0)\mathbf{x}_0$ and $\mathbf{y}_k = \mathbf{C}\mathbf{x}_k$ stacks into

$$\begin{bmatrix}\mathbf{y}_1\\ \vdots\\ \mathbf{y}_n\end{bmatrix} = \begin{bmatrix}\mathbf{C}\,\mathbf{\Phi}(1,0)\\ \vdots\\ \mathbf{C}\,\mathbf{\Phi}(n,0)\end{bmatrix}\mathbf{x}_0, \tag{65}$$

which relates the measured outputs to the initial state (an observability-type relation).

1.5.4 Full LTV solution

For the complete system $\dot{\mathbf{x}} = \mathbf{A}(t)\mathbf{x} + \mathbf{B}(t)\mathbf{u}(t)$, write $\dot{\mathbf{x}} - \mathbf{A}(t)\mathbf{x} = \mathbf{B}(t)\mathbf{u}(t)$ and use the integrating factor $e^{-\int_0^t \mathbf{A}(\tau)\mathrm{d}\tau}$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(e^{-\int_0^t \mathbf{A}(\tau)\mathrm{d}\tau}\mathbf{x}\Big) = e^{-\int_0^t \mathbf{A}(\tau)\mathrm{d}\tau}\big(\dot{\mathbf{x}} - \mathbf{A}(t)\mathbf{x}\big) \tag{66}$$

$$= e^{-\int_0^t \mathbf{A}(\tau)\mathrm{d}\tau}\mathbf{B}(t)\mathbf{u}(t). \tag{67}$$

Integrating,

$$e^{-\int_0^t \mathbf{A}(\tau)\mathrm{d}\tau}\mathbf{x}(t) - \mathbf{x}_0 = \int_0^t e^{-\int_0^\tau \mathbf{A}(s)\mathrm{d}s}\mathbf{B}(\tau)\mathbf{u}(\tau)\,\mathrm{d}\tau, \tag{68}$$

so that, with $\mathbf{\Phi}(t,0) = e^{\int_0^t \mathbf{A}(\tau)\mathrm{d}\tau}$ and $\mathbf{\Phi}(t,\tau) = e^{\int_\tau^t \mathbf{A}(s)\mathrm{d}s}$,

$$\boxed{\begin{aligned} \mathbf{x}(t) &= \mathbf{\Phi}(t,0)\,\mathbf{x}_0 + \int_0^t \mathbf{\Phi}(t,\tau)\,\mathbf{B}(\tau)\,\mathbf{u}(\tau)\,\mathrm{d}\tau, && (69)\\ \mathbf{y}(t) &= \mathbf{C}(t)\Big(\mathbf{\Phi}(t,0)\mathbf{x}_0 + \int_0^t \mathbf{\Phi}(t,\tau)\mathbf{B}(\tau)\mathbf{u}(\tau)\,\mathrm{d}\tau\Big) + \mathbf{D}(t)\mathbf{u}(t). && (70)\end{aligned}}$$

This is one of the most important properties of linear systems.

1.6 Homework Problems and Solutions

1.6.1 Variance identity

Problem. Show that the variance of a scalar random variable can be written as $\sigma_x^2 = \mathbb{E}(\mathrm{x}^2) - \big(\mathbb{E}(\mathrm{x})\big)^2$, and discuss when it vanishes.

Solution. Expanding the definition with $m_x = \mathbb{E}(\mathrm{x})$,

$$\sigma_x^2 = \mathbb{E}\big((\mathrm{x} - m_x)^2\big) \tag{71}$$

$$= \mathbb{E}\big(\mathrm{x}^2 - 2m_x\mathrm{x} + m_x^2\big) \tag{72}$$

$$= \mathbb{E}(\mathrm{x}^2) - 2m_x\mathbb{E}(\mathrm{x}) + m_x^2 \tag{73}$$

$$= \mathbb{E}(\mathrm{x}^2) - m_x^2. \tag{74}$$

Since $\sigma_x^2 \geq 0$, we have $\mathbb{E}(\mathrm{x}^2) \geq \big(\mathbb{E}(\mathrm{x})\big)^2$, with equality if and only if $\sigma_x^2 = 0$, i.e. $\mathrm{x} = m_x$ almost surely (a deterministic variable) — a degenerate special case.

1.6.2 Covariance matrix identity

Problem. For a random vector $\mathbf{x}$ with mean $\mathbf{m}_x$, show that $\mathbf{P} = \mathbb{E}\big((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top\big) = \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbb{E}(\mathbf{x}^\top)$.

Solution. Expanding the outer product and using linearity of expectation, together with $\mathbb{E}(\mathbf{x}) = \mathbf{m}_x$,

$$\mathbf{P} = \mathbb{E}\big(\mathbf{x}\mathbf{x}^\top - \mathbf{x}\mathbf{m}_x^\top - \mathbf{m}_x\mathbf{x}^\top + \mathbf{m}_x\mathbf{m}_x^\top\big) \tag{75}$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbf{m}_x^\top - \mathbf{m}_x\mathbb{E}(\mathbf{x}^\top) + \mathbf{m}_x\mathbf{m}_x^\top \tag{76}$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbf{m}_x\mathbf{m}_x^\top - \mathbf{m}_x\mathbf{m}_x^\top + \mathbf{m}_x\mathbf{m}_x^\top \tag{77}$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbf{m}_x\mathbf{m}_x^\top \tag{78}$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbb{E}(\mathbf{x}^\top). \tag{79}$$

1.6.3 Scalar covariance identity

Problem. For two random variables x_i, x_j, show that $\sigma_{ij} = \mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j)$.

Solution. With $m_i = \mathbb{E}(x_i)$, $m_j = \mathbb{E}(x_j)$, expand

$$
\begin{aligned}
\sigma_{ij} &= \mathbb{E}\big((x_i - m_i)(x_j - m_j)\big) && (80)\\
&= \iint (x_i - m_i)(x_j - m_j)\, p(x_i, x_j)\, \mathrm{d}x_i\, \mathrm{d}x_j && (81)\\
&= \iint x_i x_j\, p(x_i, x_j)\, \mathrm{d}x_i \mathrm{d}x_j - m_j \iint x_i\, p(x_i, x_j)\, \mathrm{d}x_i \mathrm{d}x_j \\
&\quad - m_i \iint x_j\, p(x_i, x_j)\, \mathrm{d}x_i \mathrm{d}x_j + m_i m_j \iint p(x_i, x_j)\, \mathrm{d}x_i \mathrm{d}x_j. && (82)
\end{aligned}
$$

Using the marginalization identities (and symmetrically for x_j)

$$
\int p(x_i, x_j)\, \mathrm{d}x_i = p(x_j), \qquad \iint p(x_i, x_j)\, \mathrm{d}x_i \mathrm{d}x_j = 1, \tag{83}
$$

we get

$$
\begin{aligned}
m_i \iint x_j\, p(x_i, x_j)\, \mathrm{d}x_i \mathrm{d}x_j &= m_i \int x_j\, p(x_j)\, \mathrm{d}x_j && (84)\\
&= m_i m_j, && (85)
\end{aligned}
$$

and likewise for the other term, so

$$
\begin{aligned}
\sigma_{ij} &= \mathbb{E}(x_i x_j) - m_i m_j - m_i m_j + m_i m_j && (86)\\
&= \mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j). && (87)
\end{aligned}
$$

1.6.4 Independence versus uncorrelatedness

Problem. Resolve the three discussion questions: (1) does independence imply uncorrelatedness? (2) does uncorrelatedness imply independence? (3) for jointly Gaussian variables, are the two equivalent?

Solution. *(1) Independent $\Rightarrow$ uncorrelated.* If $p(x_i, x_j) = p(x_i)p(x_j)$, then

$$
\begin{aligned}
\mathbb{E}(x_i x_j) &= \iint x_i x_j\, p(x_i, x_j)\, \mathrm{d}x_i \mathrm{d}x_j && (88)\\
&= \int x_i\, p(x_i)\, \mathrm{d}x_i \cdot \int x_j\, p(x_j)\, \mathrm{d}x_j && (89)\\
&= \mathbb{E}(x_i)\mathbb{E}(x_j), && (90)
\end{aligned}
$$

so $\sigma_{ij} = 0$.

(2) Uncorrelated $\not\Rightarrow$ independent. Uncorrelatedness gives only

$$
\begin{aligned}
\mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j) &= \iint x_i x_j \big(p(x_i, x_j) - p(x_i)p(x_j)\big)\, \mathrm{d}x_i \mathrm{d}x_j && (91)\\
&= 0, && (92)
\end{aligned}
$$

which does *not* force $p(x_i, x_j) = p(x_i)p(x_j)$ in general. Hence uncorrelated does not imply independent.

(3) Jointly Gaussian. For $x_i \sim \mathcal{N}(m_i, \sigma_i^2)$ and $x_j \sim \mathcal{N}(m_j, \sigma_j^2)$, the product of marginals is

$$p(x_i)p(x_j) = \frac{1}{2\pi\sigma_i\sigma_j} \exp\left\{-\tfrac{1}{2} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}^\top \begin{bmatrix} \sigma_i^2 & 0 \\ 0 & \sigma_j^2 \end{bmatrix}^{-1} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}\right\}. \tag{93}$$

The joint Gaussian $[x_i, x_j]^\top \sim \mathcal{N}\big([m_i, m_j]^\top, \mathbf{P}_{xx}\big)$ with $\mathbf{P}_{xx} = \begin{bmatrix} \sigma_i^2 & \sigma_{ij} \\ \sigma_{ji} & \sigma_j^2 \end{bmatrix}$ is

$$p(x_i, x_j) = \frac{1}{2\pi\sqrt{|\mathbf{P}_{xx}|}} \exp\left\{-\tfrac{1}{2} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}^\top \mathbf{P}_{xx}^{-1} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}\right\}. \tag{94}$$

If $\sigma_{ij} = \sigma_{ji} = 0$, then $\mathbf{P}_{xx}$ is diagonal and $p(x_i, x_j) = p(x_i)p(x_j)$. Hence, for (jointly) Gaussian variables, *independent* $\Leftrightarrow$ *uncorrelated.*

1.6.5 The χ^2 distribution

Problem. Let $\mathbf{x} = [x_1, \ldots, x_n]^\top$ with $p(\mathbf{x}) = \mathcal{N}(\mathbf{m}_x, \mathbf{P}_{xx})$ and define $q = (\mathbf{x} - \mathbf{m}_x)^\top \mathbf{P}_{xx}^{-1} (\mathbf{x} - \mathbf{m}_x)$. Find the mean and variance of q.

Solution. Factor $\mathbf{P}_{xx} = \mathbf{L}\mathbf{L}^\top$ (Cholesky), so $\mathbf{P}_{xx}^{-1} = \mathbf{L}^{-\top}\mathbf{L}^{-1}$, and define the whitened variable $\mathbf{y} = \mathbf{L}^{-1}(\mathbf{x} - \mathbf{m}_x)$. Then

$$\begin{aligned} q &= (\mathbf{x} - \mathbf{m}_x)^\top \mathbf{L}^{-\top}\mathbf{L}^{-1}(\mathbf{x} - \mathbf{m}_x) && (95) \\ &= \mathbf{y}^\top \mathbf{y} && (96) \\ &= \sum_{i=1}^{n} y_i^2. && (97) \end{aligned}$$

The whitened variable has zero mean and identity covariance,

$$\begin{aligned} \mathbf{m}_y &= \mathbb{E}(\mathbf{y}) && (98) \\ &= \mathbf{L}^{-1}\big(\mathbb{E}(\mathbf{x}) - \mathbf{m}_x\big) && (99) \\ &= \mathbf{0}, && (100) \\ \mathbf{P}_{yy} &= \mathbb{E}(\mathbf{y}\mathbf{y}^\top) && (101) \\ &= \mathbf{L}^{-1}\, \mathbb{E}\big((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top\big)\, \mathbf{L}^{-\top} && (102) \\ &= \mathbf{L}^{-1}\mathbf{P}_{xx}\mathbf{L}^{-\top} && (103) \\ &= \mathbf{L}^{-1}\mathbf{L}\mathbf{L}^\top\mathbf{L}^{-\top} && (104) \\ &= \mathbf{I}_n, && (105) \end{aligned}$$

so $y_i \sim \mathcal{N}(0, 1)$ are independent. Therefore the mean of q is

$$\begin{aligned} \mathbb{E}(q) &= \mathbb{E}(\mathbf{y}^\top \mathbf{y}) && (106) \\ &= \sum_{i=1}^{n} \mathbb{E}(y_i^2) && (107) \\ &= n. && (108) \end{aligned}$$

For the variance, with $\mathbb{E}(y_i^2) = 1$ the cross terms vanish by independence, giving

$$\begin{aligned}
\sigma_{qq}^2 &= \mathbb{E}\big((q-n)^2\big) && (109)\\
&= \mathbb{E}\Big(\big(\textstyle\sum_i (y_i^2 - 1)\big)^2\Big) && (110)\\
&= \sum_i \mathbb{E}\big((y_i^2 - 1)^2\big) && (111)\\
&= \sum_i \big(\mathbb{E}(y_i^4) - 2\mathbb{E}(y_i^2) + 1\big). && (112)
\end{aligned}$$

The fourth moment of a standard normal follows from integration by parts (with $u = y_i^3$, $\mathrm{d}v = y_i\, p(y_i)\,\mathrm{d}y_i$, $v = -p(y_i)$):

$$\begin{aligned}
\mathbb{E}(y_i^4) &= \int_{-\infty}^{+\infty} y_i^4\, p(y_i)\,\mathrm{d}y_i && (113)\\
&= \underbrace{\big[-y_i^3 p(y_i)\big]_{-\infty}^{+\infty}}_{=0} + \int_{-\infty}^{+\infty} 3y_i^2\, p(y_i)\,\mathrm{d}y_i && (114)\\
&= 3\,\mathbb{E}(y_i^2) && (115)\\
&= 3. && (116)
\end{aligned}$$

Hence each term equals $3 - 2 + 1 = 2$, and

$$\sigma_{qq}^2 = \sum_{i=1}^{n} 2 = 2n. \tag{117}$$

So $q \sim \chi^2(n)$ has mean n and variance $2n$.

2 Lecture 2: Rotation, Lie Groups, SE(3), SE$_2$(3), and Quaternions

Further reading. This lecture draws on several standard references: matrix Lie groups in Barfoot [1, Ch. 7]; the Lie-group treatment of $SO(3)/SE(3)$ in the GTSAM notes [2], Atanasov's ECE276A slides [3], and the micro Lie theory of Solà, Deray & Atchuthan [4]; and quaternion algebra/kinematics in Trawny & Roumeliotis [5, Sec. 1] and Solà [6, Sec. 1–3].

2.1 Preliminaries: useful series

The matrix exponential and the sine/cosine series underpin everything that follows:

$$e^x = \exp(x) = \sum_{k=0}^{\infty} \frac{1}{k!} x^k \tag{118}$$

$$= I + x + \tfrac{1}{2!}x^2 + \tfrac{1}{3!}x^3 + \tfrac{1}{4!}x^4 + \cdots, \tag{119}$$

$$\sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} \tag{120}$$

$$= x - \tfrac{1}{3!}x^3 + \tfrac{1}{5!}x^5 - \cdots, \tag{121}$$

$$\cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} \tag{122}$$

$$= 1 - \tfrac{1}{2!}x^2 + \tfrac{1}{4!}x^4 - \cdots. \tag{123}$$

For small x (so that $x^2, x^3 \to 0$), the important first-order approximations are $e^x \approx I + x$, $\sin(x) \approx x$, and $\cos(x) \approx 1$.

2.2 Frames, notation, and transformations

We work with coordinate frames $\{A\}$ and $\{B\}$. A point/vector $\mathbf{L}$ expressed in $\{A\}$ is ${}^{A}\mathbf{L}$, and in $\{B\}$ is ${}^{B}\mathbf{L}$. The angle $\theta_{A\to B}$ is the rotation about the z-axis taking $\{A\}$ to $\{B\}$ (Fig. 1).

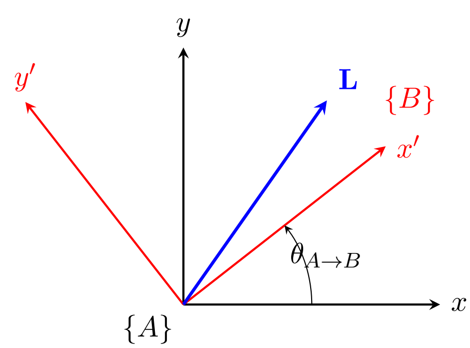


Figure 1: Frame $\{B\}$ is frame $\{A\}$ rotated by $\theta_{A\to B}$ about the z-axis; the vector $\mathbf{L}$ can be expressed in either frame.

The rotation matrix ${}^{A}_{B}\mathbf{R}$ maps a vector from $\{B\}$ to $\{A\}$:

$${}^{A}\mathbf{L} = {}^{A}_{B}\mathbf{R}\, {}^{B}\mathbf{L} \tag{124}$$

$$= {}^{A}_{B}\mathbf{R}(\theta_{A\to B})\, {}^{B}\mathbf{L}. \tag{125}$$

Adding the translation ${}^{A}\mathbf{p}_B$ (the origin of $\{B\}$ expressed in $\{A\}$, Fig. 2) gives the full rigid-body transformation

$$\{A\},\{B\} \;\Rightarrow\; \left\{ {}^{A}_{B}\mathbf{R},\ {}^{A}\mathbf{p}_B \right\} \;\Rightarrow\; {}^{A}_{B}\mathbf{T} = \begin{bmatrix} {}^{A}_{B}\mathbf{R} & {}^{A}\mathbf{p}_B \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix}. \tag{126}$$

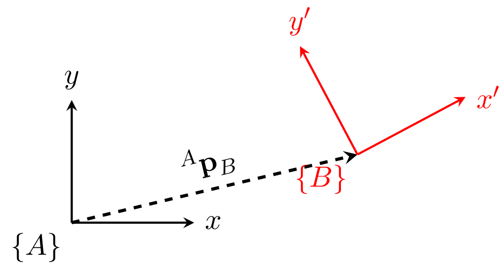


Figure 2: A rigid-body transformation combines the rotation ${}^{A}_{B}\mathbf{R}$ with the translation ${}^{A}\mathbf{p}_B$.

2.3 Defining rotation

2.3.1 Euler angles

With roll θ (about x), pitch r (about y), yaw ψ (about z),

$$\mathbf{R}(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}, \; \mathbf{R}(r) = \begin{bmatrix} \cos r & 0 & \sin r \\ 0 & 1 & 0 \\ -\sin r & 0 & \cos r \end{bmatrix}, \; \mathbf{R}(\psi) = \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \tag{127}$$

and the yaw–pitch–roll composition is

$$\mathbf{R} = \mathbf{R}(\psi)\,\mathbf{R}(r)\,\mathbf{R}(\theta) \tag{128}$$

$$= \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos r & 0 & \sin r \\ 0 & 1 & 0 \\ -\sin r & 0 & \cos r \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}. \tag{129}$$

2.3.2 Infinitesimal rotation and the skew-symmetric matrix

For small angles $\delta\theta, \delta r, \delta\psi \to 0$ each elementary rotation linearizes ($\cos \to 1$, $\sin \to$ angle):

$$\mathbf{R}(\delta\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\delta\theta \\ 0 & \delta\theta & 1 \end{bmatrix}, \quad \mathbf{R}(\delta r) = \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & 0 \\ -\delta r & 0 & 1 \end{bmatrix}, \quad \mathbf{R}(\delta\psi) = \begin{bmatrix} 1 & -\delta\psi & 0 \\ \delta\psi & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{130}$$

Multiplying the last two and dropping all second-order products ($\delta r\,\delta\theta \approx 0$ etc.),

$$\mathbf{R}(\delta r)\mathbf{R}(\delta\theta) = \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & 0 \\ -\delta r & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\delta\theta \\ 0 & \delta\theta & 1 \end{bmatrix} \tag{131}$$

$$= \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & -\delta\theta \\ -\delta r & \delta\theta & 1 \end{bmatrix}, \tag{132}$$

and then

$$\mathbf{R} = \mathbf{R}(\delta\psi)\mathbf{R}(\delta r)\mathbf{R}(\delta\theta) = \begin{bmatrix} 1 & -\delta\psi & 0 \\ \delta\psi & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & -\delta\theta \\ -\delta r & \delta\theta & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -\delta\psi & \delta r \\ \delta\psi & 1 & -\delta\theta \\ -\delta r & \delta\theta & 1 \end{bmatrix} \tag{133}$$

$$= \mathbf{I} + \begin{bmatrix} 0 & -\delta\psi & \delta r \\ \delta\psi & 0 & -\delta\theta \\ -\delta r & \delta\theta & 0 \end{bmatrix} \tag{134}$$

$$= \mathbf{I} + \lfloor \boldsymbol{\theta} \rfloor_\times , \tag{135}$$

where the small-angle vector is $\boldsymbol{\theta} = [\delta\theta,\ \delta r,\ \delta\psi]^\top$. The matrix $\lfloor \boldsymbol{\theta} \rfloor_\times$ is skew-symmetric, $\lfloor \boldsymbol{\theta} \rfloor_\times = -\lfloor \boldsymbol{\theta} \rfloor_\times^\top$. This motivates writing a finite rotation as $\mathbf{R}(\boldsymbol{\theta}) \approx \mathbf{I} + \lfloor \boldsymbol{\theta} \rfloor_\times \approx \exp(\lfloor \boldsymbol{\theta} \rfloor_\times)$, and defining $\mathrm{Exp}(\boldsymbol{\theta}) \triangleq \exp(\lfloor \boldsymbol{\theta} \rfloor_\times)$. The question is then whether $\mathbf{R}(\boldsymbol{\theta}) = \mathrm{Exp}(\boldsymbol{\theta})$.

2.4 Axis–angle and Rodrigues' formula

Let $\mathbf{k}$ be the unit rotation axis and θ the rotation angle, with $\boldsymbol{\theta} = \theta\mathbf{k} = [\theta_1, \theta_2, \theta_3]^\top$, so

$$\mathbf{k} = \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix}, \quad \lfloor \mathbf{k} \rfloor_\times = \begin{bmatrix} 0 & -k_3 & k_2 \\ k_3 & 0 & -k_1 \\ -k_2 & k_1 & 0 \end{bmatrix}, \quad \lfloor \boldsymbol{\theta} \rfloor_\times = \begin{bmatrix} 0 & -\theta_3 & \theta_2 \\ \theta_3 & 0 & -\theta_1 \\ -\theta_2 & \theta_1 & 0 \end{bmatrix}. \tag{136}$$

The skew operator on a unit vector satisfies the cyclic power identities

$$\lfloor \mathbf{k} \rfloor_\times^2 = \mathbf{k}\mathbf{k}^\top - \mathbf{I}_3, \qquad \lfloor \mathbf{k} \rfloor_\times^3 = -\lfloor \mathbf{k} \rfloor_\times , \qquad \lfloor \mathbf{k} \rfloor_\times^4 = -\lfloor \mathbf{k} \rfloor_\times^2 , \tag{137}$$

$$\lfloor \mathbf{k} \rfloor_\times^5 = \lfloor \mathbf{k} \rfloor_\times , \qquad \lfloor \mathbf{k} \rfloor_\times^6 = \lfloor \mathbf{k} \rfloor_\times^2 , \qquad \lfloor \mathbf{k} \rfloor_\times^7 = -\lfloor \mathbf{k} \rfloor_\times . \tag{138}$$

Substituting $\lfloor \boldsymbol{\theta} \rfloor_\times = \theta \lfloor \mathbf{k} \rfloor_\times$ into the exponential series term by term,

$$\begin{aligned} \mathrm{Exp}(\boldsymbol{\theta}) = \exp(\theta \lfloor \mathbf{k} \rfloor_\times) &= \mathbf{I} + \theta \lfloor \mathbf{k} \rfloor_\times + \tfrac{\theta^2}{2!} \lfloor \mathbf{k} \rfloor_\times^2 + \tfrac{\theta^3}{3!} \lfloor \mathbf{k} \rfloor_\times^3 + \tfrac{\theta^4}{4!} \lfloor \mathbf{k} \rfloor_\times^4 \\ &\quad + \tfrac{\theta^5}{5!} \lfloor \mathbf{k} \rfloor_\times^5 + \tfrac{\theta^6}{6!} \lfloor \mathbf{k} \rfloor_\times^6 + \tfrac{\theta^7}{7!} \lfloor \mathbf{k} \rfloor_\times^7 + \cdots \\ &= \mathbf{I} + \theta \lfloor \mathbf{k} \rfloor_\times + \tfrac{\theta^2}{2!} \lfloor \mathbf{k} \rfloor_\times^2 - \tfrac{\theta^3}{3!} \lfloor \mathbf{k} \rfloor_\times - \tfrac{\theta^4}{4!} \lfloor \mathbf{k} \rfloor_\times^2 \\ &\quad + \tfrac{\theta^5}{5!} \lfloor \mathbf{k} \rfloor_\times + \tfrac{\theta^6}{6!} \lfloor \mathbf{k} \rfloor_\times^2 - \tfrac{\theta^7}{7!} \lfloor \mathbf{k} \rfloor_\times + \cdots \\ &= \mathbf{I} + \underbrace{\left(\theta - \tfrac{\theta^3}{3!} + \tfrac{\theta^5}{5!} - \tfrac{\theta^7}{7!} + \cdots\right)}_{\sin\theta} \lfloor \mathbf{k} \rfloor_\times \\ &\quad + \underbrace{\left(\tfrac{\theta^2}{2!} - \tfrac{\theta^4}{4!} + \tfrac{\theta^6}{6!} - \cdots\right)}_{1-\cos\theta} \lfloor \mathbf{k} \rfloor_\times^2 , \end{aligned} \tag{139}$$

which gives **Rodrigues' rotation formula**

$$\boxed{\mathbf{R}(\boldsymbol{\theta}) = \mathrm{Exp}(\boldsymbol{\theta}) = \exp(\theta \lfloor \mathbf{k} \rfloor_\times) = \mathbf{I} + \sin\theta\ \lfloor \mathbf{k} \rfloor_\times + (1 - \cos\theta)\ \lfloor \mathbf{k} \rfloor_\times^2 .} \tag{140}$$

For a small angle $\delta\theta \to 0$, $\mathbf{R}(\delta\boldsymbol{\theta}) \approx \mathbf{I} + \lfloor \delta\boldsymbol{\theta} \rfloor_\times$ (small-angle approximation).

2.4.1 Logarithmic map

The log map is the inverse of the exponential map:

$$\log(\mathbf{R}) = \lfloor \boldsymbol{\theta} \rfloor_\times, \quad \mathrm{Log}(\mathbf{R}) = \boldsymbol{\theta}_{3\times1}, \qquad \theta = \arccos\Big(\tfrac{\operatorname{tr}\mathbf{R}-1}{2}\Big), \quad \lfloor \mathbf{k} \rfloor_\times = \frac{\mathbf{R}-\mathbf{R}^\top}{2\sin\theta}. \tag{141}$$

2.5 SO(3) kinematics

The two directions of the maps are: from the vector space $\mathbb{R}^3$ via the Lie algebra to the Lie group, $\boldsymbol{\theta} \to \lfloor \boldsymbol{\theta} \rfloor_\times \to \mathbf{R} = \mathrm{Exp}(\boldsymbol{\theta})$; and back, $\mathbf{R} \to \lfloor \boldsymbol{\theta} \rfloor_\times \to \boldsymbol{\theta} = \mathrm{Log}(\mathbf{R})$.

(1) Angular-velocity kinematics. Differentiating the orthogonality constraint $\mathbf{R}\mathbf{R}^\top = \mathbf{I}_3$,

$$\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{R}\mathbf{R}^\top) = \mathbf{0} \;\Rightarrow\; \dot{\mathbf{R}}\mathbf{R}^\top + \mathbf{R}\dot{\mathbf{R}}^\top = \mathbf{0} \;\Rightarrow\; \dot{\mathbf{R}}\mathbf{R}^\top = -\mathbf{R}\dot{\mathbf{R}}^\top = -(\dot{\mathbf{R}}\mathbf{R}^\top)^\top, \tag{142}$$

so $\dot{\mathbf{R}}\mathbf{R}^\top$ is skew-symmetric. Writing it as $\lfloor \boldsymbol{\omega} \rfloor_\times$ with $\lfloor \boldsymbol{\omega} \rfloor_\times = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}$,

$$\dot{\mathbf{R}}\mathbf{R}^\top = \lfloor \boldsymbol{\omega} \rfloor_\times \;\Rightarrow\; \dot{\mathbf{R}} = \lfloor \boldsymbol{\omega} \rfloor_\times \mathbf{R}. \tag{143}$$

(2) Equivalent limit form. With $\mathbf{R}(t+\Delta t) = \mathbf{R}(t)\exp(\lfloor \boldsymbol{\omega} \rfloor_\times \Delta t)$,

$$\begin{aligned} \dot{\mathbf{R}}(t) &= \lim_{\Delta t \to 0} \frac{\mathbf{R}(t+\Delta t) - \mathbf{R}(t)}{\Delta t} && (144)\\ &= \lim_{\Delta t \to 0} \frac{\mathbf{R}(t)\exp(\lfloor \boldsymbol{\omega} \rfloor_\times \Delta t) - \mathbf{R}(t)}{\Delta t} && (145)\\ &= \mathbf{R}(t) \lim_{\Delta t \to 0} \frac{\exp(\lfloor \boldsymbol{\omega} \rfloor_\times \Delta t) - \mathbf{I}}{\Delta t} && (146)\\ &= \mathbf{R}(t) \lfloor \boldsymbol{\omega} \rfloor_\times. && (147) \end{aligned}$$

(3) Integration. With $\mathbf{R}_i = \exp(\lfloor \boldsymbol{\omega}_i \rfloor_\times \Delta t_i)$, composing increments gives

$$\begin{aligned} \mathbf{R}_{1\to k} &= \mathbf{R}_k \mathbf{R}_{k-1} \cdots \mathbf{R}_2 \mathbf{R}_1 && (148)\\ &= \exp(\lfloor \boldsymbol{\omega}_k \rfloor_\times \Delta t_k) \cdots \exp(\lfloor \boldsymbol{\omega}_1 \rfloor_\times \Delta t_1). && (149) \end{aligned}$$

(4) Interpolation. For ${}^G_{t_1}\mathbf{R}, {}^G_{t_2}\mathbf{R}$ and $t \in [t_1, t_2]$, define the relative rotation and its angle, and the interpolation ratio:

$$\begin{aligned} {}^{t_1}_{t_2}\mathbf{R} &= {}^G_{t_1}\mathbf{R}^\top \, {}^G_{t_2}\mathbf{R}, && (150)\\ {}^{t_1}_{t_2}\boldsymbol{\theta} &= \mathrm{Log}({}^{t_1}_{t_2}\mathbf{R}) = \mathrm{Log}({}^G_{t_1}\mathbf{R}^\top \, {}^G_{t_2}\mathbf{R}), && (151)\\ \lambda &= \frac{t - t_1}{t_2 - t_1}. && (152) \end{aligned}$$

Then

$$\begin{aligned} {}^G_t\mathbf{R} &= {}^G_{t_1}\mathbf{R} \, {}^{t_1}_t\mathbf{R} && (153)\\ &= {}^G_{t_1}\mathbf{R} \; \mathrm{Exp}\big(\lambda \, {}^{t_1}_{t_2}\boldsymbol{\theta}\big). && (154) \end{aligned}$$

(5) Right Jacobian. For a rotation $\mathbf{R}(\boldsymbol{\theta}) = \mathrm{Exp}(\boldsymbol{\theta})$, perturb the angle by $\delta\boldsymbol{\theta} \in \mathbb{R}^3$ and let $\delta\boldsymbol{\varphi}$ be the corresponding small rotation in $\mathfrak{so}(3)$:

$$\mathbf{R}(\boldsymbol{\theta})\ \mathrm{Exp}(\delta\boldsymbol{\varphi}) = \mathbf{R}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) \ \Rightarrow\ \mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}(\delta\boldsymbol{\varphi}) = \mathrm{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}). \tag{155}$$

Solving for $\delta\boldsymbol{\varphi}$,

$$\mathrm{Exp}(\delta\boldsymbol{\varphi}) = \mathrm{Exp}(\boldsymbol{\theta})^{-1}\,\mathrm{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) \ \Rightarrow\ \delta\boldsymbol{\varphi} = \mathrm{Log}\big(\,\mathrm{Exp}(\boldsymbol{\theta})^{-1}\,\mathrm{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta})\big), \tag{156}$$

and the *right Jacobian* is the derivative of $\delta\boldsymbol{\varphi}$ with respect to $\delta\boldsymbol{\theta}$:

$$\mathbf{J}_r(\boldsymbol{\theta}) \triangleq \frac{\partial\,\delta\boldsymbol{\varphi}}{\partial\,\delta\boldsymbol{\theta}} = \lim_{\delta\boldsymbol{\theta}\to 0} \frac{\mathrm{Log}\big(\,\mathrm{Exp}(\boldsymbol{\theta})^{-1}\,\mathrm{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta})\big)}{\delta\boldsymbol{\theta}}. \tag{157}$$

Equivalently, $\mathrm{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) = \mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}\big(\mathbf{J}_r(\boldsymbol{\theta})\,\delta\boldsymbol{\theta}\big)$, and inverting,

$$\mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}(\delta\boldsymbol{\varphi}) = \mathrm{Exp}\big(\boldsymbol{\theta} + \mathbf{J}_r^{-1}(\boldsymbol{\theta})\,\delta\boldsymbol{\varphi}\big) \ \Rightarrow\ \mathrm{Log}\big(\,\mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}(\delta\boldsymbol{\varphi})\big) = \boldsymbol{\theta} + \mathbf{J}_r^{-1}(\boldsymbol{\theta})\,\delta\boldsymbol{\varphi}, \tag{158}$$

with the closed form

$$\mathbf{J}_r(\boldsymbol{\theta}) = \mathbf{I} - \frac{1 - \cos\|\boldsymbol{\theta}\|}{\|\boldsymbol{\theta}\|^2}\,\lfloor\boldsymbol{\theta}\rfloor_\times + \frac{\|\boldsymbol{\theta}\| - \sin\|\boldsymbol{\theta}\|}{\|\boldsymbol{\theta}\|^3}\,\lfloor\boldsymbol{\theta}\rfloor_\times^2\,. \tag{159}$$

(6) Left Jacobian. Applying the perturbation on the left instead defines the left Jacobian:

$$\mathrm{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) = \mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}\big(\mathbf{J}_r(\boldsymbol{\theta})\,\delta\boldsymbol{\theta}\big) = \mathrm{Exp}\big(\mathbf{J}_l(\boldsymbol{\theta})\,\delta\boldsymbol{\theta}\big)\,\mathrm{Exp}(\boldsymbol{\theta}). \tag{160}$$

Using the adjoint identity $\mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}(\mathbf{x})\,\mathrm{Exp}(\boldsymbol{\theta})^{-1} = \mathrm{Exp}(\mathbf{R}(\boldsymbol{\theta})\mathbf{x})$ on the middle expression,

$$\mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}\big(\mathbf{J}_r\delta\boldsymbol{\theta}\big) = \mathrm{Exp}\big(\mathbf{R}(\boldsymbol{\theta})\mathbf{J}_r\delta\boldsymbol{\theta}\big)\,\mathrm{Exp}(\boldsymbol{\theta}) \ \Rightarrow\ \mathbf{J}_l(\boldsymbol{\theta}) = \mathbf{R}(\boldsymbol{\theta})\,\mathbf{J}_r(\boldsymbol{\theta}). \tag{161}$$

(7) Sign symmetry. From $\mathrm{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) = \mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}(\mathbf{J}_r(\boldsymbol{\theta})\delta\boldsymbol{\theta})$, replacing $\boldsymbol{\theta} \to -\boldsymbol{\theta}$, $\delta\boldsymbol{\theta} \to -\delta\boldsymbol{\theta}$,

$$\mathrm{Exp}(-\boldsymbol{\theta} - \delta\boldsymbol{\theta}) = \mathrm{Exp}\big(-\mathbf{J}_l(-\boldsymbol{\theta})\delta\boldsymbol{\theta}\big)\,\mathrm{Exp}(-\boldsymbol{\theta}) = \mathrm{Exp}\big(-\mathbf{J}_r(\boldsymbol{\theta})\delta\boldsymbol{\theta}\big)\,\mathrm{Exp}(-\boldsymbol{\theta}) \ \Rightarrow\ \mathbf{J}_l(-\boldsymbol{\theta}) = \mathbf{J}_r(\boldsymbol{\theta}). \tag{162}$$

(8) Summary. Collecting (5)–(7),

$$\mathrm{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) = \mathrm{Exp}(\boldsymbol{\theta})\,\mathrm{Exp}\big(\mathbf{J}_r(\boldsymbol{\theta})\delta\boldsymbol{\theta}\big), \qquad \mathbf{J}_l(\boldsymbol{\theta}) = \mathbf{R}(\boldsymbol{\theta})\mathbf{J}_r(\boldsymbol{\theta}), \qquad \mathbf{J}_l(-\boldsymbol{\theta}) = \mathbf{J}_r(\boldsymbol{\theta}). \tag{163}$$

(9) Identity $\lfloor\mathbf{R}\boldsymbol{\omega}\rfloor_\times = \mathbf{R}\,\lfloor\boldsymbol{\omega}\rfloor_\times\,\mathbf{R}^\top$. For any vector $\mathbf{v}$, using $\mathbf{R}(\mathbf{u} \times \mathbf{v}) = (\mathbf{R}\mathbf{u}) \times (\mathbf{R}\mathbf{v})$ and $\mathbf{R}^\top\mathbf{R} = \mathbf{I}$,

$$\begin{aligned}
\lfloor\mathbf{R}\mathbf{u}\rfloor_\times\,(\mathbf{R}\mathbf{v}) &= \mathbf{R}\big(\,\lfloor\mathbf{u}\rfloor_\times\,\mathbf{v}\big) && (164)\\
&= \mathbf{R}\big(\,\lfloor\mathbf{u}\rfloor_\times\,\mathbf{R}^\top\mathbf{R}\mathbf{v}\big) && (165)\\
&= \mathbf{R}\,\lfloor\mathbf{u}\rfloor_\times\,\mathbf{R}^\top\,(\mathbf{R}\mathbf{v}). && (166)
\end{aligned}$$

Since this holds for all $\mathbf{R}\mathbf{v}$, we conclude $\lfloor\mathbf{R}\mathbf{u}\rfloor_\times = \mathbf{R}\,\lfloor\mathbf{u}\rfloor_\times\,\mathbf{R}^\top$.

Homework. Prove $\lfloor\mathbf{R}\boldsymbol{\omega}\rfloor_\times = \mathbf{R}\,\lfloor\boldsymbol{\omega}\rfloor_\times\,\mathbf{R}^\top$.

(10) Identity $\operatorname{Exp}(\mathbf{R}\boldsymbol{\varphi}) = \mathbf{R}\operatorname{Exp}(\boldsymbol{\varphi})\,\mathbf{R}^\top$. Using $\lfloor \mathbf{R}\boldsymbol{\varphi} \rfloor_\times^k = \mathbf{R}\lfloor \boldsymbol{\varphi} \rfloor_\times^k \mathbf{R}^\top$ (which follows from (9), since $\lfloor \mathbf{R}\boldsymbol{\varphi} \rfloor_\times^2 = \mathbf{R}\lfloor \boldsymbol{\varphi} \rfloor_\times \mathbf{R}^\top \mathbf{R} \lfloor \boldsymbol{\varphi} \rfloor_\times \mathbf{R}^\top = \mathbf{R}\lfloor \boldsymbol{\varphi} \rfloor_\times^2 \mathbf{R}^\top$, etc.),

$$\begin{aligned}
\operatorname{Exp}(\mathbf{R}\boldsymbol{\varphi}) &= \mathbf{I} + \lfloor \mathbf{R}\boldsymbol{\varphi} \rfloor_\times + \tfrac{1}{2!}\lfloor \mathbf{R}\boldsymbol{\varphi} \rfloor_\times^2 + \tfrac{1}{3!}\lfloor \mathbf{R}\boldsymbol{\varphi} \rfloor_\times^3 + \cdots \\
&= \mathbf{R}\mathbf{R}^\top + \mathbf{R}\lfloor \boldsymbol{\varphi} \rfloor_\times \mathbf{R}^\top + \tfrac{1}{2!}\mathbf{R}\lfloor \boldsymbol{\varphi} \rfloor_\times^2 \mathbf{R}^\top + \cdots \\
&= \mathbf{R}\big(\mathbf{I} + \lfloor \boldsymbol{\varphi} \rfloor_\times + \tfrac{1}{2!}\lfloor \boldsymbol{\varphi} \rfloor_\times^2 + \tfrac{1}{3!}\lfloor \boldsymbol{\varphi} \rfloor_\times^3 + \cdots\big)\mathbf{R}^\top && (167) \\
&= \mathbf{R}\operatorname{Exp}(\boldsymbol{\varphi})\,\mathbf{R}^\top. && (168)
\end{aligned}$$

Homework. Prove $\operatorname{Exp}(\mathbf{R}\boldsymbol{\varphi}) = \mathbf{R}\operatorname{Exp}(\boldsymbol{\varphi})\,\mathbf{R}^\top$.

2.5.1 Worked applications

(11) Recovering ${}^C_I\mathbf{R}$ from vector pairs. Given camera frame $\{C\}$ and IMU frame $\{I\}$ with ${}^C\mathbf{v}_1 = {}^C_I\mathbf{R}\,{}^I\mathbf{v}_1$ and ${}^C\mathbf{v}_2 = {}^C_I\mathbf{R}\,{}^I\mathbf{v}_2$, the trick is to stack three independent directions (the third from the cross product) into an invertible matrix:

$$\begin{bmatrix} {}^C\mathbf{v}_1 & {}^C\mathbf{v}_2 & {}^C\mathbf{v}_1 \times {}^C\mathbf{v}_2 \end{bmatrix} = {}^C_I\mathbf{R}\begin{bmatrix} {}^I\mathbf{v}_1 & {}^I\mathbf{v}_2 & {}^I\mathbf{v}_1 \times {}^I\mathbf{v}_2 \end{bmatrix}. \tag{169}$$

(12) Given instead relative rotations ${}^{C_1}_{C_2}\mathbf{R} = {}^C_I\mathbf{R}\,{}^{I_1}_{I_2}\mathbf{R}\,{}^C_I\mathbf{R}^\top$, recover ${}^C_I\mathbf{R}$ from two such pairs.

> **(13) Discussion.** For (11)/(12): What is the minimum number of data pairs needed? What requirements must the pairs satisfy? What if more than the minimum is available?

(14) Averaging noisy rotations. Given n measurements $\mathbf{R}_{m_i} = \mathbf{R}\exp(\lfloor \mathbf{n}_i \rfloor_\times)$ with $\mathbf{n}_i \sim \mathcal{N}(\mathbf{0}_{3\times 1}, \mathbf{P}_i)$, estimate $\mathbf{R}$ and its covariance.

2.6 Groups: SO(3) and SE(3)

SO(3) — special orthogonal group.

$$SO(3) = \left\{\mathbf{R} \in \mathbb{R}^{3\times 3} \;\middle|\; \mathbf{R}^\top\mathbf{R} = \mathbf{I}_3,\ \det(\mathbf{R}) = 1\right\}. \tag{170}$$

It satisfies the four group axioms: *closure* ($\mathbf{R}_1\mathbf{R}_2 \in SO(3)$), *identity* ($\mathbf{I}_3 \in SO(3)$), *inverse* ($\mathbf{R}^{-1} = \mathbf{R}^\top \in SO(3)$), and *associativity* ($(\mathbf{R}_1\mathbf{R}_2)\mathbf{R}_3 = \mathbf{R}_1(\mathbf{R}_2\mathbf{R}_3)$).

SE(3) — special Euclidean group.

$$SE(3) = \left\{\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix} \;\middle|\; \mathbf{R} \in SO(3),\ \mathbf{p} \in \mathbb{R}^3\right\}. \tag{171}$$

The four axioms: *closure*

$$\begin{aligned}
\mathbf{T}_1\mathbf{T}_2 &= \begin{bmatrix} \mathbf{R}_1 & \mathbf{p}_1 \\ \mathbf{0} & 1 \end{bmatrix}\begin{bmatrix} \mathbf{R}_2 & \mathbf{p}_2 \\ \mathbf{0} & 1 \end{bmatrix} && (172) \\
&= \begin{bmatrix} \mathbf{R}_1\mathbf{R}_2 & \mathbf{R}_1\mathbf{p}_2 + \mathbf{p}_1 \\ \mathbf{0} & 1 \end{bmatrix} \in SE(3), && (173)
\end{aligned}$$

identity $\begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_{3\times 1} \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix}$, *inverse*

$$\begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{R}^\top & -\mathbf{R}^\top\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \in SE(3), \tag{174}$$

and *associativity* $(\mathbf{T}_1\mathbf{T}_2)\mathbf{T}_3 = \mathbf{T}_1(\mathbf{T}_2\mathbf{T}_3)$.

2.6.1 Matrix Lie groups

A *group* is a set with an operation satisfying closure, associativity, identity, and invertibility. A *Lie group* is a group that is also a smooth (differentiable) manifold whose group operation is smooth. A *matrix Lie group* has matrices as its elements. The maps between the vector space, the Lie algebra, and the Lie group are

$$\mathbb{R}^m \xrightarrow{\wedge\,\text{(wedge)}} \mathfrak{so}(n) \xrightarrow{\exp} SO(n), \qquad SO(n) \xrightarrow{\log} \mathfrak{so}(n) \xrightarrow{\vee\,\text{(vee)}} \mathbb{R}^m, \tag{175}$$

and for short we write $\mathbf{X} = \mathrm{Exp}(\mathbf{x}) = \exp(\mathbf{x}^\wedge)$ and $\mathbf{x} = \mathrm{Log}(\mathbf{X}) = \log(\mathbf{X})^\vee$. For $SO(3)$ specifically, $\boldsymbol{\theta} \in \mathbb{R}^3 \to \lfloor \boldsymbol{\theta} \rfloor_\times = \boldsymbol{\theta}^\wedge \to \mathbf{R} = \exp(\boldsymbol{\theta}^\wedge)$, and $\mathbf{R} \in SO(3) \to \lfloor \boldsymbol{\theta} \rfloor_\times = \log(\mathbf{R}) \to \boldsymbol{\theta} = \log(\mathbf{R})^\vee$.

2.6.2 Why this (manifold) machinery?

It *simplifies* estimation: the core idea is to use a smooth manifold to model parameters that are awkward in plain vector space, which makes optimization, interpolation, integration, and differentiation easier or even possible (Fig. 3). For example, a raw angle $\theta \in [-\pi, \pi]$ has a wrap-around cut; the circle/sphere is smooth, and the tangent line at a point models the small change / error state used in optimization.

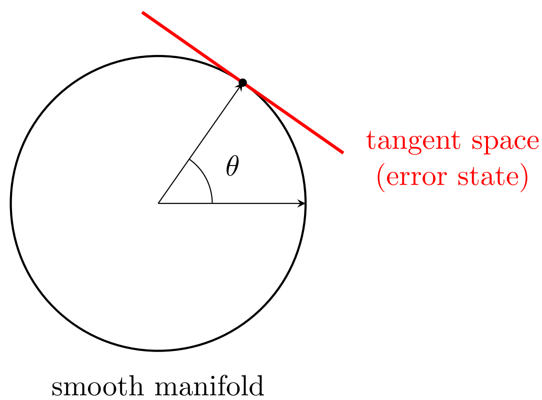


Figure 3: A smooth manifold (here a circle/sphere) avoids the wrap-around of a raw angle; the red tangent line is the local vector space used to model derivatives and error states for optimization.

For $SO(3)$ the left Jacobian also appears directly in the exponential expansion:

$$\begin{aligned} \mathbf{R}(\boldsymbol{\theta}) &= \mathrm{Exp}(\boldsymbol{\theta}) && (176)\\ &= \mathbf{I}_3 + \lfloor \boldsymbol{\theta} \rfloor_\times \mathbf{J}_l(\boldsymbol{\theta}) = \mathbf{I}_3 + \boldsymbol{\theta}^\wedge \mathbf{J}_l(\boldsymbol{\theta}), && (177)\\ \mathbf{J}_l(\boldsymbol{\theta}) &= \sum_{n=0}^{\infty} \frac{1}{(n+1)!} (\boldsymbol{\theta}^\wedge)^n. && (178) \end{aligned}$$

2.7 SE(3)

Exp / Log. The SE(3) tangent vector uses $\boldsymbol{\rho}$ for its translation component, $\boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{\rho} \end{bmatrix} \in \mathbb{R}^6$, which is *distinct* from the group translation $\mathbf{p}$ in $\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}$; the two are related by $\mathbf{p} = \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\rho}$. With the 4×4 algebra element $\boldsymbol{\xi}^\wedge = \begin{bmatrix} \lfloor\boldsymbol{\theta}\rfloor_\times & \boldsymbol{\rho} \\ \mathbf{0}_{1\times3} & 0 \end{bmatrix}$,

$$\mathbf{T} = \mathrm{Exp}(\boldsymbol{\xi}) = \exp\begin{bmatrix} \lfloor\boldsymbol{\theta}\rfloor_\times & \boldsymbol{\rho} \\ \mathbf{0} & 0 \end{bmatrix} \tag{179}$$

$$= \begin{bmatrix} \exp(\lfloor\boldsymbol{\theta}\rfloor_\times) & \mathbf{J}_l(\boldsymbol{\theta})\,\boldsymbol{\rho} \\ \mathbf{0}_{1\times3} & 1 \end{bmatrix} \tag{180}$$

$$= \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}, \qquad \mathbf{p} = \mathbf{J}_l(\boldsymbol{\theta})\,\boldsymbol{\rho}, \tag{181}$$

and inverting,

$$\boldsymbol{\xi} = \mathrm{Log}(\mathbf{T})^\vee, \qquad \log(\mathbf{T}) = \log\begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \tag{182}$$

$$= \begin{bmatrix} \log(\mathbf{R}) & \mathbf{J}_l^{-1}(\boldsymbol{\theta})\mathbf{p} \\ \mathbf{0} & 0 \end{bmatrix} \;\Rightarrow\; \boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{\theta} \\ \mathbf{J}_l^{-1}(\boldsymbol{\theta})\mathbf{p} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{\rho} \end{bmatrix}. \tag{183}$$

Discussion. Prove $\exp\left(\begin{bmatrix} \lfloor\boldsymbol{\theta}\rfloor_\times & \boldsymbol{\rho} \\ \mathbf{0} & 0 \end{bmatrix} \right) = \begin{bmatrix} \mathbf{R} & \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix}$.

Left/right Jacobians. With $\mathrm{Exp}(\boldsymbol{\xi} + \delta\boldsymbol{\xi}) = \mathrm{Exp}(\boldsymbol{\xi})\,\mathrm{Exp}(\mathbf{J}_R(\boldsymbol{\xi})\delta\boldsymbol{\xi}) = \mathrm{Exp}(\mathbf{J}_L(\boldsymbol{\xi})\delta\boldsymbol{\xi})\,\mathrm{Exp}(\boldsymbol{\xi})$,

$$\mathbf{J}_R(\boldsymbol{\xi}) = \begin{bmatrix} \mathbf{J}_R(\boldsymbol{\theta}) & \mathbf{0} \\ \mathbf{Q}_R(\boldsymbol{\xi}) & \mathbf{J}_R(\boldsymbol{\theta}) \end{bmatrix}_{6\times6}, \quad \mathbf{J}_L(\boldsymbol{\xi}) = \begin{bmatrix} \mathbf{J}_L(\boldsymbol{\theta}) & \mathbf{0} \\ \mathbf{Q}_L(\boldsymbol{\xi}) & \mathbf{J}_L(\boldsymbol{\theta}) \end{bmatrix}_{6\times6}, \tag{184}$$

and to first order, with $\boldsymbol{\xi}^\curlywedge = \begin{bmatrix} \lfloor\boldsymbol{\theta}\rfloor_\times & \mathbf{0} \\ \lfloor\boldsymbol{\rho}\rfloor_\times & \lfloor\boldsymbol{\theta}\rfloor_\times \end{bmatrix}_{6\times6}$,

$$\mathbf{J}_L(\boldsymbol{\xi}) \approx \mathbf{I}_6 + \tfrac{1}{2}\boldsymbol{\xi}^\curlywedge, \qquad \mathbf{J}_R(\boldsymbol{\xi}) \approx \mathbf{I}_6 - \tfrac{1}{2}\boldsymbol{\xi}^\curlywedge. \tag{185}$$

Homework. Find $\mathbf{Q}_L$ (and $\mathbf{Q}_R$).

Error states: SE(3) vs SO(3) $+\mathbb{R}^3$. Perturb $\mathbf{T}$ on the right by the SE(3) tangent $\delta\boldsymbol{\xi} = [\delta\boldsymbol{\theta}^\top,\ \delta\boldsymbol{\rho}^\top]^\top$. Expanding with $\mathbf{J}_l(\delta\boldsymbol{\theta}) \approx \mathbf{I} + \frac{1}{2}\lfloor\delta\boldsymbol{\theta}\rfloor_\times$ and $\delta\mathbf{R} = \mathrm{Exp}(\delta\boldsymbol{\theta})$,

$$\mathbf{T}\,\delta\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \mathrm{Exp}(\delta\boldsymbol{\theta}) & \mathbf{J}_l(\delta\boldsymbol{\theta})\delta\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix} \tag{186}$$

$$= \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & (\mathbf{I} + \frac{1}{2}\lfloor\delta\boldsymbol{\theta}\rfloor_\times)\delta\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix} \tag{187}$$

$$\approx \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \delta\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix} \tag{188}$$

$$= \begin{bmatrix} \mathbf{R}\delta\mathbf{R} & \mathbf{R}\delta\boldsymbol{\rho} + \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}. \tag{189}$$

If instead we treat the pose as $SO(3) + \mathbb{R}^3$ (rotation and translation perturbed separately, $\mathbf{p} \to \mathbf{p} + \delta\mathbf{p}$),

$$\mathbf{T} = \begin{bmatrix} \mathbf{R}\delta\mathbf{R} & \mathbf{p} + \delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \tag{190}$$

$$= \begin{bmatrix} \mathbf{R}\delta\mathbf{R} & \mathbf{p} + \mathbf{R}\mathbf{R}^\top\delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \tag{191}$$

$$= \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \mathbf{R}^\top\delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}. \tag{192}$$

Comparing the two conventions gives the relation between the error states (SE(3) tangent translation $\delta\boldsymbol{\rho}$ vs. naive position error $\delta\mathbf{p}$),

$$\begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}_{SO(3)+\mathbb{R}^3} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \delta\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix}_{SE(3)} \Rightarrow \begin{cases} \delta\boldsymbol{\rho} = \mathbf{R}^\top\delta\mathbf{p} \\ \delta\mathbf{R}_{SE(3)} = \delta\mathbf{R} \end{cases}. \tag{193}$$

Adjoint. The adjoint transfers algebra elements across the group:

$$SO(3): \quad \mathbf{R}\,\mathrm{Exp}(\boldsymbol{\varphi})\,\mathbf{R}^\top = \mathrm{Exp}(\mathbf{R}\boldsymbol{\varphi}) \;\Leftrightarrow\; \mathrm{Ad}_{\mathbf{R}} = \mathbf{R}, \qquad SE(3): \quad \mathbf{T}\,\mathrm{Exp}(\boldsymbol{\xi})\,\mathbf{T}^{-1} = \mathrm{Exp}(\mathrm{Ad}_{\mathbf{T}}\,\boldsymbol{\xi}), \tag{194}$$

with $\mathrm{Ad}_{\mathbf{T}} = \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ \lfloor \mathbf{p} \rfloor_\times \mathbf{R} & \mathbf{R} \end{bmatrix}_{6\times 6}$. The adjoint is itself a Lie group, with $\mathrm{Ad}_{\mathbf{T}} = \exp(\boldsymbol{\xi}^\curlywedge)$ and $\log(\mathrm{Ad}_{\mathbf{T}})^\vee = \boldsymbol{\xi}$.

Homework. Prove $\mathbf{T}\,\mathrm{Exp}(\boldsymbol{\xi})\,\mathbf{T}^{-1} = \mathrm{Exp}(\mathrm{Ad}_{\mathbf{T}}\,\boldsymbol{\xi})$.

2.7.1 More operations

(1) Interpolation. For ${}^G_{t_1}\mathbf{T}, {}^G_{t_2}\mathbf{T}$, $t \in [t_1, t_2]$, $\lambda = \frac{t-t_1}{t_2-t_1}$: in $SE(3)$, with ${}^{t_1}_{t_2}\boldsymbol{\xi} = \mathrm{Log}({}^G_{t_1}\mathbf{T}^{-1}\,{}^G_{t_2}\mathbf{T})$,

$${}^G_t\mathbf{T} = {}^G_{t_1}\mathbf{T}\;\mathrm{Exp}\big(\lambda\,{}^{t_1}_{t_2}\boldsymbol{\xi}\big); \tag{195}$$

in $SO(3) + \mathbb{R}^3$, the rotation interpolates on the manifold and the position linearly:

$${}^G_t\mathbf{R} = {}^G_{t_1}\mathbf{R}\;\mathrm{Exp}\big(\lambda\,\mathrm{Log}({}^G_{t_1}\mathbf{R}^\top\,{}^G_{t_2}\mathbf{R})\big), \qquad {}^G\mathbf{p}_t = {}^G\mathbf{p}_{t_1} + \lambda({}^G\mathbf{p}_{t_2} - {}^G\mathbf{p}_{t_1}). \tag{196}$$

(2) Time derivative. For $SE(3)$, with $\mathbf{T} = \mathbf{T}\,\mathrm{Exp}\big([\delta\boldsymbol{\theta};\delta\boldsymbol{\rho}]^\wedge\big)$ and body velocity, expand the limit using $\exp(\lfloor\delta\boldsymbol{\theta}\rfloor_\times) - \mathbf{I}_3 \approx \lfloor\delta\boldsymbol{\theta}\rfloor_\times$ and $\mathbf{J}_l(\delta\boldsymbol{\theta}) \approx \mathbf{I} + \frac{1}{2}\lfloor\delta\boldsymbol{\theta}\rfloor_\times$:

$$\begin{aligned}
\frac{\mathrm{d}\mathbf{T}}{\mathrm{d}t} &= \lim_{\delta t\to 0}\frac{\mathbf{T}(t+\delta t)-\mathbf{T}(t)}{\delta t} && (197)\\
&= \lim_{\delta t\to 0}\frac{1}{\delta t}\left(\begin{bmatrix}\mathbf{R} & \mathbf{p}\\ \mathbf{0} & 1\end{bmatrix}\begin{bmatrix}\exp(\lfloor\delta\boldsymbol{\theta}\rfloor_\times) & \mathbf{J}_l(\delta\boldsymbol{\theta})\delta\boldsymbol{\rho}\\ \mathbf{0} & 1\end{bmatrix} - \begin{bmatrix}\mathbf{R} & \mathbf{p}\\ \mathbf{0} & 1\end{bmatrix}\right) && (198)\\
&= \lim_{\delta t\to 0}\frac{1}{\delta t}\begin{bmatrix}\mathbf{R} & \mathbf{p}\\ \mathbf{0} & 1\end{bmatrix}\begin{bmatrix}\exp(\lfloor\delta\boldsymbol{\theta}\rfloor_\times)-\mathbf{I}_3 & \mathbf{J}_l(\delta\boldsymbol{\theta})\delta\boldsymbol{\rho}\\ \mathbf{0} & 0\end{bmatrix} && (199)\\
&= \lim_{\delta t\to 0}\frac{1}{\delta t}\begin{bmatrix}\mathbf{R} & \mathbf{p}\\ \mathbf{0} & 1\end{bmatrix}\begin{bmatrix}\lfloor\delta\boldsymbol{\theta}\rfloor_\times & \delta\boldsymbol{\rho}\\ \mathbf{0} & 0\end{bmatrix} && (200)\\
&= \begin{bmatrix}\mathbf{R} & \mathbf{p}\\ \mathbf{0} & 1\end{bmatrix}\begin{bmatrix}\lfloor\boldsymbol{\omega}\rfloor_\times & \mathbf{v}\\ \mathbf{0} & 0\end{bmatrix} && (201)\\
&= \begin{bmatrix}\mathbf{R}\lfloor\boldsymbol{\omega}\rfloor_\times & \mathbf{R}\mathbf{v}\\ \mathbf{0} & 0\end{bmatrix}, && (202)
\end{aligned}$$

where $\mathbf{v}$ is the body velocity. For $SO(3)+\mathbb{R}^3$ the same computation with

$$\mathbf{T} = \begin{bmatrix}\mathbf{R}\exp(\lfloor\delta\boldsymbol{\theta}\rfloor_\times) & \mathbf{p}+\delta\mathbf{p}\\ \mathbf{0} & 1\end{bmatrix} \tag{203}$$

gives

$$\frac{\mathrm{d}\mathbf{T}}{\mathrm{d}t} = \begin{bmatrix}\mathbf{R}\lfloor\boldsymbol{\omega}\rfloor_\times & {}^{G}\mathbf{v}\\ \mathbf{0} & 0\end{bmatrix}, \tag{204}$$

with ${}^{G}\mathbf{v}$ the global velocity.

(3) Calibration. For IMU frame $\{I\}$ and camera frame $\{C\}$,

$$ {}^{C_1}_{C_2}\mathbf{T} = {}^{C}_{I}\mathbf{T}\,{}^{I_1}_{I_2}\mathbf{T}\,{}^{C}_{I}\mathbf{T}^{-1}, \qquad \mathrm{Exp}({}^{C_1}_{C_2}\boldsymbol{\xi}) = \mathrm{Exp}\big(\mathrm{Ad}_{({}^{C}_{I}\mathbf{T})}\,{}^{I_1}_{I_2}\boldsymbol{\xi}\big) \;\Rightarrow\; {}^{C_1}_{C_2}\boldsymbol{\xi} = \mathrm{Ad}_{({}^{C}_{I}\mathbf{T})}\,{}^{I_1}_{I_2}\boldsymbol{\xi}. \tag{205}$$

Homework. Compute the Jacobians $\partial\,{}^{C_1}_{C_2}\boldsymbol{\xi}/\partial\,{}^{I_1}_{I_2}\boldsymbol{\xi}$ and $\partial\,{}^{C_1}_{C_2}\boldsymbol{\xi}/\partial\,{}^{C}_{I}\boldsymbol{\xi}$ (and the corresponding $SO(3)+\mathbb{R}^3$ versions).

2.8 SE$_2$(3)

Augmenting the pose with a velocity column gives the 5×5 group element

$$\boldsymbol{\Upsilon} = \begin{bmatrix}\mathbf{R} & \mathbf{p} & \mathbf{v}\\ \mathbf{0}_{1\times 3} & 1 & 0\\ \mathbf{0}_{1\times 3} & 0 & 1\end{bmatrix}, \qquad \boldsymbol{\xi} = \begin{bmatrix}\boldsymbol{\theta}\\ \boldsymbol{\rho}\\ \boldsymbol{\nu}\end{bmatrix} \to \boldsymbol{\xi}^\wedge = \begin{bmatrix}\lfloor\boldsymbol{\theta}\rfloor_\times & \boldsymbol{\rho} & \boldsymbol{\nu}\\ \mathbf{0} & 0 & 0\\ \mathbf{0} & 0 & 0\end{bmatrix}_{5\times 5}, \tag{206}$$

where (as in SE(3)) the tangent components $\boldsymbol{\rho},\boldsymbol{\nu}$ are distinct from the group position/velocity $\mathbf{p} = \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\rho}$, $\mathbf{v} = \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\nu}$. The exponential, logarithm, and adjoint follow the same pattern as $SE(3)$:

$$\mathrm{Exp}(\boldsymbol{\xi}) = \begin{bmatrix}\mathbf{R} & \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\rho} & \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\nu}\\ \mathbf{0} & 1 & 0\\ \mathbf{0} & 0 & 1\end{bmatrix} = \begin{bmatrix}\mathbf{R} & \mathbf{p} & \mathbf{v}\\ \mathbf{0} & 1 & 0\\ \mathbf{0} & 0 & 1\end{bmatrix}, \tag{207}$$

$$\mathrm{Log}(\boldsymbol{\Upsilon})^\vee = \begin{bmatrix}\boldsymbol{\theta}\\ \mathbf{J}_l^{-1}(\boldsymbol{\theta})\mathbf{p}\\ \mathbf{J}_l^{-1}(\boldsymbol{\theta})\mathbf{v}\end{bmatrix}, \qquad \mathrm{Ad}_{\boldsymbol{\Upsilon}} = \begin{bmatrix}\mathbf{R} & \mathbf{0} & \mathbf{0}\\ \lfloor\mathbf{p}\rfloor_\times\mathbf{R} & \mathbf{R} & \mathbf{0}\\ \lfloor\mathbf{v}\rfloor_\times\mathbf{R} & \mathbf{0} & \mathbf{R}\end{bmatrix}. \tag{208}$$

2.9 Quaternions

Quaternions are presented last because $SO(3)$, $SE(3)$, and the Lie-group view already cover most needs. There are two main conventions — *Hamilton* (right-handed) and *JPL* (left-handed); we use the **Hamilton** convention. For a thorough treatment see Solà [6] (Hamilton) and Trawny & Roumeliotis [5] (JPL).

A quaternion and its imaginary units satisfy

$$Q = q_w + q_x i + q_y j + q_z k, \qquad i^2 = j^2 = k^2 = ijk = -1, \quad ij = k,\ jk = i,\ ki = j, \tag{209}$$

written as the 4×1 vector $\bar{q} = [q_w,\ q_x,\ q_y,\ q_z]^\top = [q_w,\ \mathbf{q}_v^\top]^\top$.

Product. The quaternion product is bilinear and can be written with left/right matrices:

$$\bar{p} \otimes \bar{q} = \mathcal{L}(\bar{p})\,\bar{q} = \mathcal{R}(\bar{q})\,\bar{p}, \tag{210}$$

$$\mathcal{L}(\bar{p}) = p_w \mathbf{I}_4 + \begin{bmatrix} 0 & -\mathbf{p}_v^\top \\ \mathbf{p}_v & \lfloor \mathbf{p}_v \rfloor_\times \end{bmatrix}, \tag{211}$$

$$\mathcal{R}(\bar{q}) = q_w \mathbf{I}_4 + \begin{bmatrix} 0 & -\mathbf{q}_v^\top \\ \mathbf{q}_v & -\lfloor \mathbf{q}_v \rfloor_\times \end{bmatrix}. \tag{212}$$

Identity, unit quaternion, inverse. The identity is $\bar{q} = [1,\ \mathbf{0}_{3\times1}^\top]^\top$. A rotation by angle θ about unit axis $\mathbf{k}$ is the unit quaternion

$$\bar{q} = \begin{bmatrix} \cos\frac{\theta}{2} \\ \sin\frac{\theta}{2}\,\mathbf{k} \end{bmatrix}, \qquad \bar{q}^{-1} = \begin{bmatrix} \cos\frac{\theta}{2} \\ -\sin\frac{\theta}{2}\,\mathbf{k} \end{bmatrix}, \qquad \mathbf{R}_{12} = \mathbf{R}_1\mathbf{R}_2 \ \leftrightarrow\ \bar{q}_{12} = \bar{q}_1 \otimes \bar{q}_2. \tag{213}$$

Why $\theta/2$? — rotating a vector. With global frame $\{G\}$ and IMU frame $\{I\}$, rotate ${}^I\mathbf{p}$ via the sandwich product and convert to matrices with $\bar{p} \otimes \bar{q} = \mathcal{R}(\bar{q})\bar{p}$ and $\bar{p} \otimes \bar{q} = \mathcal{L}(\bar{p})\bar{q}$:

$$\begin{bmatrix} 0 \\ {}^G\mathbf{p} \end{bmatrix} = {}^G_I\bar{q} \otimes \begin{bmatrix} 0 \\ {}^I\mathbf{p} \end{bmatrix} \otimes {}^G_I\bar{q}^{-1} \tag{214}$$

$$= \mathcal{R}({}^G_I\bar{q}^{-1})\,\mathcal{L}({}^G_I\bar{q}) \begin{bmatrix} 0 \\ {}^I\mathbf{p} \end{bmatrix}. \tag{215}$$

Writing $\bar{q} = [q_w, \mathbf{q}_v^\top]^\top$ and multiplying the two matrices,

$$\begin{aligned} \mathcal{R}(\bar{q}^{-1})\mathcal{L}(\bar{q}) &= \left(q_w\mathbf{I}_4 + \begin{bmatrix} 0 & \mathbf{q}_v^\top \\ -\mathbf{q}_v & \lfloor \mathbf{q}_v \rfloor_\times \end{bmatrix} \right) \left(q_w\mathbf{I}_4 + \begin{bmatrix} 0 & -\mathbf{q}_v^\top \\ \mathbf{q}_v & \lfloor \mathbf{q}_v \rfloor_\times \end{bmatrix} \right) \\ &= q_w^2 \mathbf{I}_4 + 2q_w \begin{bmatrix} 0 & \mathbf{0} \\ \mathbf{0} & \lfloor \mathbf{q}_v \rfloor_\times \end{bmatrix} + \begin{bmatrix} \mathbf{q}_v^\top\mathbf{q}_v & \mathbf{0} \\ \mathbf{0} & -\mathbf{q}_v\mathbf{q}_v^\top + \lfloor \mathbf{q}_v \rfloor_\times^2 \end{bmatrix}, \end{aligned} \tag{216}$$

whose lower-right 3×3 block (acting on ${}^I\mathbf{p}$) gives

$${}^G\mathbf{p} = \left(q_w^2\mathbf{I}_3 + 2q_w \lfloor \mathbf{q}_v \rfloor_\times + \mathbf{q}_v\mathbf{q}_v^\top + \lfloor \mathbf{q}_v \rfloor_\times^2 \right) {}^I\mathbf{p}. \tag{217}$$

Using $\lfloor \mathbf{q}_v \rfloor_\times^2 = \mathbf{q}_v\mathbf{q}_v^\top - (\mathbf{q}_v^\top\mathbf{q}_v)\mathbf{I}_3$ and the unit-norm condition $q_w^2 + \mathbf{q}_v^\top\mathbf{q}_v = 1$,

$$q_w^2\mathbf{I}_3 + \mathbf{q}_v\mathbf{q}_v^\top + \lfloor \mathbf{q}_v \rfloor_\times^2 = q_w^2\mathbf{I}_3 + (\mathbf{q}_v^\top\mathbf{q}_v)\mathbf{I}_3 + 2\lfloor \mathbf{q}_v \rfloor_\times^2 \tag{218}$$

$$= \mathbf{I}_3 + 2\lfloor \mathbf{q}_v \rfloor_\times^2, \tag{219}$$

so

$$ {}^{G}\mathbf{p} = \big(\mathbf{I}_3 + 2q_w \lfloor \mathbf{q}_v \rfloor_\times + 2\lfloor \mathbf{q}_v \rfloor_\times^2\big)\, {}^{I}\mathbf{p}. \tag{220} $$

Finally substitute $q_w = \cos\frac{\theta}{2}$, $\mathbf{q}_v = \sin\frac{\theta}{2}\mathbf{k}$, so $2q_w \lfloor \mathbf{q}_v \rfloor_\times = 2\cos\frac{\theta}{2}\sin\frac{\theta}{2}\lfloor \mathbf{k} \rfloor_\times = \sin\theta \lfloor \mathbf{k} \rfloor_\times$ and $2\lfloor \mathbf{q}_v \rfloor_\times^2 = 2\sin^2\frac{\theta}{2}\lfloor \mathbf{k} \rfloor_\times^2 = (1-\cos\theta)\lfloor \mathbf{k} \rfloor_\times^2$:

$$ \begin{aligned} {}^{G}\mathbf{p} &= \big(\mathbf{I}_3 + \sin\theta \lfloor \mathbf{k} \rfloor_\times + (1-\cos\theta)\lfloor \mathbf{k} \rfloor_\times^2\big)\, {}^{I}\mathbf{p} && (221)\\ &= {}^{G}_{I}\mathbf{R}\, {}^{I}\mathbf{p} && (222)\\ &= \mathrm{Exp}(\theta\mathbf{k})\, {}^{I}\mathbf{p}, && (223) \end{aligned} $$

recovering Rodrigues' formula — which is why the half-angle $\theta/2$ appears.

Derivative. With $\delta\boldsymbol{\theta} = \delta\theta\,\mathbf{k}$ and small-angle $\delta\bar{q} = [\cos\frac{\delta\theta}{2},\ \sin\frac{\delta\theta}{2}\mathbf{k}^\top]^\top \approx [1,\ \frac{1}{2}\delta\boldsymbol{\theta}^\top]^\top$,

$$ \begin{aligned} {}^{G}_{t}\dot{\bar{q}} &= \lim_{\Delta t\to 0} \frac{{}^{G}_{t+\Delta t}\bar{q} - {}^{G}_{t}\bar{q}}{\Delta t} && (224)\\ &= \lim_{\Delta t\to 0} \frac{{}^{G}_{t}\bar{q} \otimes \begin{bmatrix} \cos\frac{\delta\theta}{2} \\ \sin\frac{\delta\theta}{2}\mathbf{k} \end{bmatrix} - {}^{G}_{t}\bar{q} \otimes \begin{bmatrix} 1 \\ \mathbf{0} \end{bmatrix}}{\Delta t} && (225)\\ &= \lim_{\Delta t\to 0} \frac{{}^{G}_{t}\bar{q} \otimes \begin{bmatrix} 0 \\ \frac{1}{2}\delta\boldsymbol{\theta} \end{bmatrix}}{\Delta t} && (226)\\ &= {}^{G}_{t}\bar{q} \otimes \begin{bmatrix} 0 \\ \frac{1}{2}\boldsymbol{\omega} \end{bmatrix} && (227)\\ &= \frac{1}{2}\,\mathcal{R}\left(\begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix}\right) {}^{G}_{t}\bar{q} && (228)\\ &= \frac{1}{2}\begin{bmatrix} 0 & -\boldsymbol{\omega}^\top \\ \boldsymbol{\omega} & -\lfloor \boldsymbol{\omega} \rfloor_\times \end{bmatrix} {}^{G}_{t}\bar{q} && (229)\\ &= \frac{1}{2}\,\Omega(\boldsymbol{\omega})\, {}^{G}_{t}\bar{q}, && (230) \end{aligned} $$

where $\Omega(\boldsymbol{\omega}) \triangleq \begin{bmatrix} 0 & -\boldsymbol{\omega}^\top \\ \boldsymbol{\omega} & -\lfloor \boldsymbol{\omega} \rfloor_\times \end{bmatrix}$.

Integration. Given ${}^{G}_{t}\bar{q}$, $\boldsymbol{\omega}$, δt, with $\delta\boldsymbol{\theta} = \boldsymbol{\omega}\,\delta t = \delta\theta\,\mathbf{k}$ and $\delta\bar{q} = [\cos\frac{\delta\theta}{2},\ \sin\frac{\delta\theta}{2}\mathbf{k}^\top]^\top$,

$$ {}^{G}_{t+\delta t}\bar{q} = {}^{G}_{t}\bar{q} \otimes \delta\bar{q}. \tag{231} $$

3 Lecture 3: Camera Model, Triangulation, Point/Line/Plane

Topics. (1) camera measurements; (2) essential matrix / triangulation; (3) point / line / plane representations.

3.1 The camera measurement model

A camera produces pixel measurements that depend on the camera pose and the observed scene:

$$\mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n}_c, \tag{232}$$

where $\mathbf{z}_c$ are the pixel measurements, $\mathbf{x}$ is the state vector (camera pose and point feature), $\mathbf{h}_c(\cdot)$ is the camera model, and $\mathbf{n}_c$ is the pixel noise. The geometry is shown in Fig. 4.

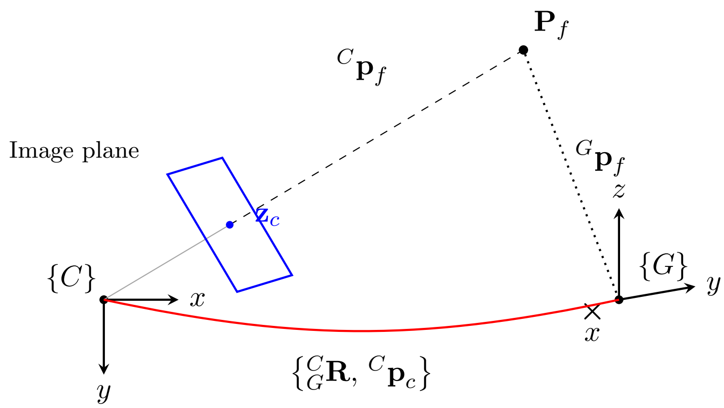


Figure 4: Camera measurement geometry: the feature $\mathbf{P}_f$ projects through the image plane to the pixel $\mathbf{z}_c$ in the camera frame $\{C\}$; the global frame $\{G\}$ and the extrinsic $\{{}^C_G\mathbf{R}, {}^C\mathbf{p}_c\}$ relate the two views of the same point (${}^C\mathbf{p}_f$ vs. ${}^G\mathbf{p}_f$).

Simplest pinhole model. With the feature in the camera frame ${}^C\mathbf{p}_f = [x,\ y,\ z]^\top$, the normalized projection is

$$\mathbf{z}_n = \begin{bmatrix} u_n \\ v_n \end{bmatrix} = \begin{bmatrix} x/z \\ y/z \end{bmatrix}, \qquad \mathbf{z}_n = \mathbf{h}_p({}^C\mathbf{p}_f) + \mathbf{n}_c, \quad {}^C\mathbf{p}_f = {}^C_G\mathbf{R}^\top\big({}^G\mathbf{p}_f - {}^G\mathbf{p}_c\big). \tag{233}$$

Linearizing the measurement, $\tilde{\mathbf{z}}_n = \mathbf{H}_c\,\tilde{\mathbf{x}} + \mathbf{n}_c$, with the residual

$$\tilde{\mathbf{z}}_n = \mathbf{z}_n - \mathbf{h}_p({}^C\hat{\mathbf{p}}_f) = \begin{bmatrix} u_n \\ v_n \end{bmatrix} - \begin{bmatrix} \hat{x}/\hat{z} \\ \hat{y}/\hat{z} \end{bmatrix}. \tag{234}$$

The state and its error state are

$$\mathbf{x} : \{{}^C_G\mathbf{R},\ {}^G\mathbf{p}_c,\ {}^G\mathbf{p}_f\}, \quad {}^C_G\mathbf{R} = {}^C_G\hat{\mathbf{R}}\ \mathrm{Exp}(\delta\boldsymbol{\theta}), \quad {}^G\mathbf{p}_c = {}^G\hat{\mathbf{p}}_c + {}^G\tilde{\mathbf{p}}_c, \quad {}^G\mathbf{p}_f = {}^G\hat{\mathbf{p}}_f + {}^G\tilde{\mathbf{p}}_f, \tag{235}$$

with error state $\tilde{\mathbf{x}} = [\,\delta\boldsymbol{\theta}^\top,\ {}^G\tilde{\mathbf{p}}_c^\top,\ {}^G\tilde{\mathbf{p}}_f^\top\,]^\top$.

3.2 The measurement Jacobian

By the chain rule, factoring every block through ${}^{C}\tilde{\mathbf{p}}_f$,

$$\begin{aligned}
\tilde{\mathbf{z}}_n &= \mathbf{H}_c\,\tilde{\mathbf{x}} + \mathbf{n}_c && (236)\\
&= \begin{bmatrix} \dfrac{\partial \tilde{\mathbf{z}}_n}{\partial \delta\boldsymbol{\theta}} & \dfrac{\partial \tilde{\mathbf{z}}_n}{\partial {}^{G}\tilde{\mathbf{p}}_c} & \dfrac{\partial \tilde{\mathbf{z}}_n}{\partial {}^{G}\tilde{\mathbf{p}}_f} \end{bmatrix} \tilde{\mathbf{x}} + \mathbf{n}_c && (237)\\
&= \frac{\partial \tilde{\mathbf{z}}_n}{\partial {}^{C}\tilde{\mathbf{p}}_f} \begin{bmatrix} \dfrac{\partial {}^{C}\tilde{\mathbf{p}}_f}{\partial \delta\boldsymbol{\theta}} & \dfrac{\partial {}^{C}\tilde{\mathbf{p}}_f}{\partial {}^{G}\tilde{\mathbf{p}}_c} & \dfrac{\partial {}^{C}\tilde{\mathbf{p}}_f}{\partial {}^{G}\tilde{\mathbf{p}}_f} \end{bmatrix} \tilde{\mathbf{x}} + \mathbf{n}_c. && (238)
\end{aligned}$$

The projection Jacobian, writing ${}^{C}\mathbf{p}_f = [c_x,\, c_y,\, c_z]^\top$, is

$$\mathbf{z}_n = \begin{bmatrix} c_x/c_z \\ c_y/c_z \end{bmatrix}, \qquad \frac{\partial \mathbf{z}_n}{\partial {}^{C}\mathbf{p}_f} = \begin{bmatrix} \dfrac{1}{c_z} & 0 & -\dfrac{c_x}{c_z^2} \\ 0 & \dfrac{1}{c_z} & -\dfrac{c_y}{c_z^2} \end{bmatrix} = \frac{1}{c_z^2} \begin{bmatrix} c_z & 0 & -c_x \\ 0 & c_z & -c_y \end{bmatrix}. \tag{239}$$

Perturbing ${}^{C}\mathbf{p}_f$. From ${}^{C}\mathbf{p}_f = {}^{C}_{G}\mathbf{R}^\top({}^{G}\mathbf{p}_f - {}^{G}\mathbf{p}_c)$ with ${}^{C}_{G}\mathbf{R} = {}^{C}_{G}\hat{\mathbf{R}}\,\mathrm{Exp}(\delta\boldsymbol{\theta})$ and $\mathrm{Exp}(\delta\boldsymbol{\theta})^\top \approx \mathbf{I} - \lfloor \delta\boldsymbol{\theta} \rfloor_\times$,

$$\begin{aligned}
{}^{C}\hat{\mathbf{p}}_f + {}^{C}\tilde{\mathbf{p}}_f &= \big({}^{C}_{G}\hat{\mathbf{R}}\,\mathrm{Exp}(\delta\boldsymbol{\theta})\big)^\top \big({}^{G}\hat{\mathbf{p}}_f + {}^{G}\tilde{\mathbf{p}}_f - {}^{G}\hat{\mathbf{p}}_c - {}^{G}\tilde{\mathbf{p}}_c\big) && (240)\\
&= \big(\mathbf{I} - \lfloor \delta\boldsymbol{\theta} \rfloor_\times\big)\, {}^{C}_{G}\hat{\mathbf{R}}^\top \big({}^{G}\hat{\mathbf{p}}_f + {}^{G}\tilde{\mathbf{p}}_f - {}^{G}\hat{\mathbf{p}}_c - {}^{G}\tilde{\mathbf{p}}_c\big) && (241)\\
&\approx {}^{C}_{G}\hat{\mathbf{R}}^\top \big({}^{G}\hat{\mathbf{p}}_f - {}^{G}\hat{\mathbf{p}}_c\big) - \lfloor \delta\boldsymbol{\theta} \rfloor_\times\, {}^{C}_{G}\hat{\mathbf{R}}^\top \big({}^{G}\hat{\mathbf{p}}_f - {}^{G}\hat{\mathbf{p}}_c\big) \\
&\quad + {}^{C}_{G}\hat{\mathbf{R}}^\top \big({}^{G}\tilde{\mathbf{p}}_f - {}^{G}\tilde{\mathbf{p}}_c\big). && (242)
\end{aligned}$$

Using $-\lfloor \delta\boldsymbol{\theta} \rfloor_\times\, \mathbf{a} = \lfloor \mathbf{a} \rfloor_\times\, \delta\boldsymbol{\theta}$ with $\mathbf{a} = {}^{C}\hat{\mathbf{p}}_f = {}^{C}_{G}\hat{\mathbf{R}}^\top({}^{G}\hat{\mathbf{p}}_f - {}^{G}\hat{\mathbf{p}}_c)$ and matching hat/tilde terms,

$$\begin{aligned}
{}^{C}\hat{\mathbf{p}}_f + {}^{C}\tilde{\mathbf{p}}_f &= \underbrace{{}^{C}_{G}\hat{\mathbf{R}}^\top \big({}^{G}\hat{\mathbf{p}}_f - {}^{G}\hat{\mathbf{p}}_c\big)}_{{}^{C}\hat{\mathbf{p}}_f} + \Big\lfloor {}^{C}_{G}\hat{\mathbf{R}}^\top ({}^{G}\hat{\mathbf{p}}_f - {}^{G}\hat{\mathbf{p}}_c) \Big\rfloor_\times\, \delta\boldsymbol{\theta} \\
&\quad - {}^{C}_{G}\hat{\mathbf{R}}^\top\, {}^{G}\tilde{\mathbf{p}}_c + {}^{C}_{G}\hat{\mathbf{R}}^\top\, {}^{G}\tilde{\mathbf{p}}_f, && (243)
\end{aligned}$$

so that, with ${}^{C}\hat{\mathbf{p}}_f = {}^{C}_{G}\hat{\mathbf{R}}^\top({}^{G}\hat{\mathbf{p}}_f - {}^{G}\hat{\mathbf{p}}_c)$, the three blocks are

$$\frac{\partial {}^{C}\tilde{\mathbf{p}}_f}{\partial \delta\boldsymbol{\theta}} = \big\lfloor {}^{C}\hat{\mathbf{p}}_f \big\rfloor_\times, \qquad \frac{\partial {}^{C}\tilde{\mathbf{p}}_f}{\partial {}^{G}\tilde{\mathbf{p}}_c} = -{}^{C}_{G}\hat{\mathbf{R}}^\top, \qquad \frac{\partial {}^{C}\tilde{\mathbf{p}}_f}{\partial {}^{G}\tilde{\mathbf{p}}_f} = {}^{C}_{G}\hat{\mathbf{R}}^\top. \tag{244}$$

Therefore $\tilde{\mathbf{z}}_n = \mathbf{H}_c\,\tilde{\mathbf{x}} + \mathbf{n}_c$ with

$$\mathbf{H}_c = \frac{1}{c_z^2} \begin{bmatrix} c_z & 0 & -c_x \\ 0 & c_z & -c_y \end{bmatrix} \begin{bmatrix} \lfloor {}^{C}\hat{\mathbf{p}}_f \rfloor_\times & -{}^{C}_{G}\hat{\mathbf{R}}^\top & {}^{C}_{G}\hat{\mathbf{R}}^\top \end{bmatrix}. \tag{245}$$

3.3 A few important discussions

(1) SLAM. The pose $\{{}^{C}_{G}\mathbf{R}, {}^{G}\mathbf{p}_c\}$ is the localization of the camera, and ${}^{G}\mathbf{p}_f$ is a 3D feature of the environment. Solving them together means we solve localization and mapping together $\Rightarrow$ *Simultaneous Localization And Mapping* (SLAM).

(2) PnP / P3P — known feature, solve pose. If ${}^{G}\mathbf{p}_f$ is known, given (u, v), how do we solve $\{{}^{C}_{G}\mathbf{R}, {}^{C}\mathbf{p}_c\}$? Here $\mathbf{x} : \{{}^{C}_{G}\mathbf{R}, {}^{C}\mathbf{p}_c\}$ is 6-dof, while each measurement $\mathbf{z}_n = \mathbf{h}({}^{C}_{G}\mathbf{R}, {}^{C}\mathbf{p}_c) + \mathbf{n}_c$ contributes 2 dof, so we need at least 3 points ${}^{C}\mathbf{p}_{f_i} \leftrightarrow (u_i, v_i)$ to solve $\{{}^{C}\mathbf{R}, {}^{C}\mathbf{p}_c\}$.

- 3 pairs ${}^{C}\mathbf{p}_{f_i} \leftrightarrow (u_i, v_i)$, **P3P** (perspective-3-point), i.e. 2+1 points: use 3D↔3D correspondences to determine the same 3D pose.
- n pairs of 3D↔3D correspondences, **PnP** (perspective-n-point).
- **RANSAC** (random sample consensus) for outlier rejection.

This is the *structure* → *motion* direction (known structure, solve motion).

(3) Triangulation — known poses, solve feature. If $\{{}^{C}_{G}\mathbf{R}, {}^{C}\mathbf{p}_c\}$ is known, given (u_1, v_1), (u_2, v_2), solve $\mathbf{P}_f$ (Fig. 5). Each pixel gives a unit bearing and an unknown range r_i:

$$(u_1, v_1) \to {}^{C_1}\mathbf{b}_1 = \frac{[u_1, v_1, 1]^\top}{\|[u_1, v_1, 1]^\top\|}, \quad (u_2, v_2) \to {}^{C_2}\mathbf{b}_2 = \frac{[u_2, v_2, 1]^\top}{\|[u_2, v_2, 1]^\top\|}, \tag{246}$$

$${}^{C_1}\mathbf{p}_f = {}^{C_1}\mathbf{b}_1\, r_1, \quad {}^{C_2}\mathbf{p}_f = {}^{C_2}\mathbf{b}_2\, r_2, \tag{247}$$

with r_1, r_2 unknown. From ${}^{C_1}\mathbf{p}_f = {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{p}_f + {}^{C_1}\mathbf{p}_{C_2}$ we obtain ${}^{C_1}\mathbf{b}_1\, r_1 = {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2\, r_2 + {}^{C_1}\mathbf{p}_{C_2}$.

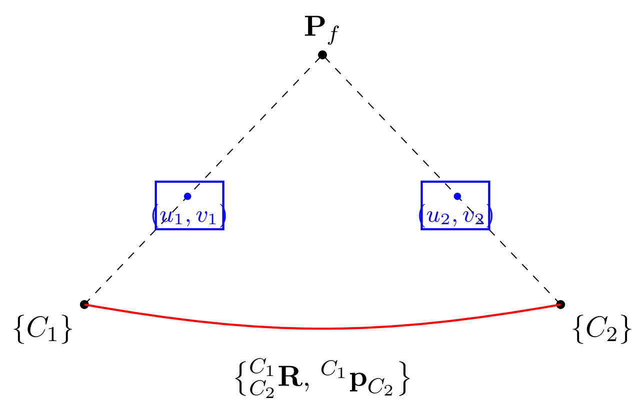


Figure 5: Two-view triangulation: bearings ${}^{C_1}\mathbf{b}_1, {}^{C_2}\mathbf{b}_2$ from pixels $(u_1, v_1), (u_2, v_2)$ and the known relative pose $\{{}^{C_1}_{C_2}\mathbf{R}, {}^{C_1}\mathbf{p}_{C_2}\}$ fix the feature $\mathbf{P}_f$ up to the two ranges r_1, r_2.

Way 1 (stacked linear system). Collect the two ranges:

$${}^{C_1}\mathbf{b}_1\, r_1 - {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2\, r_2 = {}^{C_1}\mathbf{p}_{C_2} \;\Rightarrow\; \underbrace{\begin{bmatrix} {}^{C_1}\mathbf{b}_1 & -{}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2 \end{bmatrix}}_{3\times 2} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = {}^{C_1}\mathbf{p}_{C_2}, \tag{248}$$

a 3×2 system; solving r_1, r_2 gives ${}^{C_1}\mathbf{p}_f, {}^{C_2}\mathbf{p}_f$.

> **Degenerate case.** If the motion is pure rotation (no translation), then ${}^{C_1}\mathbf{b}_1 = {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2$ and ${}^{C_1}\mathbf{p}_{C_2} = \mathbf{0}_{3\times 1}$, so $r_1 = r_2$ and there is no way to solve r. For a monocular camera, pure rotation is dangerous.

Way 2 (left null space). With ${}^{C_1}\mathbf{b}_1\, r_1 = {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2\, r_2 + {}^{C_1}\mathbf{p}_{C_2}$, collect into $\mathbf{B}^\perp = [\,\mathbf{b}^{\perp_1} \;\; \mathbf{b}^{\perp_2}\,]$ two directions orthogonal to ${}^{C_1}\mathbf{b}_1$ (${}^{C_1}\mathbf{b}_1^\top \mathbf{b}^{\perp_1} = 0$, ${}^{C_1}\mathbf{b}_1^\top \mathbf{b}^{\perp_2} = 0$); projecting removes r_1:

$$\mathbf{B}^{\perp\top}\,{}^{C_1}\mathbf{b}_1\, r_1 = \mathbf{B}^{\perp\top}\big({}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2\, r_2 + {}^{C_1}\mathbf{p}_{C_2}\big) = \mathbf{0}_{2\times 1} \;\Rightarrow\; \underbrace{\mathbf{B}^{\perp\top}\,{}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2}_{2\times 1}\, r_2 = -\underbrace{\mathbf{B}^{\perp\top}\,{}^{C_1}\mathbf{p}_{C_2}}_{2\times 1}. \tag{249}$$

(4) Monocular triangulation is up to scale. If $\{{}^{C}_{G}\mathbf{R}\}$ is known but for ${}^{C_1}\mathbf{p}_{C_2}$ we only know the direction $\overrightarrow{{}^{C_1}\mathbf{p}_{C_2}}$ (with $\|{}^{C_1}\mathbf{p}_{C_2}\|$ unknown), can we triangulate ${}^{C}\mathbf{p}_f$? Let $\|{}^{C_1}\mathbf{p}_{C_2}\| = \rho$, then ${}^{C_1}\mathbf{p}_f = {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{p}_f + {}^{C_1}\mathbf{p}_{C_2}$ gives

$$
{}^{C_1}\mathbf{b}_1\, r_1 = {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2\, r_2 + \rho\, \overrightarrow{{}^{C_1}\mathbf{p}_{C_2}}. \tag{250}
$$

With $\mathbf{A} = [\mathbf{v}^{\perp_1}\ \ \mathbf{v}^{\perp_2}]^\top$ built from $\overrightarrow{{}^{C_1}\mathbf{p}_{C_2}}^\top \mathbf{v}^{\perp_1} = 0$, $\overrightarrow{{}^{C_1}\mathbf{p}_{C_2}}^\top \mathbf{v}^{\perp_2} = 0$, projecting kills the translation term:

$$
\mathbf{A}\,{}^{C_1}\mathbf{b}_1\, r_1 = \mathbf{A}\,{}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b}_2\, r_2 + \rho\, \underbrace{\mathbf{A}\,\overrightarrow{{}^{C_1}\mathbf{p}_{C_2}}}_{\mathbf{0}} \tag{251}
$$

$$
\Rightarrow\quad \begin{bmatrix} \mathbf{A}\,{}^{C_1}\mathbf{b} & -\mathbf{A}\,{}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b} \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \mathbf{0}. \tag{252}
$$

Hence r_1, r_2 are up to scale: a monocular camera does not have scale. The above uses *motion to infer structure.*

(5) Essential matrix — solve the motion. Given $\{u_1, v_1\}, \{u_2, v_2\}$, can we solve $\{{}^{C}_{G}\mathbf{R}, {}^{C}\mathbf{p}_c\}$? The three vectors ${}^{C_1}\mathbf{p}_f$, ${}^{C_2}\mathbf{p}_f$, ${}^{C_1}\mathbf{p}_{C_2}$ are coplanar (Fig. 6), so ${}^{C_1}\mathbf{p}_{C_2} \times {}^{C_1}\mathbf{p}_f \perp {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{p}_f$:

$$
\left(\lfloor {}^{C_1}\mathbf{p}_{C_2} \rfloor_\times\, {}^{C_1}\mathbf{p}_f\right)^\top {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{p}_f = 0 \;\Rightarrow\; {}^{C_1}\mathbf{p}_f^\top\, \lfloor {}^{C_1}\mathbf{p}_{C_2} \rfloor_\times\, {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{p}_f = 0. \tag{253}
$$

With ${}^{C_1}\mathbf{p}_f = {}^{C_1}\mathbf{b}\, r_1$, ${}^{C_2}\mathbf{p}_f = {}^{C_2}\mathbf{b}\, r_2$,

$$
\boxed{{}^{C_1}\mathbf{b}^\top\, \lfloor {}^{C_1}\mathbf{p}_{C_2} \rfloor_\times\, {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b} = 0, \qquad \mathbf{E} = \lfloor {}^{C_1}\mathbf{p}_{C_2} \rfloor_\times\, {}^{C_1}_{C_2}\mathbf{R} \quad \text{(essential matrix)}.} \tag{254}
$$

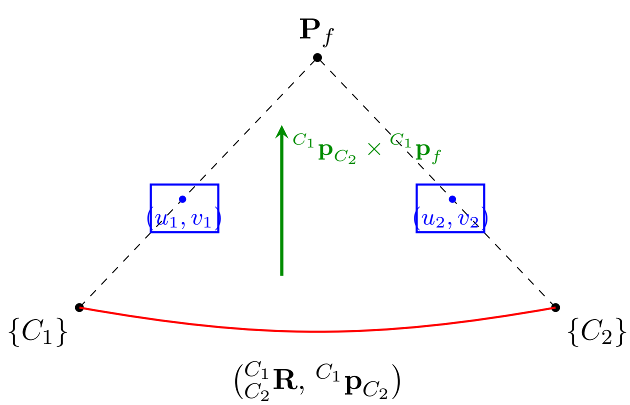


Figure 6: Epipolar geometry: ${}^{C_1}\mathbf{p}_f$, ${}^{C_2}\mathbf{p}_f$, and the baseline ${}^{C_1}\mathbf{p}_{C_2}$ are coplanar, giving the essential-matrix constraint ${}^{C_1}\mathbf{b}^\top \mathbf{E}\,{}^{C_2}\mathbf{b} = 0$.

Why is $\mathbf{E}$ 5-dof? Each pair $(u_1, v_1), (u_2, v_2)$ provides 1 constraint, so we need at least **5 pairs of points** to solve $\mathbf{E} \Rightarrow {}^{C_1}_{C_2}\mathbf{R}$ (3 dof) and $\overrightarrow{{}^{C_1}\mathbf{p}_{C_2}}$ (2 dof, direction only). When selecting pairs of points, RANSAC (random sample consensus) comes in. Finally:

$$
\text{normalized coordinates: } {}^{C_1}\mathbf{b}^\top \lfloor {}^{C_1}\mathbf{p}_{C_2} \rfloor_\times\, {}^{C_1}_{C_2}\mathbf{R}\,{}^{C_2}\mathbf{b} = 0 \ \text{ (essential matrix)}, \tag{255}
$$

$$
\text{pixel coordinates: fundamental matrix.} \tag{256}
$$

This is the *motion from structure* direction.

3.4 The full intrinsic calibration model

The calibration geometry is the same as Fig. 4 (repeated here as Fig. 7). The camera model is a composition of four functions.

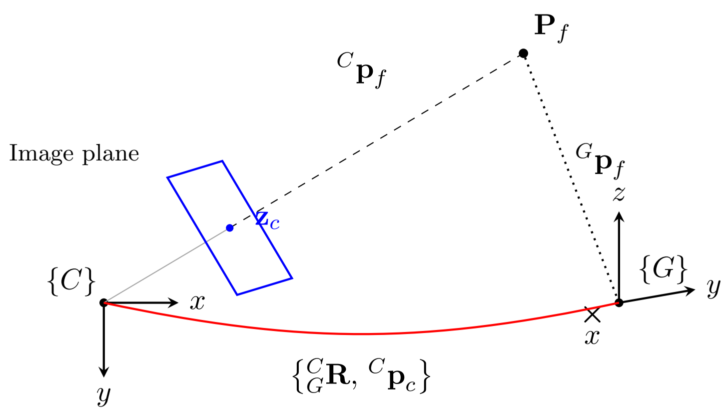


Figure 7: The intrinsic calibration sits inside the same projection geometry: the 3D feature is transformed into $\{C\}$, projected, distorted, and mapped to pixels.

Projection and transform.

$$\mathbf{z}_n = \begin{bmatrix} c_x/c_z \\ c_y/c_z \end{bmatrix} \triangleq \mathbf{h}_p({}^C\mathbf{p}_f), \qquad {}^C\mathbf{p}_f = {}^C_G\mathbf{R}^\top({}^G\mathbf{p}_f - {}^G\mathbf{p}_c) \triangleq \mathbf{h}_t({}^C_G\mathbf{R}, {}^G\mathbf{p}_c, {}^G\mathbf{p}_f), \tag{257}$$

$$\mathbf{z}_n = \mathbf{h}_p\big(\mathbf{h}_t({}^C_G\mathbf{R}, {}^G\mathbf{p}_c, {}^G\mathbf{p}_f)\big) + \mathbf{n}. \tag{258}$$

Distortion. Radial-tangential (for example); fisheye uses the equidistant model. With $r^2 = u_n^2 + v_n^2$ and distortion parameters $\mathbf{x}_d = [k_1, k_2, p_1, p_2]^\top$,

$$\begin{bmatrix} u_d \\ v_d \end{bmatrix} = \begin{bmatrix} u_n \\ v_n \end{bmatrix} \left(1 + k_1 r^2 + k_2 r^4\right) + \begin{bmatrix} 2p_1 u_n v_n + p_2\left(r^2 + 2u_n^2\right) \\ p_1\left(r^2 + 2v_n^2\right) + 2p_2 u_n v_n \end{bmatrix}, \qquad \mathbf{z}_d = \begin{bmatrix} u_d \\ v_d \end{bmatrix} = \mathbf{h}_d(\mathbf{z}_n). \tag{259}$$

Intrinsics / homography. With $\mathbf{x}_{in} = [f_u, f_v, c_u, c_v]$,

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} f_u & 0 & c_u \\ 0 & f_v & c_v \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{K}} \begin{bmatrix} u_d \\ v_d \\ 1 \end{bmatrix}, \qquad \mathbf{z} = \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} f_u & 0 \\ 0 & f_v \end{bmatrix} \begin{bmatrix} u_d \\ v_d \end{bmatrix} + \begin{bmatrix} c_u \\ c_v \end{bmatrix} \triangleq \mathbf{h}_k(\mathbf{z}_d, \mathbf{x}_{in}). \tag{260}$$

Full composition.

$$\begin{aligned} \mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n}_c &= \mathbf{h}_k(\mathbf{z}_d, \mathbf{x}_{in}) + \mathbf{n}_c && (261) \\ &= \mathbf{h}_k\big(\mathbf{h}_d(\mathbf{z}_n, \mathbf{x}_d), \mathbf{x}_{in}\big) + \mathbf{n}_c && (262) \\ &= \mathbf{h}_k\big(\mathbf{h}_d(\mathbf{h}_p({}^C\mathbf{p}_f), \mathbf{x}_d), \mathbf{x}_{in}\big) + \mathbf{n}_c && (263) \\ &= \mathbf{h}_k\Big(\mathbf{h}_d\big(\mathbf{h}_p(\mathbf{h}_t({}^C_G\mathbf{R}, {}^G\mathbf{p}_c, {}^G\mathbf{p}_f)), \mathbf{x}_d\big), \mathbf{x}_{in}\Big) + \mathbf{n}_c, && (264) \end{aligned}$$

where $\mathbf{h}_k(\cdot)$ is the camera-intrinsics homography, $\mathbf{h}_d(\cdot)$ the distortion function, $\mathbf{h}_p(\cdot)$ the projection function, $\mathbf{h}_t(\cdot)$ the transform function, $\mathbf{x}_d$ the distortion parameters, and $\mathbf{x}_{in} = \{f_u, f_v, c_u, c_v\}$ the intrinsics. The Jacobians for calibration follow by the chain rule.

3.5 The extrinsic (camera–IMU) calibration

The camera and IMU sensor frames are rigidly connected (Fig. 8).

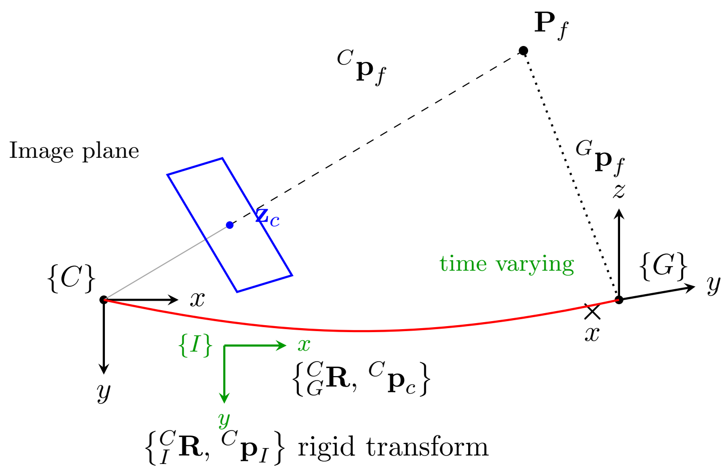


Figure 8: Extrinsic calibration: the IMU frame $\{I\}$ is rigidly fixed to the camera $\{C\}$ via $\{{}^{C}_{I}\mathbf{R}, {}^{C}\mathbf{p}_I\}$, while the camera-to-global transform is time varying.

$$\begin{bmatrix} {}^{C}\mathbf{p}_f \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} {}^{C}_{I}\mathbf{R} & {}^{C}\mathbf{p}_I \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix}}_{{}^{C}_{I}\mathbf{T}} \underbrace{\begin{bmatrix} {}^{G}_{I}\mathbf{R} & {}^{G}\mathbf{p}_I \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix}^{-1}}_{{}^{G}_{I}\mathbf{T}^{-1}} \begin{bmatrix} {}^{G}\mathbf{p}_f \\ 1 \end{bmatrix}, \tag{265}$$

$$ {}^{C}\mathbf{p}_f = {}^{C}_{I}\mathbf{R}\,{}^{I}\mathbf{p}_f + {}^{C}\mathbf{p}_I = {}^{C}_{I}\mathbf{R}\big({}^{I}_{G}\mathbf{R}({}^{G}\mathbf{p}_f - {}^{G}\mathbf{p}_I)\big) + {}^{C}\mathbf{p}_I. \tag{266}$$

Time offset. How do we estimate the time offset between camera and IMU? With $t_I = t_c + t_d$,

$$\begin{bmatrix} {}^{C_{t_c}}\mathbf{p}_f \\ 1 \end{bmatrix} = {}^{C}_{I}\mathbf{T} \cdot \big({}^{G}_{I(t+t_d)}\mathbf{T}\big)^{-1} \begin{bmatrix} {}^{G}\mathbf{p}_f \\ 1 \end{bmatrix}, \tag{267}$$

$$ {}^{G}_{I(t+t_d)}\dot{\mathbf{T}} = \begin{bmatrix} {}^{G}_{I}\mathbf{R} & {}^{G}\mathbf{p}_I \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix} \begin{bmatrix} \lfloor {}^{I}\boldsymbol{\omega} \rfloor_\times & {}^{I}\mathbf{v} \\ \mathbf{0}_{1\times 3} & 0 \end{bmatrix}. \tag{268}$$

which is the Jacobian for t_d. This is then the full calibration for the vision system.

3.6 Point / line / plane representation

3.6.1 Point representation

In $\mathbb{R}^3$ (closest point),

$$\mathbf{P}_f = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = z \begin{bmatrix} x/z \\ y/z \\ 1 \end{bmatrix} = z \begin{bmatrix} u_n \\ v_n \\ 1 \end{bmatrix}. \tag{269}$$

Inverse depth. Store the inverse of the depth, $\lambda \triangleq 1/z$:

$$\mathbf{x}_f = \begin{bmatrix} x \\ y \\ 1/z \end{bmatrix} = \begin{bmatrix} x \\ y \\ \lambda \end{bmatrix}. \tag{270}$$

What is the motivation to use inverse depth? Transform the feature into a second view and divide by z:

$$^{C_2}\mathbf{p}_f = {}^{C_2}_{C_1}\mathbf{R}\big({}^{C_1}\mathbf{p}_f - {}^{C_1}\mathbf{p}_{C_2}\big) = {}^{C_2}_{C_1}\mathbf{R}\Bigg(\begin{bmatrix} x \\ y \\ z \end{bmatrix} - {}^{C_1}\mathbf{p}_{C_2}\Bigg) = {}^{C_2}_{C_1}\mathbf{R}\Bigg(z\begin{bmatrix} x/z \\ y/z \\ 1 \end{bmatrix} - {}^{C_1}\mathbf{p}_{C_2}\Bigg), \tag{271}$$

$$\frac{^{C_2}\mathbf{p}_f}{z} = {}^{C_2}_{C_1}\mathbf{R}\Bigg(\begin{bmatrix} x/z \\ y/z \\ 1 \end{bmatrix} - \frac{1}{z}{}^{C_1}\mathbf{p}_{C_2}\Bigg) \;\Rightarrow\; \begin{bmatrix} {}^{C_2}x/z \\ {}^{C_2}y/z \\ {}^{C_2}z/z \end{bmatrix} = {}^{C_2}_{C_1}\mathbf{R}\Bigg(\begin{bmatrix} x\lambda \\ y\lambda \\ 1 \end{bmatrix} - \lambda\,{}^{C_1}\mathbf{p}_{C_2}\Bigg). \tag{272}$$

If $z \to \infty$, the feature is far away, $z \approx {}^{C_2}z \gg {}^{C_2}x, {}^{C_2}y$, then the left side tends to the bearing measurement in $\{C_2\}$:

$$\begin{bmatrix} {}^{C_2}x/z \\ {}^{C_2}y/z \\ {}^{C_2}z/z \end{bmatrix} \approx \underbrace{\begin{bmatrix} u_{n_2} \\ v_{n_2} \\ 1 \end{bmatrix}}_{\text{measurements in } C_2} \propto {}^{C_2}_{C_1}\mathbf{R}\Bigg(\begin{bmatrix} x\lambda \\ y\lambda \\ 1 \end{bmatrix} - \lambda\,{}^{C_1}\mathbf{p}_{C_2}\Bigg). \tag{273}$$

The Jacobian with respect to the inverse-depth state $\tilde{\mathbf{x}} = [\tilde{x}, \tilde{y}, \tilde{\lambda}]$ is

$$\frac{\partial[u_{n_2}; v_{n_2}]}{\partial\tilde{\mathbf{x}}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} \mathbf{H}_\theta & \mathbf{H}_p & \mathbf{H}_{p_f} \end{bmatrix}, \tag{274}$$

with the three blocks

$$\mathbf{H}_\theta = \Big\lfloor {}^{C_2}_{C_1}\mathbf{R}\big([x\lambda;\ y\lambda;\ 1] - \lambda\,{}^{C_1}\mathbf{p}_{C_2}\big)\Big\rfloor_\times, \qquad \mathbf{H}_p = \underbrace{-\lambda\,{}^{C_2}_{C_1}\mathbf{R}}_{\lambda\to 0}, \tag{275}$$

$$\mathbf{H}_{p_f} = {}^{C_2}_{C_1}\mathbf{R}\big[\ \underbrace{\lambda\mathbf{e}_1}_{0}\ \ \underbrace{\lambda\mathbf{e}_2}_{0}\ \ [x;\ y;\ 0] - {}^{C_1}\mathbf{p}_{C_2}\ \big]. \tag{276}$$

If the feature is far away, then the rotation is improved, but the Jacobian for translation will not be improved (the $-\lambda\,{}^{C_2}_{C_1}\mathbf{R}$ and $\lambda\mathbf{e}_1, \lambda\mathbf{e}_2$ columns vanish as $\lambda \to 0$).

3.6.2 Plane representation

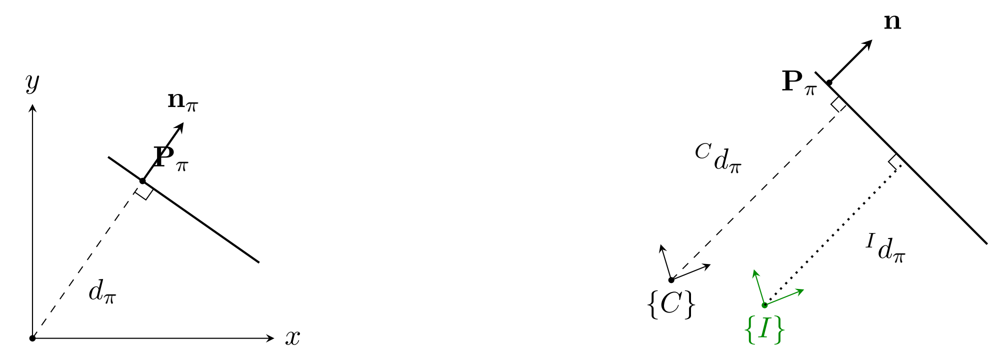
A plane is parameterized by its normal direction and distance to the origin (Fig. 9a):

$$\boldsymbol{\pi} = \begin{bmatrix} \mathbf{n}_\pi \\ d_\pi \end{bmatrix}_{4\times 1}, \qquad \mathbf{n}_\pi : \text{normal direction}, \quad d_\pi : \text{distance to origin}. \tag{277}$$

A plane is 3-dof but $\boldsymbol{\pi}$ has 4 parameters. The *closest-point representation* packs both into one vector,

$$\mathbf{P}_\pi = d_\pi\,\mathbf{n}_\pi = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \tag{278}$$

the closest point from the plane to the origin.



(a) plane: $\mathbf{n}_\pi$, d_π, foot $\mathbf{P}_\pi$ (b) transfer the plane $\{C\} \to \{I\}$

Figure 9: Plane representation (a) and its transfer between the camera and IMU frames (b).

Transfer $\{C\} \to \{I\}$. For ${}^{C}\mathbf{p}_\pi \leftrightarrow {}^{I}\mathbf{p}_\pi$ (Fig. 9b),

$$\begin{cases} {}^{I}\mathbf{n}_\pi = {}^{I}_{C}\mathbf{R}\,{}^{C}\mathbf{n}_\pi & \Rightarrow \text{normal direction} \\ {}^{I}d_\pi = {}^{C}d_\pi - {}^{C}\mathbf{n}_\pi^\top \cdot {}^{C}\mathbf{p}_I & \Rightarrow \text{origin to plane} \end{cases} \tag{279}$$

Hence, using ${}^{I}d_\pi = {}^{C}d_\pi - {}^{C}\mathbf{p}_I^\top\,{}^{C}\mathbf{n}_\pi$ and ${}^{C}\mathbf{p}_I = -{}^{C}_{I}\mathbf{R}\,{}^{I}\mathbf{p}_C$ (so $-{}^{C}\mathbf{p}_I^\top = {}^{I}\mathbf{p}_C^\top\,{}^{I}_{C}\mathbf{R}$),

$$\begin{bmatrix} {}^{I}\mathbf{n}_\pi \\ {}^{I}d_\pi \end{bmatrix} = \begin{bmatrix} {}^{I}_{C}\mathbf{R} & \mathbf{0}_{3\times1} \\ -{}^{C}\mathbf{p}_I^\top & 1 \end{bmatrix} \begin{bmatrix} {}^{C}\mathbf{n}_\pi \\ {}^{C}d_\pi \end{bmatrix} = \begin{bmatrix} {}^{I}_{C}\mathbf{R} & \mathbf{0}_{3\times1} \\ {}^{I}\mathbf{p}_C^\top\,{}^{I}_{C}\mathbf{R} & 1 \end{bmatrix} \begin{bmatrix} {}^{C}\mathbf{n}_\pi \\ {}^{C}d_\pi \end{bmatrix}. \tag{280}$$

The transformation chain is therefore

$${}^{G}\mathbf{p}_\pi \to ({}^{C}\mathbf{n}_\pi, {}^{C}d_\pi) \to ({}^{I}_{C}\mathbf{R}, {}^{I}\mathbf{p}_C) \to ({}^{I}\mathbf{n}_\pi, {}^{I}d_\pi) \to {}^{I}\mathbf{p}_\pi. \tag{281}$$

3.6.3 Line representation

How do we represent a 3D line (Fig. 10)? For a line L: $\mathbf{v}_e$ is the 3D direction, d_L is the distance from the origin to the line, and $\mathbf{n}_e$ is the normal direction of the plane (through the line and the origin).

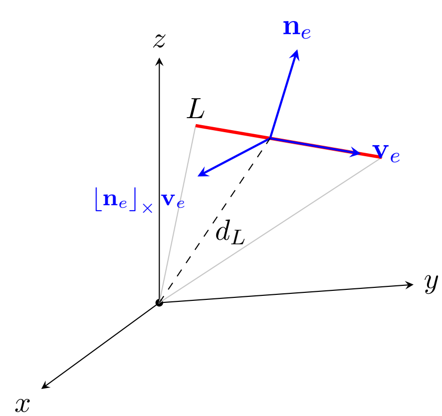


Figure 10: 3D line L: direction $\mathbf{v}_e$, plane normal $\mathbf{n}_e$, third axis $\lfloor \mathbf{n}_e \rfloor_\times \mathbf{v}_e$, and perpendicular distance d_L from the origin.

We can represent a line with $\{\mathbf{n}_e, \mathbf{v}_e, d_L\}$, where

$$\mathbf{R}_L = \begin{bmatrix}\mathbf{n}_e & \mathbf{v}_e & \lfloor \mathbf{n}_e \rfloor_\times \mathbf{v}_e\end{bmatrix} \text{ (a rotation matrix)}, \qquad d_L \text{ (a distance scalar)}. \tag{282}$$

So $L : \{\mathbf{R}_L, d_L\} : \{\bar{\mathbf{q}}_L, d_L\}$, packed as the 4D vector $\mathbf{P}_L = \bar{\mathbf{q}}_L \cdot d_L$, with the error states in Euclidean space. Then

$$\mathbf{P}_L = \hat{\mathbf{P}}_L + \tilde{\mathbf{P}}_L = \bar{\mathbf{q}}_L \cdot d_L = \hat{\bar{\mathbf{q}}}_L \otimes \delta\mathbf{q}_L \cdot (\hat{d}_L + \tilde{d}_L) \tag{283}$$

$$\Leftrightarrow \ \hat{\mathbf{P}}_L + \tilde{\mathbf{P}}_L = \hat{\bar{\mathbf{q}}}_L \otimes \begin{bmatrix} 1 \\ \frac{1}{2}\delta\boldsymbol{\theta}_L \end{bmatrix} (\hat{d}_L + \tilde{d}_L) \tag{284}$$

$$= \hat{\bar{\mathbf{q}}}_L \otimes \begin{bmatrix} \hat{d}_L + \tilde{d}_L \\ \frac{1}{2}\hat{d}_L\, \delta\boldsymbol{\theta}_L \end{bmatrix} \tag{285}$$

$$= \hat{\bar{\mathbf{q}}}_L \otimes \left(\begin{bmatrix} \hat{d}_L \\ \mathbf{0}_{3\times1} \end{bmatrix} + \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \frac{1}{2}\hat{d}_L \mathbf{I}_3 \end{bmatrix} \begin{bmatrix} \tilde{d}_L \\ \delta\boldsymbol{\theta}_L \end{bmatrix} \right), \tag{286}$$

so that, with $\lfloor \hat{\bar{\mathbf{q}}}_L \rfloor_L$ the left quaternion-product matrix,

$$\boxed{\tilde{\mathbf{P}}_L = \lfloor \hat{\bar{\mathbf{q}}}_L \rfloor_L \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \frac{1}{2}\hat{d}_L \mathbf{I}_3 \end{bmatrix} \begin{bmatrix} \tilde{d}_L \\ \delta\boldsymbol{\theta}_L \end{bmatrix}.} \tag{287}$$

A few discussions on lines with cameras. (1) How to project a 3D line to a 2D line in the image? Given a line $\{\mathbf{n}_e, \mathbf{v}_e, d_L\}$ and camera parameters f_u, f_v, c_u, c_v (Fig. 11a):

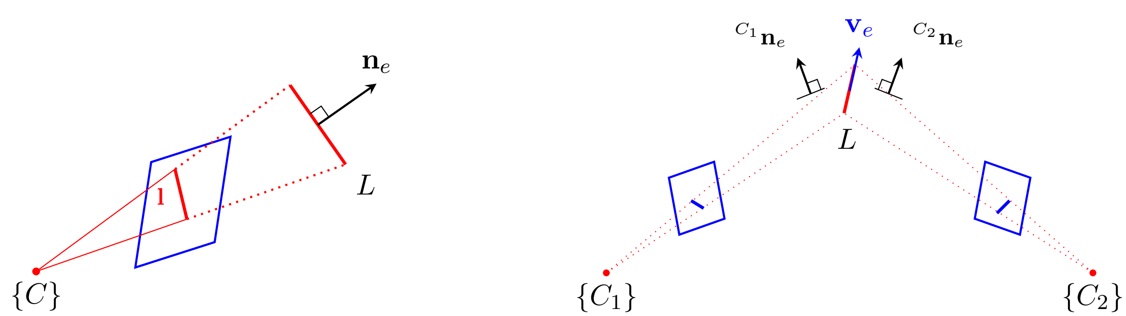
$$\begin{bmatrix} u_n \\ v_n \\ 1 \end{bmatrix} \sim \underbrace{\begin{bmatrix} f_u & 0 & c_u \\ 0 & f_v & c_v \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{K}} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\mathbf{p}_f}, \qquad \underbrace{\mathbf{p}}_{\text{pixel on image}} \sim \mathbf{K}\,\mathbf{p}_f. \tag{288}$$

Every $\mathbf{p}_f$ on the line lies in the plane through the line and the origin, so $\mathbf{p}_f^\top \mathbf{n}_e = 0$. With $\mathbf{p}_f \sim \mathbf{K}^{-1}\mathbf{p}$,

$$(\mathbf{K}^{-1}\mathbf{p})^\top \mathbf{n}_e = 0 \ \Rightarrow\ \mathbf{p}^\top \mathbf{K}^{-\top} \mathbf{n}_e = 0 \ \Rightarrow\ \mathbf{n}_e^\top \mathbf{K}^{-1} \mathbf{p} = 0 \ \Rightarrow\ (\mathbf{K}^{-\top}\mathbf{n}_e)^\top \mathbf{p} = 0, \tag{289}$$

so the 2D image line is

$$\mathbf{l} = \mathbf{K}^{-\top}\mathbf{n}_e, \qquad \mathbf{p} \text{ is every pixel on the line.} \tag{290}$$



(a) project 3D line → image line $\mathbf{l} = \mathbf{K}^{-\top}\mathbf{n}_e$

(b) transfer a line $\{C_1\} \to \{C_2\}$

Figure 11: Lines with cameras: projecting a 3D line to a 2D image line (a) and transferring a line between two camera frames (b).

(2) How to transfer a line from frame $\{C_1\}$ to $\{C_2\}$? (Fig. 11b) The line L is given by ${}^{C_1}\mathbf{n}_e$, ${}^{C_1}\mathbf{v}_e$, ${}^{C_1}d_L$ in $\{C_1\}$ and by ${}^{C_2}\mathbf{n}_e$, ${}^{C_2}\mathbf{v}_e$, ${}^{C_2}d_L$ in $\{C_2\}$. Assume ${}^{C_1}\mathbf{p}_f$ is a point on L; then its moment about the origin is

$$\left\lfloor {}^{C_1}\mathbf{p}_f \right\rfloor_\times {}^{C_1}\mathbf{v}_e = {}^{C_1}\mathbf{n}_e \, {}^{C_1}d_L. \tag{291}$$

The direction transforms by rotation only, and the point by the full rigid transform:

$${}^{C_2}\mathbf{v}_e = {}^{C_2}_{C_1}\mathbf{R} \, {}^{C_1}\mathbf{v}_e, \qquad {}^{C_2}\mathbf{p}_f = {}^{C_2}_{C_1}\mathbf{R} \, {}^{C_1}\mathbf{p}_f + {}^{C_2}\mathbf{p}_{C_1}. \tag{292}$$

Hence the moment in $\{C_2\}$ is

$$\left\lfloor {}^{C_2}\mathbf{p}_f \right\rfloor_\times {}^{C_2}\mathbf{v}_e = \left\lfloor {}^{C_2}_{C_1}\mathbf{R} \, {}^{C_1}\mathbf{p}_f + {}^{C_2}\mathbf{p}_{C_1} \right\rfloor_\times {}^{C_2}_{C_1}\mathbf{R} \, {}^{C_1}\mathbf{v}_e = {}^{C_2}\mathbf{n}_e \, {}^{C_2}d_L. \tag{293}$$

Using $\lfloor \mathbf{R}\mathbf{a} \rfloor_\times \mathbf{R} = \mathbf{R} \lfloor \mathbf{a} \rfloor_\times$ on the first term,

$${}^{C_2}_{C_1}\mathbf{R} \left\lfloor {}^{C_1}\mathbf{p}_f \right\rfloor_\times {}^{C_1}\mathbf{v}_e + \left\lfloor {}^{C_2}\mathbf{p}_{C_1} \right\rfloor_\times {}^{C_2}_{C_1}\mathbf{R} \, {}^{C_1}\mathbf{v}_e = {}^{C_2}\mathbf{n}_e \, {}^{C_2}d_L, \tag{294}$$

and substituting the moment identity ${}^{C_1}\mathbf{n}_e \, {}^{C_1}d_L = \left\lfloor {}^{C_1}\mathbf{p}_f \right\rfloor_\times {}^{C_1}\mathbf{v}_e$,

$$\begin{bmatrix} {}^{C_2}_{C_1}\mathbf{R} & \left\lfloor {}^{C_2}\mathbf{p}_{C_1} \right\rfloor_\times {}^{C_2}_{C_1}\mathbf{R} \end{bmatrix} \begin{bmatrix} {}^{C_1}\mathbf{n}_e \, {}^{C_1}d_L \\ {}^{C_1}\mathbf{v}_e \end{bmatrix} = {}^{C_2}\mathbf{n}_e \, {}^{C_2}d_L. \tag{295}$$

Stacking the moment and the direction (${}^{C_2}\mathbf{v}_e = {}^{C_2}_{C_1}\mathbf{R} \, {}^{C_1}\mathbf{v}_e$) gives the full line transfer:

$$\boxed{\begin{bmatrix} {}^{C_2}\mathbf{n}_e \, {}^{C_2}d_L \\ {}^{C_2}\mathbf{v}_e \end{bmatrix} = \begin{bmatrix} {}^{C_2}_{C_1}\mathbf{R} & \left\lfloor {}^{C_2}\mathbf{p}_{C_1} \right\rfloor_\times {}^{C_2}_{C_1}\mathbf{R} \\ \mathbf{0}_3 & {}^{C_2}_{C_1}\mathbf{R} \end{bmatrix} \begin{bmatrix} {}^{C_1}\mathbf{n}_e \, {}^{C_1}d_L \\ {}^{C_1}\mathbf{v}_e \end{bmatrix}.} \tag{296}$$

4 Lecture 4: Information, Marginalization, and Estimator Design

Topics. (1) the matrix inversion lemma; (2) covariance vs. information; (3) marginalization and the Schur complement; (4) why the information matrix reflects problem structure; (5) a typical visual-odometry / SLAM pipeline; (6) many ways to handle $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$; (7) designing a common estimator — deriving the EKF from MAP by marginalization.

4.1 The matrix inversion lemma

For a 2×2 block matrix, the inverse can be written block by block. In the general form,

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{E} & \mathbf{F} \\ \mathbf{G} & \mathbf{H} \end{bmatrix}, \qquad \begin{cases} \mathbf{E} = (\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{C})^{-1}, \\ \mathbf{F} = -\mathbf{E}\mathbf{B}\mathbf{D}^{-1}, \\ \mathbf{G} = -\mathbf{D}^{-1}\mathbf{C}\mathbf{E}, \\ \mathbf{H} = \mathbf{D}^{-1} + \mathbf{D}^{-1}\mathbf{C}\mathbf{E}\mathbf{B}\mathbf{D}^{-1}. \end{cases} \tag{297}$$

For the symmetric case ($\mathbf{C} = \mathbf{B}^\top$, so the inverse is symmetric too),

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{D} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{E} & \mathbf{F} \\ \mathbf{F}^\top & \mathbf{H} \end{bmatrix}, \qquad \begin{aligned} &\mathbf{E} = (\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{B}^\top)^{-1}, \\ &\mathbf{F} = -\mathbf{E}\mathbf{B}\mathbf{D}^{-1}, \\ &\mathbf{H} = \mathbf{D}^{-1} + \mathbf{D}^{-1}\mathbf{B}^\top\mathbf{E}\mathbf{B}\mathbf{D}^{-1} = (\mathbf{D} - \mathbf{B}^\top\mathbf{A}^{-1}\mathbf{B})^{-1}. \end{aligned} \tag{298}$$

The term $\mathbf{A} - \mathbf{B}\mathbf{D}^{-1}\mathbf{B}^\top$ (resp. $\mathbf{D} - \mathbf{B}^\top\mathbf{A}^{-1}\mathbf{B}$) is the *Schur complement.* This is one of the most important matrix operations for the derivations that follow.

4.2 Covariance and information

Recall the measurement model from Lecture 1, $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$ with $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$ and $\mathbf{R}$ the noise covariance. The MAP / weighted-least-squares estimate is

$$\max_{\mathbf{x}} \; p(\mathbf{z} \mid \mathbf{x}) \;\Leftrightarrow\; \min_{\mathbf{x}} \; \|\mathbf{z} - \mathbf{h}(\mathbf{x})\|^2_{\mathbf{R}}. \tag{299}$$

Given a linearization point $\hat{\mathbf{x}}$, $\mathbf{z} = \mathbf{h}(\hat{\mathbf{x}}) + \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n}$, so with the residual $\tilde{\mathbf{z}} = \mathbf{z} - \mathbf{h}(\hat{\mathbf{x}})$,

$$\tilde{\mathbf{z}} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n}. \tag{300}$$

The normal equations and their solution are

$$\boxed{\underbrace{\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}}_{\text{information}}\,\tilde{\mathbf{x}} = \mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}}, \qquad \tilde{\mathbf{x}} = \left(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\right)^{-1}\mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}}.} \tag{301}$$

To find the covariance of the corrected estimate $\mathbf{x}^\oplus = \hat{\mathbf{x}} + \tilde{\mathbf{x}}$, substitute $\tilde{\mathbf{z}} = \mathbf{H}(\mathbf{x} - \hat{\mathbf{x}}) + \mathbf{n}$:

$$\begin{aligned} \mathbf{x}^\oplus &= \hat{\mathbf{x}} + \left(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\right)^{-1}\mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}} && (302) \\ &= \hat{\mathbf{x}} + \left(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\right)^{-1}\mathbf{H}^\top\mathbf{R}^{-1}\left(\mathbf{H}(\mathbf{x} - \hat{\mathbf{x}}) + \mathbf{n}\right) && (303) \\ &= \mathbf{x} + \left(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\right)^{-1}\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{n}. && (304) \end{aligned}$$

Hence $\mathbf{x} - \mathbf{x}^{\oplus} = -(\mathbf{H}^\top \mathbf{R}^{-1}\mathbf{H})^{-1}\mathbf{H}^\top \mathbf{R}^{-1}\mathbf{n}$, and

$$\begin{aligned}
\mathbf{P}_{\mathbf{x}^{\oplus}\mathbf{x}^{\oplus}} &= \mathbb{E}\big[(\mathbf{x}-\mathbf{x}^{\oplus})(\mathbf{x}-\mathbf{x}^{\oplus})^\top\big] \\
&= \big(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\big)^{-1}\mathbf{H}^\top\mathbf{R}^{-1}\,\mathbb{E}(\mathbf{n}\mathbf{n}^\top)\,\mathbf{R}^{-1}\mathbf{H}\big(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\big)^{-1} && (305)\\
&= \big(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\big)^{-1}, && (306)
\end{aligned}$$

using $\mathbb{E}(\mathbf{n}\mathbf{n}^\top) = \mathbf{R}$. Therefore information and covariance are inverses of each other:

$$\boxed{\text{information} \quad \mathbf{\Sigma} = \mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}, \qquad \text{covariance} \quad \mathbf{P} = \big(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\big)^{-1}, \qquad \mathbf{\Sigma} = \mathbf{P}^{-1},\;\; \mathbf{P} = \mathbf{\Sigma}^{-1}.} \tag{307}$$

This is fundamental for what follows:

- for **optimization / least squares**, we keep track of the information matrix $\mathbf{\Sigma}$;
- for **filtering (EKF)**, we keep track of the covariance $\mathbf{P}$.

4.3 Marginalization

Let $\mathbf{x} \sim \mathcal{N}(\hat{\mathbf{x}}, \mathbf{P}_{\mathbf{xx}})$, where $\hat{\mathbf{x}}$ is the best estimate and $\mathbf{P}_{\mathbf{xx}}$ the covariance. Partition the state as $\mathbf{x} = [\mathbf{x}_A^\top, \mathbf{x}_B^\top]^\top$, with $\mathbf{x}_A$ and $\mathbf{x}_B$ both part of $\mathbf{x}$:

$$\mathbf{x} \sim \mathcal{N}\left(\begin{bmatrix}\hat{\mathbf{x}}_A\\ \hat{\mathbf{x}}_B\end{bmatrix}, \begin{bmatrix}\mathbf{P}_{AA} & \mathbf{P}_{AB}\\ \mathbf{P}_{BA} & \mathbf{P}_{BB}\end{bmatrix}\right), \qquad \mathbf{P}^\top = \mathbf{P}\ (\text{SPD}), \quad \mathbf{P}_{AB} = \mathbf{P}_{BA}^\top. \tag{308}$$

Covariance form (easy). The marginal of either block is read off directly:

$$\mathbf{x}_A \sim \mathcal{N}(\hat{\mathbf{x}}_A, \mathbf{P}_{AA}), \qquad \mathbf{x}_B \sim \mathcal{N}(\hat{\mathbf{x}}_B, \mathbf{P}_{BB}). \tag{309}$$

Marginalization in covariance form is straightforward: just keep the corresponding diagonal block.

Information form (Schur complement). For optimization we work with the information system $(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H})\tilde{\mathbf{x}} = \mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}}$, i.e.

$$\mathbf{\Sigma}\,\tilde{\mathbf{x}} = \mathbf{b}, \qquad \begin{bmatrix}\mathbf{\Sigma}_{AA} & \mathbf{\Sigma}_{AB}\\ \mathbf{\Sigma}_{AB}^\top & \mathbf{\Sigma}_{BB}\end{bmatrix}\begin{bmatrix}\tilde{\mathbf{x}}_A\\ \tilde{\mathbf{x}}_B\end{bmatrix} = \begin{bmatrix}\mathbf{b}_A\\ \mathbf{b}_B\end{bmatrix}, \tag{310}$$

where $\mathbf{\Sigma}$ is the information matrix and $\mathbf{b}$ the residual vector. We seek $\mathbf{\Sigma}'_{AA}, \mathbf{b}'_A$ (and $\mathbf{\Sigma}'_{BB}, \mathbf{b}'_B$) such that $\mathbf{\Sigma}'_{AA}\tilde{\mathbf{x}}_A = \mathbf{b}'_A$ and $\mathbf{\Sigma}'_{BB}\tilde{\mathbf{x}}_B = \mathbf{b}'_B$ hold after eliminating the other block.

Solving $\mathbf{\Sigma}\tilde{\mathbf{x}} = \mathbf{b}$ via the block inverse,

$$\begin{bmatrix}\tilde{\mathbf{x}}_A\\ \tilde{\mathbf{x}}_B\end{bmatrix} = \begin{bmatrix}\mathbf{\Sigma}_{AA} & \mathbf{\Sigma}_{AB}\\ \mathbf{\Sigma}_{AB}^\top & \mathbf{\Sigma}_{BB}\end{bmatrix}^{-1}\begin{bmatrix}\mathbf{b}_A\\ \mathbf{b}_B\end{bmatrix} = \begin{bmatrix}\mathbf{P}_{AA} & \mathbf{P}_{AB}\\ \mathbf{P}_{AB}^\top & \mathbf{P}_{BB}\end{bmatrix}\begin{bmatrix}\mathbf{b}_A\\ \mathbf{b}_B\end{bmatrix} = \begin{bmatrix}\mathbf{P}_{AA}\mathbf{b}_A + \mathbf{P}_{AB}\mathbf{b}_B\\ \mathbf{P}_{AB}^\top\mathbf{b}_A + \mathbf{P}_{BB}\mathbf{b}_B\end{bmatrix}, \tag{311}$$

with the blocks of the inverse given by the matrix inversion lemma,

$$\mathbf{P}_{AA} = \big(\mathbf{\Sigma}_{AA} - \mathbf{\Sigma}_{AB}\mathbf{\Sigma}_{BB}^{-1}\mathbf{\Sigma}_{AB}^\top\big)^{-1}, \qquad \mathbf{P}_{AB} = -\mathbf{P}_{AA}\mathbf{\Sigma}_{AB}\mathbf{\Sigma}_{BB}^{-1}, \tag{312}$$

$$\mathbf{P}_{BB} = \mathbf{\Sigma}_{BB}^{-1} + \mathbf{\Sigma}_{BB}^{-1}\mathbf{\Sigma}_{AB}^\top\mathbf{P}_{AA}\mathbf{\Sigma}_{AB}\mathbf{\Sigma}_{BB}^{-1} = \big(\mathbf{\Sigma}_{BB} - \mathbf{\Sigma}_{AB}^\top\mathbf{\Sigma}_{AA}^{-1}\mathbf{\Sigma}_{AB}\big)^{-1}. \tag{313}$$

For the A block,

$$\tilde{\mathbf{x}}_A = \mathbf{P}_{AA}\mathbf{b}_A + \mathbf{P}_{AB}\mathbf{b}_B = \mathbf{P}_{AA}\mathbf{b}_A - \mathbf{P}_{AA}\mathbf{\Sigma}_{AB}\mathbf{\Sigma}_{BB}^{-1}\mathbf{b}_B = \mathbf{P}_{AA}\big(\mathbf{b}_A - \mathbf{\Sigma}_{AB}\mathbf{\Sigma}_{BB}^{-1}\mathbf{b}_B\big), \tag{314}$$

so $\mathbf{P}_{AA}^{-1}\tilde{\mathbf{x}}_A = \mathbf{b}_A - \boldsymbol{\Sigma}_{AB}\boldsymbol{\Sigma}_{BB}^{-1}\mathbf{b}_B$. Using the identity $\mathbf{P}_{AB} = -\mathbf{P}_{AA}\boldsymbol{\Sigma}_{AB}\boldsymbol{\Sigma}_{BB}^{-1} = -\boldsymbol{\Sigma}_{AA}^{-1}\boldsymbol{\Sigma}_{AB}\mathbf{P}_{BB}$, the B block gives symmetrically

$$\tilde{\mathbf{x}}_B = \mathbf{P}_{AB}^\top\mathbf{b}_A + \mathbf{P}_{BB}\mathbf{b}_B = \mathbf{P}_{BB}\big(\mathbf{b}_B - \boldsymbol{\Sigma}_{AB}^\top\boldsymbol{\Sigma}_{AA}^{-1}\mathbf{b}_A\big), \tag{315}$$

so $\mathbf{P}_{BB}^{-1}\tilde{\mathbf{x}}_B = \mathbf{b}_B - \boldsymbol{\Sigma}_{AB}^\top\boldsymbol{\Sigma}_{AA}^{-1}\mathbf{b}_A$. Substituting the Schur-complement expressions for $\mathbf{P}_{AA}^{-1}$ and $\mathbf{P}_{BB}^{-1}$:

$$\boxed{\begin{aligned}&\text{marginalize } \mathbf{x}_B,\ \text{keep } \mathbf{x}_A: && \big(\boldsymbol{\Sigma}_{AA} - \boldsymbol{\Sigma}_{AB}\boldsymbol{\Sigma}_{BB}^{-1}\boldsymbol{\Sigma}_{AB}^\top\big)\tilde{\mathbf{x}}_A = \mathbf{b}_A - \boldsymbol{\Sigma}_{AB}\boldsymbol{\Sigma}_{BB}^{-1}\mathbf{b}_B, && (316)\\ &\text{marginalize } \mathbf{x}_A,\ \text{keep } \mathbf{x}_B: && \big(\boldsymbol{\Sigma}_{BB} - \boldsymbol{\Sigma}_{AB}^\top\boldsymbol{\Sigma}_{AA}^{-1}\boldsymbol{\Sigma}_{AB}\big)\tilde{\mathbf{x}}_B = \mathbf{b}_B - \boldsymbol{\Sigma}_{AB}^\top\boldsymbol{\Sigma}_{AA}^{-1}\mathbf{b}_A. && (317)\end{aligned}}$$

Both the new information block and the new residual are Schur complements of the eliminated variable. This is exactly how variables are dropped from a sliding window while preserving their influence on the kept states.

4.4 Why we care about the information matrix

Take visual measurements as an example. A pixel observation depends on a camera pose and a feature,

$$\mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n} = \mathbf{h}_c(\mathbf{x}_c, \mathbf{x}_f) + \mathbf{n}, \qquad \tilde{\mathbf{z}}_c = \mathbf{H}_c\tilde{\mathbf{x}}_c + \mathbf{H}_f\tilde{\mathbf{x}}_f + \mathbf{n}. \tag{318}$$

Consider three poses $\mathbf{x}_A, \mathbf{x}_B, \mathbf{x}_C$ observing two features f_1, f_2 (Fig. 12). Stacking the four measurements as $\tilde{\mathbf{z}} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n}$, with $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and state $\mathbf{x} = [\mathbf{x}_A^\top, \mathbf{x}_B^\top, \mathbf{x}_C^\top, f_1^\top, f_2^\top]^\top$:

$$\begin{bmatrix}\tilde{\mathbf{z}}_{A1}\\ \tilde{\mathbf{z}}_{B1}\\ \tilde{\mathbf{z}}_{B2}\\ \tilde{\mathbf{z}}_{C2}\end{bmatrix} = \underbrace{\begin{bmatrix}\mathbf{H}_A & \mathbf{0} & \mathbf{0} & \mathbf{H}_{A1} & \mathbf{0}\\ \mathbf{0} & \mathbf{H}_B & \mathbf{0} & \mathbf{H}_{B1} & \mathbf{0}\\ \mathbf{0} & \mathbf{H}_B & \mathbf{0} & \mathbf{0} & \mathbf{H}_{B2}\\ \mathbf{0} & \mathbf{0} & \mathbf{H}_C & \mathbf{0} & \mathbf{H}_{C2}\end{bmatrix}}_{\mathbf{H}} \begin{bmatrix}\tilde{\mathbf{x}}_A\\ \tilde{\mathbf{x}}_B\\ \tilde{\mathbf{x}}_C\\ \tilde{f}_1\\ \tilde{f}_2\end{bmatrix} + \mathbf{n}. \tag{319}$$

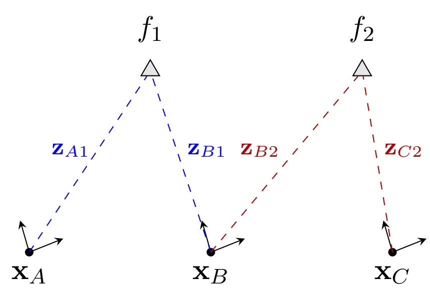


Figure 12: Visual measurement graph: poses $\mathbf{x}_A, \mathbf{x}_B, \mathbf{x}_C$ observe features f_1, f_2. Each ray $\mathbf{z}_{(\cdot)}$ couples one pose to one feature, which determines the sparsity of $\mathbf{H}$ and of $\boldsymbol{\Sigma} = \mathbf{H}^\top\mathbf{H}$.

The information matrix $\boldsymbol{\Sigma} = \mathbf{H}^\top\mathbf{H}$ has a block-sparse structure (a block (i, j) is nonzero only when states i and j are co-observed in some measurement), shown in Fig. 13. Partitioning into poses $p = \{A, B, C\}$ and features $f = \{f_1, f_2\}$,

$$\boldsymbol{\Sigma} = \mathbf{H}^\top\mathbf{H} \triangleq \begin{bmatrix}\boldsymbol{\Sigma}_{pp} & \boldsymbol{\Sigma}_{pf}\\ \boldsymbol{\Sigma}_{pf}^\top & \boldsymbol{\Sigma}_{ff}\end{bmatrix}. \tag{320}$$

$$
\begin{array}{c|ccc|cc|c}
 & A & B & C & f_1 & f_2 & \\
\hline
A & \otimes & 0 & 0 & \otimes & 0 & \\
B & 0 & \otimes & 0 & \otimes & \otimes & \boldsymbol{\Sigma}_{pp} \\
C & 0 & 0 & \otimes & 0 & \otimes & \\
\hline
f_1 & \otimes & \otimes & 0 & \otimes & 0 & \boldsymbol{\Sigma}_{ff} \\
f_2 & 0 & \otimes & \otimes & 0 & \otimes & \\
\hline
\end{array}
$$

Figure 13: Sparsity of $\boldsymbol{\Sigma} = \mathbf{H}^\top \mathbf{H}$ ($\otimes$: nonzero block). The pose–pose block $\boldsymbol{\Sigma}_{pp}$ is block-diagonal, while $\boldsymbol{\Sigma}_{pf}$ records exactly which pose sees which feature. The information matrix reflects the problem structure.

4.5 A typical visual-odometry / SLAM pipeline

How do the previous two lectures fit together into a working monocular system? A common pipeline incrementally grows poses $C_1, C_2, \ldots$ and features $f_1, f_2, \ldots$ (Fig. 14):

(1) **5-point algorithm** on C_1, C_2: recover ${}^{C_1}_{C_2}\mathbf{R}$ and ${}^{C_1}\mathbf{p}^*_{C_2}$ (up to scale).
(2) **Feature triangulation**: with the recovered relative pose, triangulate $f_1, f_2, \ldots$.
(3) **P3P / PnP** for C_3: with known features, solve ${}^{C_1}_{C_3}\mathbf{R}$, ${}^{C_1}\mathbf{p}_{C_3}$ (then triangulate new features).
(4) **5-point or P3P/PnP** for C_4: solve ${}^{C_1}_{C_4}\mathbf{R}$, ${}^{C_1}\mathbf{p}_{C_4}$.
(5) **Bundle adjustment (BA)** over C_1, C_2, C_3 and $f_1, f_2, f_3, \ldots$ (joint refinement of poses and structure).
(6) **BA** extended to C_1, C_2, C_3, C_4 and $f_1, f_2, f_3, \ldots$ as the window grows.
(7) **Marginalization**: when there are too many poses $(C_1, \ldots, C_n)$ and features $(f_1, \ldots, f_m)$, marginalize old states $C_1, f_1, \ldots$ to keep the system light-weight.

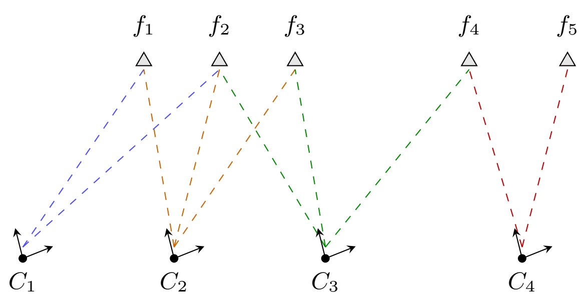


Figure 14: A monocular VO/SLAM pipeline grows poses $C_1, \ldots, C_4$ and features $f_1, \ldots, f_5$: bootstrap with the 5-point algorithm + triangulation, extend new poses with P3P/PnP, refine with bundle adjustment, and marginalize old states to bound cost. Combined with the previous two lectures, this builds up visual SLAM.

4.6 Designing estimators: many ways to handle $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$

Given $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$, $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$, there are many ways to linearize and solve, each leading to a different family of algorithms.

(1) Iterated re-linearization (GN / LM). The typical solution $\min_{\mathbf{x}} \|\mathbf{z} - \mathbf{h}(\mathbf{x})\|^2_{\mathbf{R}}$. Given a linearization $\hat{\mathbf{x}}$, $\tilde{\mathbf{z}} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n}$; update $\mathbf{x}^\oplus = \hat{\mathbf{x}} + \tilde{\mathbf{x}}$, then *redo* the linearization $\tilde{\mathbf{z}} = \mathbf{H}^\oplus \tilde{\mathbf{x}} + \mathbf{n}$ (Gauss–Newton or Levenberg–Marquardt).

(2) Linearize once + residual correction (FEJ). What if we linearize only once, at a fixed $\hat{\mathbf{x}}^0$? With $\mathbf{H}^0 = \mathbf{H}|_{\hat{\mathbf{x}}^0}$,

$$\mathbf{z} = \mathbf{h}(\hat{\mathbf{x}}^0) + \mathbf{H}^0(\mathbf{x} - \hat{\mathbf{x}}^0) + \mathbf{n} \tag{321}$$

$$\mathbf{z} - \mathbf{h}(\hat{\mathbf{x}}^0) = \mathbf{H}^0\big(\mathbf{x} - \hat{\mathbf{x}} + \underbrace{\hat{\mathbf{x}} - \hat{\mathbf{x}}^0}_{\Delta\mathbf{x}}\big) + \mathbf{n} = \mathbf{H}^0\tilde{\mathbf{x}} + \mathbf{H}^0\Delta\mathbf{x} + \mathbf{n}, \tag{322}$$

so moving the known offset to the left gives the *residual correction*

$$\underbrace{\mathbf{z} - \mathbf{h}(\hat{\mathbf{x}}^0)}_{\tilde{\mathbf{z}}^0} - \mathbf{H}^0\Delta\mathbf{x} = \mathbf{H}^0\tilde{\mathbf{x}} + \mathbf{n}. \tag{323}$$

Keeping the Jacobian fixed at the first estimate is **FEJ** (first-estimate Jacobians) [7], used e.g. in OKVIS [8]; it preserves the correct observability/nullspace structure.

(3) Partial linearization (preintegration / iSAM). For $\mathbf{x} = [\mathbf{x}_A^\top, \mathbf{x}_B^\top]^\top$, $\mathbf{z} = \mathbf{h}(\mathbf{x}_A, \mathbf{x}_B) + \mathbf{n}$, linearize only in $\mathbf{x}_B$ (about $\hat{\mathbf{x}}_B^0$) while keeping $\mathbf{x}_A$ exact:

$$\mathbf{z} = \mathbf{h}(\hat{\mathbf{x}}_A, \hat{\mathbf{x}}_B^0) + \mathbf{H}_A\tilde{\mathbf{x}}_A + \mathbf{H}_B^0\tilde{\mathbf{x}}_B + \mathbf{H}_B^0\Delta\mathbf{x}_B + \mathbf{n}, \tag{324}$$

$$\underbrace{\mathbf{z} - \mathbf{h}(\hat{\mathbf{x}}_A, \hat{\mathbf{x}}_B^0)}_{\tilde{\mathbf{z}}^0} - \mathbf{H}_B^0\Delta\mathbf{x}_B = \begin{bmatrix}\mathbf{H}_A & \mathbf{H}_B^0\end{bmatrix}\begin{bmatrix}\tilde{\mathbf{x}}_A \\ \tilde{\mathbf{x}}_B\end{bmatrix} + \mathbf{n}. \tag{325}$$

This is the idea behind IMU **pre-integration** [9] and **ACI** (analytic combined integration) [10]; **iSAM** [11] uses a similar partial-relinearization idea.

(4) Nullspace projection (MSCKF / structureless). For $\mathbf{x} = [\mathbf{x}_A^\top, \mathbf{x}_B^\top]^\top$, $\tilde{\mathbf{z}} = \mathbf{H}_A\tilde{\mathbf{x}}_A + \mathbf{H}_B\tilde{\mathbf{x}}_B + \mathbf{n}$. If we find $\mathbf{N}$ with $\mathbf{N}^\top\mathbf{H}_B = \mathbf{0}$, then projecting onto the left nullspace removes $\mathbf{x}_B$:

$$\mathbf{N}^\top\tilde{\mathbf{z}} = \mathbf{N}^\top\mathbf{H}_A\tilde{\mathbf{x}}_A + \cancel{\mathbf{N}^\top\mathbf{H}_B\tilde{\mathbf{x}}_B} + \mathbf{N}^\top\mathbf{n}. \tag{326}$$

This is the **MSCKF** [12] (multi-state constraint Kalman filter), a.k.a. the smart / structure-less vision factor: the features are eliminated without ever being added to the state.

(5) Invariant errors. Instead of $\mathbf{x} = \hat{\mathbf{x}} + \tilde{\mathbf{x}}$, put $\mathbf{x}$ on a group $SE(3)$ / $SE_2(3)$ and use a right-invariant error,

$$\mathbf{x} = \hat{\mathbf{x}} \boxplus \tilde{\mathbf{x}} = \mathrm{Exp}(\tilde{\mathbf{x}}) \cdot \hat{\mathbf{x}} \;\Rightarrow\; \text{invariant SLAM / VIO.} \tag{327}$$

(6) Decoupled (right-)invariant errors. For $\mathbf{x} = [\mathbf{x}_A^\top, \mathbf{x}_B^\top]^\top$, mix error types per block:

$$\begin{cases} \mathbf{x}_A = \hat{\mathbf{x}}_A \boxplus \tilde{\mathbf{x}}_A = \mathrm{Exp}(\tilde{\mathbf{x}}_A) \cdot \hat{\mathbf{x}}_A & \text{(group / invariant error)} \\ \mathbf{x}_B = \hat{\mathbf{x}}_B + \tilde{\mathbf{x}}_B & \text{(vector error)} \end{cases} \tag{328}$$

which gives decoupled right-invariant SLAM / VIO.

> Use your imagination to deal with $\big\{\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n},\; \tilde{\mathbf{z}} = \mathbf{H}\tilde{\mathbf{x}} + \mathbf{n}\big\}$ to design different algorithms.

4.7 Designing a common system: from MAP to the EKF

Consider the canonical prior + motion + measurement system

$$\begin{cases} \mathbf{x}_0 \sim \mathcal{N}(\hat{\mathbf{x}}_0, \mathbf{P}_0) & \to \text{prior} \\ \mathbf{x}_1 = \mathbf{f}(\mathbf{x}_0) + \mathbf{w}_0, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) & \to \text{motion model} \\ \mathbf{z}_1 = \mathbf{h}(\mathbf{x}_1) + \mathbf{n}_1, \quad \mathbf{n}_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{R}) & \to \text{measurement model} \end{cases} \tag{329}$$

The MAP estimate factorizes by Bayes' rule,

$$\max_{\mathbf{x}} p(\mathbf{x} \mid \mathbf{z}) = \max_{\mathbf{x}} \frac{p(\mathbf{z} \mid \mathbf{x})\, p(\mathbf{x})}{p(\mathbf{z})} = \max_{\mathbf{x}} \frac{p(\mathbf{z}_1 \mid \mathbf{x}_1)\, p(\mathbf{x}_1 \mid \mathbf{x}_0)\, p(\mathbf{x}_0)}{p(\mathbf{z})}, \tag{330}$$

which is equivalent to the nonlinear least-squares cost

$$\min_{\mathbf{x}} \|\mathbf{z}_1 - \mathbf{h}(\mathbf{x}_1)\|^2_{\mathbf{R}} + \|\mathbf{x}_1 - \mathbf{f}(\mathbf{x}_0)\|^2_{\mathbf{Q}} + \|\mathbf{x}_0 - \hat{\mathbf{x}}_0\|^2_{\mathbf{P}_0}. \tag{331}$$

There are two ways to solve it:

1. **Batch:** minimize all three terms jointly $\Rightarrow \hat{\mathbf{x}}_1, \mathbf{P}_1$.
2. **Incremental:**
   (i) handle the prior + motion terms $\|\mathbf{x}_0 - \hat{\mathbf{x}}_0\|^2_{\mathbf{P}_0} + \|\mathbf{x}_1 - \mathbf{f}(\mathbf{x}_0)\|^2_{\mathbf{Q}}$;
   (ii) marginalize $\mathbf{x}_0$, keep $\mathbf{x}_1$, giving $(\hat{\mathbf{x}}_{1|0}, \mathbf{P}_{1|0})$;
   (iii) add the measurement $\left\|\mathbf{x}_1 - \hat{\mathbf{x}}_{1|0}\right\|^2_{\mathbf{P}_{1|0}} + \|\mathbf{z}_1 - \mathbf{h}(\mathbf{x}_1)\|^2_{\mathbf{R}} \Rightarrow \hat{\mathbf{x}}_1, \mathbf{P}_1$.

The incremental route is exactly the EKF predict/update, as we now derive.

Propagation (marginalize $\mathbf{x}_0$). Linearize the motion model, $\mathbf{x}_1 = \mathbf{f}(\mathbf{x}_0) + \mathbf{w}_0 \Rightarrow \hat{\mathbf{x}}_1 + \tilde{\mathbf{x}}_1 = \mathbf{f}(\hat{\mathbf{x}}_0) + \mathbf{F}\tilde{\mathbf{x}}_0 + \mathbf{w}_0$, and define the motion residual

$$\mathbf{r}_{\mathbf{x}} \triangleq \hat{\mathbf{x}}_1 - \mathbf{f}(\hat{\mathbf{x}}_0) = \mathbf{F}\tilde{\mathbf{x}}_0 - \tilde{\mathbf{x}}_1 + \mathbf{w}_0, \qquad \mathbf{w}_0 = \mathbf{r}_{\mathbf{x}} - \begin{bmatrix}\mathbf{F} & -\mathbf{I}\end{bmatrix} \begin{bmatrix}\tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1\end{bmatrix}. \tag{332}$$

The prior + motion cost (writing $\mathbf{Q}_0 = \mathbf{Q}$) is

$$\min \|\tilde{\mathbf{x}}_0\|^2_{\mathbf{P}_0} + \left\| \mathbf{r}_{\mathbf{x}} - \begin{bmatrix}\mathbf{F} & -\mathbf{I}\end{bmatrix} \begin{bmatrix}\tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1\end{bmatrix} \right\|^2_{\mathbf{Q}_0}. \tag{333}$$

Collecting quadratic and linear terms, the cost C is

$$C = \begin{bmatrix}\tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1\end{bmatrix}^\top \begin{bmatrix}\mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F} & -\mathbf{F}^\top\mathbf{Q}_0^{-1} \\ -\mathbf{Q}_0^{-1}\mathbf{F} & \mathbf{Q}_0^{-1}\end{bmatrix} \begin{bmatrix}\tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1\end{bmatrix} - 2 \begin{bmatrix}\tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1\end{bmatrix}^\top \begin{bmatrix}\mathbf{F}^\top \\ -\mathbf{I}\end{bmatrix} \mathbf{Q}_0^{-1}\mathbf{r}_{\mathbf{x}} + \mathbf{r}_{\mathbf{x}}^\top\mathbf{Q}_0^{-1}\mathbf{r}_{\mathbf{x}}. \tag{334}$$

Setting $\partial C / \partial[\tilde{\mathbf{x}}_0; \tilde{\mathbf{x}}_1] = \mathbf{0}$ gives the information system

$$\begin{bmatrix}\mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F} & -\mathbf{F}^\top\mathbf{Q}_0^{-1} \\ -\mathbf{Q}_0^{-1}\mathbf{F} & \mathbf{Q}_0^{-1}\end{bmatrix} \begin{bmatrix}\tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1\end{bmatrix} = \begin{bmatrix}\mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{r}_{\mathbf{x}} \\ -\mathbf{Q}_0^{-1}\mathbf{r}_{\mathbf{x}}\end{bmatrix}. \tag{335}$$

Marginalizing $\mathbf{x}_0$ (Schur complement onto the $\mathbf{x}_1$ block) and applying the matrix inversion lemma,

$$\begin{aligned} \left[\mathbf{Q}_0^{-1} - \mathbf{Q}_0^{-1}\mathbf{F}\big(\mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F}\big)^{-1}\mathbf{F}^\top\mathbf{Q}_0^{-1}\right]\tilde{\mathbf{x}}_1 &= -\mathbf{Q}_0^{-1}\mathbf{r}_{\mathbf{x}} + \mathbf{Q}_0^{-1}\mathbf{F}\big(\mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F}\big)^{-1}\mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{r}_{\mathbf{x}}, \\ \underbrace{\big(\mathbf{Q}_0 + \mathbf{F}\mathbf{P}_0\mathbf{F}^\top\big)^{-1}}_{\mathbf{P}_{1|0}^{-1}} \tilde{\mathbf{x}}_1 &= -\big(\mathbf{Q}_0 + \mathbf{F}\mathbf{P}_0\mathbf{F}^\top\big)^{-1}\mathbf{r}_{\mathbf{x}}. \end{aligned} \tag{336}$$

This is the EKF **propagation**: the predicted covariance and mean are

$$\boxed{\mathbf{P}_{1|0} = \mathbf{F}\mathbf{P}_0\mathbf{F}^\top + \mathbf{Q}_0, \qquad \hat{\mathbf{x}}_{1|0} = \mathbf{f}(\hat{\mathbf{x}}_0) \ \text{(i.e. } \mathbf{r}_{\mathbf{x}} = \mathbf{0}\text{)}.} \tag{337}$$

Update (add the measurement). Now include the measurement $\mathbf{z}_1 = \mathbf{h}(\mathbf{x}_1) + \mathbf{n} \Rightarrow \tilde{\mathbf{z}} = \mathbf{H}\tilde{\mathbf{x}}_1 + \mathbf{n}$. The full cost

$$\|\tilde{\mathbf{z}} - \mathbf{H}\tilde{\mathbf{x}}_1\|^2_{\mathbf{R}} + \left\| \mathbf{r}_{\mathbf{x}} - \begin{bmatrix} \mathbf{F} & -\mathbf{I} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix} \right\|^2_{\mathbf{Q}_0} + \|\tilde{\mathbf{x}}_0\|^2_{\mathbf{P}_0} \tag{338}$$

yields the augmented information system (the measurement adds $\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}$ to the $\mathbf{x}_1$ block and $\mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}}$ to its residual),

$$\begin{bmatrix} \mathbf{P}_0^{-1} + \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{F} & -\mathbf{F}^\top\mathbf{Q}_0^{-1} \\ -\mathbf{Q}_0^{-1}\mathbf{F} & \mathbf{Q}_0^{-1} + \mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_0 \\ \tilde{\mathbf{x}}_1 \end{bmatrix} = \begin{bmatrix} \mathbf{F}^\top\mathbf{Q}_0^{-1}\mathbf{r}_{\mathbf{x}} \\ -\mathbf{Q}_0^{-1}\mathbf{r}_{\mathbf{x}} + \mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}} \end{bmatrix}. \tag{339}$$

Marginalizing $\mathbf{x}_0$ again and recognizing $\mathbf{P}_{1|0}^{-1} = (\mathbf{Q}_0 + \mathbf{F}\mathbf{P}_0\mathbf{F}^\top)^{-1}$,

$$\underbrace{\left(\mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H} + \mathbf{P}_{1|0}^{-1}\right)}_{\boldsymbol{\Sigma} = \mathbf{P}_{1|1}^{-1}} \tilde{\mathbf{x}}_1 = \mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}} - \underbrace{\mathbf{P}_{1|0}^{-1}\mathbf{r}_{\mathbf{x}}}_{\to \mathbf{0}}. \tag{340}$$

After propagation $\mathbf{r}_{\mathbf{x}} = \mathbf{0}$, so the second term drops. Applying the matrix inversion lemma to the updated covariance,

$$\mathbf{P}_{1|1} = \left(\mathbf{P}_{1|0}^{-1} + \mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\right)^{-1} = \mathbf{P}_{1|0} - \mathbf{P}_{1|0}\mathbf{H}^\top\left(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\right)^{-1}\mathbf{H}\mathbf{P}_{1|0}, \tag{341}$$

and for the mean correction, using the same lemma to convert information → covariance form,

$$\begin{aligned}
\tilde{\mathbf{x}}_1 &= \left(\mathbf{P}_{1|0}^{-1} + \mathbf{H}^\top\mathbf{R}^{-1}\mathbf{H}\right)^{-1}\mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}} && (342)\\
&= \Big(\mathbf{P}_{1|0} - \mathbf{P}_{1|0}\mathbf{H}^\top\left(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\right)^{-1}\mathbf{H}\mathbf{P}_{1|0}\Big)\mathbf{H}^\top\mathbf{R}^{-1}\tilde{\mathbf{z}} && (343)\\
&= \mathbf{P}_{1|0}\mathbf{H}^\top\Big(\mathbf{R}^{-1} - \left(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\right)^{-1}\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top\mathbf{R}^{-1}\Big)\tilde{\mathbf{z}} && (344)\\
&= \mathbf{P}_{1|0}\mathbf{H}^\top\left(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\right)^{-1}\Big(\left(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\right)\mathbf{R}^{-1} - \mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top\mathbf{R}^{-1}\Big)\tilde{\mathbf{z}} && (345)\\
&= \mathbf{P}_{1|0}\mathbf{H}^\top\left(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\right)^{-1}\tilde{\mathbf{z}}. && (346)
\end{aligned}$$

Finally, with the Kalman gain $\mathbf{K} = \mathbf{P}_{1|0}\mathbf{H}^\top(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R})^{-1}$, we recover the EKF **update**:

$$\boxed{\tilde{\mathbf{x}}_1 = \mathbf{P}_{1|0}\mathbf{H}^\top\left(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\right)^{-1}\tilde{\mathbf{z}}, \qquad \mathbf{P}_{1|1} = \mathbf{P}_{1|0} - \mathbf{P}_{1|0}\mathbf{H}^\top\left(\mathbf{H}\mathbf{P}_{1|0}\mathbf{H}^\top + \mathbf{R}\right)^{-1}\mathbf{H}\mathbf{P}_{1|0}.} \tag{347}$$

The extended Kalman filter is thus the incremental (marginalize-then-update) solution of the same MAP problem that batch least squares solves jointly — the two views are connected by the matrix inversion lemma and the Schur complement.

4.8 The relationship between Schur complement and nullspace projection

The Schur-complement marginalization (§4.3) and the nullspace projection (§4.6, item 4, the MSCKF) both eliminate $\mathbf{x}_B$ — it turns out they are the *same* operation. Take $\mathbf{z} = \mathbf{h}(\mathbf{x}_A, \mathbf{x}_B) + \mathbf{n}$, $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$, with $\tilde{\mathbf{z}} = \mathbf{H}_A\tilde{\mathbf{x}}_A + \mathbf{H}_B\tilde{\mathbf{x}}_B + \mathbf{n}$.

(1) Information form. The normal equations are

$$\begin{bmatrix} \mathbf{H}_A^\top \mathbf{R}^{-1}\mathbf{H}_A & \mathbf{H}_A^\top \mathbf{R}^{-1}\mathbf{H}_B \\ \mathbf{H}_B^\top \mathbf{R}^{-1}\mathbf{H}_A & \mathbf{H}_B^\top \mathbf{R}^{-1}\mathbf{H}_B \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_A \\ \tilde{\mathbf{x}}_B \end{bmatrix} = \begin{bmatrix} \mathbf{H}_A^\top \\ \mathbf{H}_B^\top \end{bmatrix} \mathbf{R}^{-1}\tilde{\mathbf{z}}. \tag{348}$$

Marginalizing $\mathbf{x}_B$ and keeping $\mathbf{x}_A$ via the Schur complement gives

$$\boxed{\begin{aligned} &\underbrace{\left(\mathbf{H}_A^\top \mathbf{R}^{-1}\mathbf{H}_A - \mathbf{H}_A^\top \mathbf{R}^{-1}\mathbf{H}_B(\mathbf{H}_B^\top \mathbf{R}^{-1}\mathbf{H}_B)^{-1}\mathbf{H}_B^\top \mathbf{R}^{-1}\mathbf{H}_A\right)}_{\text{information}} \tilde{\mathbf{x}}_A \\ &\qquad = \underbrace{\mathbf{H}_A^\top \mathbf{R}^{-1}\tilde{\mathbf{z}} - \mathbf{H}_A^\top \mathbf{R}^{-1}\mathbf{H}_B(\mathbf{H}_B^\top \mathbf{R}^{-1}\mathbf{H}_B)^{-1}\mathbf{H}_B^\top \mathbf{R}^{-1}\tilde{\mathbf{z}}}_{\text{residual}}. \end{aligned}} \tag{Eq. 1}$$

(2) Nullspace form. Take a (thin) QR-type factorization of $\mathbf{H}_B$,

$$\mathbf{H}_B = \begin{bmatrix} \mathbf{Q}_e & \mathbf{Q}_N \end{bmatrix} \begin{bmatrix} \mathbf{U} \\ \mathbf{0} \end{bmatrix} = \mathbf{Q}_e\mathbf{U} + \mathbf{Q}_N \cdot \mathbf{0} = \mathbf{Q}_e\mathbf{U}, \tag{349}$$

with $\mathbf{U}$ upper triangular and $[\mathbf{Q}_e\ \mathbf{Q}_N]$ orthogonal, so $\mathbf{Q}_e\mathbf{Q}_e^\top + \mathbf{Q}_N\mathbf{Q}_N^\top = \mathbf{I}$, $\mathbf{Q}_e^\top\mathbf{Q}_e = \mathbf{I}$, $\mathbf{Q}_N^\top\mathbf{Q}_N = \mathbf{I}$, and the left nullspace satisfies $\mathbf{Q}_N^\top\mathbf{H}_B = \mathbf{Q}_N^\top\mathbf{Q}_e\mathbf{U} = \mathbf{0}$. Projecting the residual onto $\mathbf{Q}_N$ removes $\mathbf{x}_B$:

$$\mathbf{Q}_N^\top\tilde{\mathbf{z}} = \mathbf{Q}_N^\top\mathbf{H}_A\tilde{\mathbf{x}}_A + \cancel{\mathbf{Q}_N^\top\mathbf{H}_B\tilde{\mathbf{x}}_B} + \mathbf{Q}_N^\top\mathbf{n} = \mathbf{Q}_N^\top\mathbf{H}_A\tilde{\mathbf{x}}_A + \mathbf{Q}_N^\top\mathbf{n}, \tag{350}$$

with projected noise covariance $\mathbf{Q}_N^\top\mathbf{R}\mathbf{Q}_N$. The resulting normal equations are

$$\boxed{(\mathbf{Q}_N^\top\mathbf{H}_A)^\top(\mathbf{Q}_N^\top\mathbf{R}\mathbf{Q}_N)^{-1}(\mathbf{Q}_N^\top\mathbf{H}_A)\,\tilde{\mathbf{x}}_A = (\mathbf{Q}_N^\top\mathbf{H}_A)^\top(\mathbf{Q}_N^\top\mathbf{R}\mathbf{Q}_N)^{-1}\mathbf{Q}_N^\top\tilde{\mathbf{z}}.} \tag{Eq. 2}$$

Are Eq. 1 and Eq. 2 equivalent? Consider the isotropic case $\mathbf{R} = \sigma^2\mathbf{I}$. With $\mathbf{H}_B = \mathbf{Q}_e\mathbf{U}$,

$$\mathbf{H}_B^\top\mathbf{R}^{-1}\mathbf{H}_B = (\mathbf{Q}_e\mathbf{U})^\top\frac{\mathbf{I}}{\sigma^2}\mathbf{Q}_e\mathbf{U} = \frac{\mathbf{U}^\top\mathbf{U}}{\sigma^2}, \tag{351}$$

so the projector that appears in the Schur complement of Eq. 1 is

$$\begin{aligned} \mathbf{H}_A^\top\mathbf{R}^{-1}\mathbf{H}_B(\mathbf{H}_B^\top\mathbf{R}^{-1}\mathbf{H}_B)^{-1}\mathbf{H}_B^\top\mathbf{R}^{-1}\mathbf{H}_A &= \mathbf{H}_A^\top\frac{\mathbf{I}}{\sigma^2}\mathbf{Q}_e\mathbf{U}\,\sigma^2(\mathbf{U}^\top\mathbf{U})^{-1}\mathbf{U}^\top\mathbf{Q}_e^\top\frac{\mathbf{I}}{\sigma^2}\mathbf{H}_A \\ &= \frac{1}{\sigma^2}\mathbf{H}_A^\top\mathbf{Q}_e\mathbf{Q}_e^\top\mathbf{H}_A. \end{aligned} \tag{352}$$

Hence the Eq. 1 information becomes, using $\mathbf{Q}_e\mathbf{Q}_e^\top = \mathbf{I} - \mathbf{Q}_N\mathbf{Q}_N^\top$,

$$\frac{1}{\sigma^2}\mathbf{H}_A^\top\mathbf{H}_A - \frac{1}{\sigma^2}\mathbf{H}_A^\top\mathbf{Q}_e\mathbf{Q}_e^\top\mathbf{H}_A = \frac{1}{\sigma^2}\mathbf{H}_A^\top\left(\mathbf{I} - \mathbf{Q}_e\mathbf{Q}_e^\top\right)\mathbf{H}_A = \frac{1}{\sigma^2}\mathbf{H}_A^\top\mathbf{Q}_N\mathbf{Q}_N^\top\mathbf{H}_A. \tag{353}$$

The Eq. 2 information, using $\mathbf{Q}_N^\top\mathbf{R}\mathbf{Q}_N = \sigma^2\mathbf{Q}_N^\top\mathbf{Q}_N = \sigma^2\mathbf{I}$, is

$$(\mathbf{Q}_N^\top\mathbf{H}_A)^\top(\sigma^2\mathbf{I})^{-1}(\mathbf{Q}_N^\top\mathbf{H}_A) = \frac{1}{\sigma^2}\mathbf{H}_A^\top\mathbf{Q}_N\mathbf{Q}_N^\top\mathbf{H}_A, \tag{354}$$

which is identical (the residuals match the same way). Therefore, for isotropic noise, Schur-complement marginalization and left-nullspace projection produce exactly the same information and residual — the MSCKF's structureless update is precisely the marginalization of the feature.

5 Lecture 5: The IMU Model and IMU Integration

Topics. (1) the IMU measurement model (gyroscope, accelerometer, biases, noise); (2) continuous vs. discrete noise; (3) the IMU kinematics as a first-order ODE; (4) the big picture: IMU integration in a visual-inertial navigation system; (5) exact integration from t_k to t_{k+1}; (6) computing $\Delta\mathbf{R}, \Delta\mathbf{p}, \Delta\mathbf{v}$ — preintegration and three methods; (7) mean and covariance propagation with $SO(3) + \mathbb{R}^3$ errors; (8) the linearization using $SE_2(3)$.

5.1 The IMU model

The IMU (inertial measurement unit) is one of the most important sensors for robotics. We use two frames:

- $\{I\}$: the IMU frame (attached to the sensor, see Fig. 15);
- $\{G\}$: the global frame.

The IMU provides two readings: the **gyroscope** reads angular velocity and the **accelerometer** reads acceleration,

$$\boxed{\begin{aligned} {}^I\boldsymbol{\omega}_m &= {}^I\boldsymbol{\omega} + \mathbf{b}_g + \mathbf{n}_g, \\ {}^I\mathbf{a}_m &= {}^I\mathbf{a} + \mathbf{b}_a + \mathbf{n}_a, \end{aligned}} \tag{355, 356}$$

where

- ${}^I\boldsymbol{\omega}$ is the true angular-rate reading and ${}^I\mathbf{a}$ the true acceleration reading;
- $\mathbf{b}_g$ is the gyro bias, driven by a random walk $\dot{\mathbf{b}}_g = \mathbf{n}_{wg}$; $\mathbf{b}_a$ is the accel bias, driven by a random walk $\dot{\mathbf{b}}_a = \mathbf{n}_{wa}$;
- $\mathbf{n}_g \sim \mathcal{N}(\mathbf{0}, \sigma_g^2\mathbf{I}_3)$ and $\mathbf{n}_a \sim \mathcal{N}(\mathbf{0}, \sigma_a^2\mathbf{I}_3)$ are Gaussian noises.

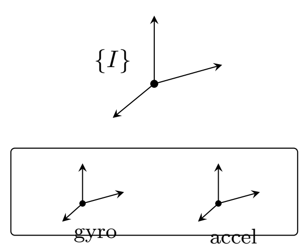


Figure 15: The IMU carries its own frame $\{I\}$; onboard, the gyroscope reads the angular velocity and the accelerometer reads the acceleration.

Another way of modeling the accelerometer makes gravity explicit:

$${}^I\mathbf{a}_m = \left({}^I\mathbf{a}_b - {}^I_G\mathbf{R}\,{}^G\mathbf{g}\right) + \mathbf{b}_a + \mathbf{n}_a, \qquad {}^G\mathbf{g} = \begin{bmatrix} 0 \\ 0 \\ -9.8 \end{bmatrix}, \tag{357}$$

where ${}^I\mathbf{a}_b$ is the body acceleration and ${}^G\mathbf{g}$ is gravity in $\{G\}$.

We use (355) and (356) as our measurements. Inverting them gives the true readings:

$$\text{true gyro reading:} \quad {}^I\boldsymbol{\omega} = {}^I\boldsymbol{\omega}_m - \mathbf{b}_g - \mathbf{n}_g, \tag{358}$$

$$\text{true accel reading:} \quad {}^I\mathbf{a} = {}^I\mathbf{a}_m - \mathbf{b}_a - \mathbf{n}_a. \tag{359}$$

The IMU state. When we model the IMU, we use the following parameters:

- $\{{}^G_I\mathbf{R}, {}^G\mathbf{p}_I\}$: IMU orientation and position — 6 DoF;
- ${}^G\mathbf{v}_I$: IMU velocity — 3 DoF;
- $\mathbf{b}_g$: gyro bias — 3 DoF; $\mathbf{b}_a$: accel bias — 3 DoF.

Then our state vector for the IMU is

$$\mathbf{x}_I = \left\{{}^G_I\mathbf{R},\ {}^G\mathbf{p}_I,\ {}^G\mathbf{v}_I,\ \mathbf{b}_g,\ \mathbf{b}_a\right\}, \qquad 15 \text{ DoF.} \tag{360}$$

Some places use the quaternion ${}^G_I\mathbf{R} \Leftrightarrow {}^G_I\bar{q}$ (Lecture 2); then

$$\mathbf{x}_I = \begin{bmatrix} {}^G_I\bar{q}^\top & {}^G\mathbf{p}_I^\top & {}^G\mathbf{v}_I^\top & \mathbf{b}_g^\top & \mathbf{b}_a^\top \end{bmatrix}^\top, \qquad 16 \times 1 \ \Rightarrow\ 15 \text{ DoF.} \tag{361}$$

5.2 Continuous and discrete noise

For the IMU we specify *continuous-time* noise, $\mathbf{n}_c \sim \mathcal{N}(\mathbf{0}, \sigma_c^2\mathbf{I}_3)$, characterized by

$$\mathbb{E}\big[\mathbf{n}_c(t)\big] = \mathbf{0}_{3\times 1}, \qquad \mathbb{E}\big[\mathbf{n}_c(t+\tau)\,\mathbf{n}_c^\top(t)\big] = \sigma_c^2\mathbf{I}_3\,\delta(\tau), \tag{362}$$

where $\delta(\tau)$ is the impulse function: $\delta(\tau) = 1$ if $\tau = 0$ and $\delta(\tau) = 0$ if $\tau \neq 0$.

Consider a random-walk bias driven by this noise, $\dot{\mathbf{b}}(t) = \mathbf{n}_c(t)$. Integrating from t_k to t_{k+1},

$$\int_{t_k}^{t_{k+1}} \dot{\mathbf{b}}(t)\,dt = \int_{t_k}^{t_{k+1}} \mathbf{n}_c(t)\,dt = \mathbf{b}_{k+1} - \mathbf{b}_k \ \ \Rightarrow\ \ \Delta\mathbf{b} \triangleq \mathbf{b}_{k+1} - \mathbf{b}_k = \int_{t_k}^{t_{k+1}} \mathbf{n}_c(t)\,dt. \tag{363}$$

Its mean is $\mathbb{E}(\Delta\mathbf{b}) = \mathbb{E}\big(\int_{t_k}^{t_{k+1}} \mathbf{n}_c(t)\,dt\big) = \mathbf{0}_{3\times1}$, and its covariance is

$$\begin{aligned}
\operatorname{Cov}(\Delta\mathbf{b}) \triangleq \mathbf{P}_{\Delta\mathbf{b}} = \mathbb{E}\big(\Delta\mathbf{b}\,\Delta\mathbf{b}^\top\big) &= \mathbb{E}\left(\int_{t_k}^{t_{k+1}} \mathbf{n}_c(t)\,dt \cdot \int_{t_k}^{t_{k+1}} \mathbf{n}_c^\top(t)\,dt\right) && (364)\\
&= \mathbb{E}\left(\int_{t_k}^{t_{k+1}}\int_{t_k}^{t_{k+1}} \mathbf{n}_c(t)\,\mathbf{n}_c^\top(s)\,dt\,ds\right) && (365)\\
&= \int_{t_k}^{t_{k+1}} \mathbb{E}\big(\mathbf{n}_c(t)\,\mathbf{n}_c^\top(t)\big)\,dt && (366)\\
&= \int_{t_k}^{t_{k+1}} \sigma_c^2\mathbf{I}_3\,dt = \delta t\,\sigma_c^2\,\mathbf{I}_3, && (367)
\end{aligned}$$

where $\delta t = t_{k+1} - t_k$ (the impulse $\delta(\cdot)$ collapses the double integral). Hence we can **discretize $\mathbf{n}_c$** in $[t_k, t_{k+1}]$ as

$$\boxed{\mathbf{n}_d \sim \mathcal{N}\left(\mathbf{0},\ \frac{\sigma_c^2}{\delta t}\,\mathbf{I}_3\right), \qquad \delta t = t_{k+1} - t_k.} \tag{368}$$

Then $\Delta\mathbf{b} = \int_{t_k}^{t_{k+1}} \mathbf{n}_c\,dt = \mathbf{n}_d \cdot \delta t$, and the discrete covariance is consistent with the continuous one:

$$\operatorname{Cov}(\Delta\mathbf{b})_d = \mathbb{E}\big(\mathbf{n}_d\mathbf{n}_d^\top\big)\,\delta t^2 = \delta t^2 \cdot \frac{\sigma_c^2}{\delta t}\,\mathbf{I}_3 = \sigma_c^2\,\delta t\,\mathbf{I}_3 = \operatorname{Cov}(\Delta\mathbf{b})_c. \checkmark \tag{369}$$

5.3 The IMU kinematics: a first-order ODE

The IMU state evolves by the first-order ordinary differential equation

$$\begin{cases} {}^G_I\dot{\mathbf{R}} = {}^G_I\mathbf{R} \left\lfloor {}^I\boldsymbol{\omega} \right\rfloor_\times = {}^G_I\mathbf{R} \left\lfloor {}^I\boldsymbol{\omega}_m - \mathbf{b}_g - \mathbf{n}_g \right\rfloor_\times, \\ {}^G\dot{\mathbf{p}}_I = {}^G\mathbf{v}_I, \\ {}^G\dot{\mathbf{v}}_I = {}^G\mathbf{a} + {}^G\mathbf{g} = {}^G_I\mathbf{R}\left({}^I\mathbf{a}_m - \mathbf{b}_a - \mathbf{n}_a\right) + {}^G\mathbf{g}, \\ {}^I\dot{\mathbf{b}}_g = \mathbf{n}_{wg}, \qquad {}^I\dot{\mathbf{b}}_a = \mathbf{n}_{wa}. \end{cases} \tag{370}$$

Stacking, we obtain the compact form

$$\boxed{\dot{\mathbf{x}}_I = \mathbf{f}(\mathbf{x}_I, \mathbf{n}_I), \qquad \mathbf{n}_I = \begin{bmatrix} \mathbf{n}_g \\ \mathbf{n}_a \\ \mathbf{n}_{wg} \\ \mathbf{n}_{wa} \end{bmatrix}, \quad \begin{array}{cc} \mathbf{n}_g \sim \mathcal{N}(\mathbf{0}, \sigma_g^2\mathbf{I}_3), & \mathbf{n}_a \sim \mathcal{N}(\mathbf{0}, \sigma_a^2\mathbf{I}_3), \\ \mathbf{n}_{wg} \sim \mathcal{N}(\mathbf{0}, \sigma_{wg}^2\mathbf{I}_3), & \mathbf{n}_{wa} \sim \mathcal{N}(\mathbf{0}, \sigma_{wa}^2\mathbf{I}_3), \end{array}} \tag{371}$$

and, per the previous subsection, the discretized noise over one interval is

$$\mathbf{n}_{Id} = \begin{bmatrix} \mathbf{n}_{gd} \\ \mathbf{n}_{ad} \\ \mathbf{n}_{wgd} \\ \mathbf{n}_{wad} \end{bmatrix}, \qquad \sigma_{gd}^2 = \frac{\sigma_g^2}{\delta t}, \quad \sigma_{ad}^2 = \frac{\sigma_a^2}{\delta t}, \quad \sigma_{wgd}^2 = \frac{\sigma_{wg}^2}{\delta t}, \quad \sigma_{wad}^2 = \frac{\sigma_{wa}^2}{\delta t}. \tag{372}$$

The IMU model is done. There are various IMU models, but the basic idea is the same.

5.4 IMU integration: the big picture

Recall the typical linear system from Lecture 1 and put it side by side with our sensor models (Fig. 16):

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u} & \to \text{state evolution} \to \text{IMU model:} \quad \dot{\mathbf{x}}_I = \mathbf{f}(\mathbf{x}_I, \mathbf{n}_I), \\ \mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u} & \to \text{observation / measurements} \to \text{vision / camera model:} \quad \mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n}_c. \end{cases} \tag{373}$$

Together they form a **VINS** (visual-inertial navigation system). These two models are the focus of our tutorials: how to process the camera measurement $\mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n}_c$ was covered in Lectures 3 and 4; we will learn how to process the IMU in this Lecture 5.

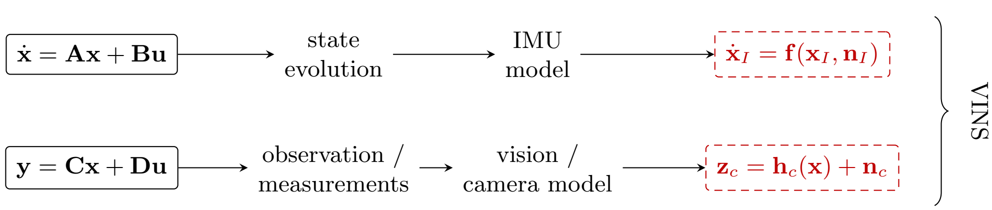


Figure 16: The big picture: the Lecture-1 linear system maps onto the two VINS models — the IMU model plays the state evolution (this lecture) and the camera model plays the observation (Lectures 3–4).

The problem formulation. Given $\mathbf{x}_k \sim \mathcal{N}(\hat{\mathbf{x}}_k, \mathbf{P}_k)$ and the IMU readings $({}^I\boldsymbol{\omega}_{m_k}, {}^I\mathbf{a}_{m_k})$, solve

$$\boxed{\mathbf{x}_{k+1} \sim \mathcal{N}(\hat{\mathbf{x}}_{k+1}, \mathbf{P}_{k+1}) \quad \Rightarrow \quad \text{two targets: } \hat{\mathbf{x}}_{k+1} \text{ and } \mathbf{P}_{k+1}.} \tag{374}$$

That is, the ODE $\dot{\mathbf{x}}_I = \mathbf{f}(\mathbf{x}_I, \mathbf{n}_I)$ carries the *current state distribution* $\mathbf{x}_k \sim \mathcal{N}(\hat{\mathbf{x}}_k, \mathbf{P}_k)$ into the *next state distribution* $\mathbf{x}_{k+1} \sim \mathcal{N}(\hat{\mathbf{x}}_{k+1}, \mathbf{P}_{k+1})$.

How to solve it? From $\dot{\mathbf{x}}_I = \mathbf{f}(\mathbf{x}_I, \mathbf{n}_I)$ we can get the *integrated function*

$$\mathbf{x}_{k+1} = \mathbf{f}_{\text{int}}\big(\mathbf{x}_k,\, {}^I\boldsymbol{\omega}_m,\, {}^I\mathbf{a}_m,\, \mathbf{n}_k\big), \qquad \begin{cases} \mathbf{x}_{k+1} = \mathbf{x}_{I_{k+1}}, \\ \mathbf{x}_k = \mathbf{x}_{I_k}, \\ \mathbf{n}_k = \mathbf{n}_{I_k} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_{I_k}). \end{cases} \tag{375}$$

Hence the **mean** is propagated with zero noise,

$$\hat{\mathbf{x}}_{k+1} = \mathbf{f}_{\text{int}}\big(\hat{\mathbf{x}}_k,\, {}^I\boldsymbol{\omega}_m,\, {}^I\mathbf{a}_m,\, \mathbf{0}\big) \quad \rightarrow \quad \text{solved } \hat{\mathbf{x}}_{k+1}. \tag{376}$$

Then, linearizing $\mathbf{f}_{\text{int}}$, we get the error-state propagation with the discretized noise $\mathbf{n}_{kd}$,

$$\tilde{\mathbf{x}}_{k+1} = \boldsymbol{\Phi}_{(k+1,k)}\, \tilde{\mathbf{x}}_k + \mathbf{G}_k\, \mathbf{n}_{kd}, \tag{377}$$

and therefore the **covariance**

$$\mathbf{P}_{k+1} = \mathbb{E}\big(\tilde{\mathbf{x}}_{k+1}\tilde{\mathbf{x}}_{k+1}^\top\big) = \boldsymbol{\Phi}_{(k+1,k)}\, \mathbf{P}_k\, \boldsymbol{\Phi}_{(k+1,k)}^\top + \mathbf{G}_k\, \mathbf{R}_{Id}\, \mathbf{G}_k^\top \quad \rightarrow \quad \text{solved } \mathbf{P}_{k+1}. \tag{378}$$

Given the big picture, let's explore the detailed derivations.

5.5 Exact integration from t_k to t_{k+1}

The input. Split each true reading into its estimate and error,

$${}^I\boldsymbol{\omega} = {}^I\boldsymbol{\omega}_m - \mathbf{b}_g - \mathbf{n}_g = \underbrace{\big({}^I\boldsymbol{\omega}_m - \hat{\mathbf{b}}_g\big)}_{{}^I\hat{\boldsymbol{\omega}}} + \underbrace{\big(-\tilde{\mathbf{b}}_g - \mathbf{n}_g\big)}_{{}^I\tilde{\boldsymbol{\omega}}} = {}^I\hat{\boldsymbol{\omega}} + {}^I\tilde{\boldsymbol{\omega}}, \tag{379}$$

$${}^I\mathbf{a} = {}^I\mathbf{a}_m - \mathbf{b}_a - \mathbf{n}_a = \underbrace{\big({}^I\mathbf{a}_m - \hat{\mathbf{b}}_a\big)}_{{}^I\hat{\mathbf{a}}} + \underbrace{\big(-\tilde{\mathbf{b}}_a - \mathbf{n}_a\big)}_{{}^I\tilde{\mathbf{a}}} = {}^I\hat{\mathbf{a}} + {}^I\tilde{\mathbf{a}}. \tag{380}$$

We integrate from k to $k+1$, with $\delta t = t_{k+1} - t_k$.

For rotation.

$${}^G_{I_{k+1}}\mathbf{R} = {}^G_{I_k}\mathbf{R} \cdot {}^{I_k}_{I_{k+1}}\mathbf{R}, \qquad \Delta\mathbf{R} \triangleq {}^{I_k}_{I_{k+1}}\mathbf{R}, \tag{381}$$

where $\Delta\mathbf{R}$ is the rotation integration from ${}^I\boldsymbol{\omega}$.

For velocity. From ${}^G\dot{\mathbf{v}}_{I_\tau} = {}^G_{I_\tau}\mathbf{R}\,{}^{I_\tau}\mathbf{a} + {}^G\mathbf{g}$,

$$\int_{t_k}^{t_{k+1}} {}^G\dot{\mathbf{v}}_{I_\tau}\, d\tau = \int_{t_k}^{t_{k+1}} \left({}^G_{I_\tau}\mathbf{R}\,{}^{I_\tau}\mathbf{a} + {}^G\mathbf{g}\right) d\tau. \tag{382}$$

The left-hand side is $\int_{t_k}^{t_{k+1}} {}^G\dot{\mathbf{v}}_{I_\tau}\, d\tau = {}^G\mathbf{v}_{I_{k+1}} - {}^G\mathbf{v}_{I_k}$, and the right-hand side is

$$\begin{aligned}
\int_{t_k}^{t_{k+1}} \left({}^G_{I_\tau}\mathbf{R}\,{}^{I_\tau}\mathbf{a} + {}^G\mathbf{g}\right) d\tau &= \int_{t_k}^{t_{k+1}} {}^G_{I_\tau}\mathbf{R}\,{}^{I_\tau}\mathbf{a}\, d\tau + {}^G\mathbf{g}\,\delta t &(383)\\
&= \int_{t_k}^{t_{k+1}} {}^G_{I_k}\mathbf{R}\,{}^{I_k}_{I_\tau}\mathbf{R}\,{}^{I_\tau}\mathbf{a}\, d\tau + {}^G\mathbf{g}\,\delta t &(384)\\
&= {}^G_{I_k}\mathbf{R} \underbrace{\int_{t_k}^{t_{k+1}} {}^{I_k}_{I_\tau}\mathbf{R}\,{}^{I_\tau}\mathbf{a}\, d\tau}_{\Delta\mathbf{v}} + {}^G\mathbf{g}\,\delta t. &(385)
\end{aligned}$$

Then

$${}^G\mathbf{v}_{I_{k+1}} = {}^G\mathbf{v}_{I_k} + {}^G_{I_k}\mathbf{R}\,\Delta\mathbf{v} + {}^G\mathbf{g}\,\delta t, \qquad \Delta\mathbf{v} \triangleq \int_{t_k}^{t_{k+1}} {}^{I_k}_{I_\tau}\mathbf{R}\,{}^{I_\tau}\mathbf{a}\, d\tau, \tag{386}$$

where $\Delta\mathbf{v}$ is the integration of ${}^I\boldsymbol{\omega}$ and ${}^I\mathbf{a}$.

For position. From ${}^G\dot{\mathbf{p}}_{I_\tau} = {}^G\mathbf{v}_{I_\tau}$,

$$\int_{t_k}^{t_{k+1}} {}^G\dot{\mathbf{p}}_{I_\tau}\, d\tau = \int_{t_k}^{t_{k+1}} {}^G\mathbf{v}_{I_\tau}\, d\tau. \tag{387}$$

Note that, by the velocity result applied up to time τ (with $\delta\tau = \tau - t_k$),

$${}^G\mathbf{v}_{I_\tau} = {}^G\mathbf{v}_{I_k} + {}^G_{I_k}\mathbf{R} \int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R}\,{}^{I_s}\mathbf{a}\, ds + {}^G\mathbf{g}\,\delta\tau. \tag{388}$$

The left-hand side is ${}^G\mathbf{p}_{I_{k+1}} - {}^G\mathbf{p}_{I_k}$, and the right-hand side becomes

$$\begin{aligned}
\int_{t_k}^{t_{k+1}} {}^G\mathbf{v}_{I_\tau}\, d\tau &= \int_{t_k}^{t_{k+1}} \left({}^G\mathbf{v}_{I_k} + {}^G_{I_k}\mathbf{R} \int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R}\,{}^{I_s}\mathbf{a}\, ds + {}^G\mathbf{g}\,\delta\tau \right) d\tau &(389)\\
&= \int_{t_k}^{t_{k+1}} {}^G\mathbf{v}_{I_k}\, d\tau + \int_{t_k}^{t_{k+1}} {}^G\mathbf{g}\,\delta\tau\, d\tau + {}^G_{I_k}\mathbf{R} \int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R}\,{}^{I_s}\mathbf{a}\, ds\, d\tau &(390)\\
&= {}^G\mathbf{v}_{I_k}\,\delta t + \frac{1}{2}{}^G\mathbf{g}\,\delta t^2 + {}^G_{I_k}\mathbf{R} \underbrace{\int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R}\,{}^{I_s}\mathbf{a}\, ds\, d\tau}_{\Delta\mathbf{p}}. &(391)
\end{aligned}$$

Then

$${}^G\mathbf{p}_{I_{k+1}} = {}^G\mathbf{p}_{I_k} + {}^G\mathbf{v}_{I_k}\,\delta t + {}^G_{I_k}\mathbf{R}\,\Delta\mathbf{p} + \frac{1}{2}{}^G\mathbf{g}\,\delta t^2, \qquad \Delta\mathbf{p} \triangleq \int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R}\,{}^{I_s}\mathbf{a}\, ds\, d\tau, \tag{392}$$

where $\Delta\mathbf{p}$ is the *double* integration of ${}^I\boldsymbol{\omega}$ and ${}^I\mathbf{a}$.

For the biases. From $\dot{\mathbf{b}}_g = \mathbf{n}_{wg}$,

$$\int_{t_k}^{t_{k+1}} \dot{\mathbf{b}}_g \, d\tau = \int_{t_k}^{t_{k+1}} \mathbf{n}_{wg} \, d\tau \quad \Rightarrow \quad \mathbf{b}_{g_{k+1}} = \mathbf{b}_{g_k} + \int_{t_k}^{t_{k+1}} \mathbf{n}_{wg} \, d\tau, \tag{393}$$

and for the accel bias, $\mathbf{b}_{a_{k+1}} = \mathbf{b}_{a_k} + \int_{t_k}^{t_{k+1}} \mathbf{n}_{wa} \, d\tau$.

Finally. Collecting everything:

$$\boxed{\begin{aligned}
{}^{G}_{I_{k+1}}\mathbf{R} &= {}^{G}_{I_k}\mathbf{R} \cdot \Delta\mathbf{R}, & \Delta\mathbf{R} &= {}^{I_k}_{I_{k+1}}\mathbf{R}, \\
{}^{G}\mathbf{p}_{I_{k+1}} &= {}^{G}\mathbf{p}_{I_k} + {}^{G}\mathbf{v}_{I_k} \, \delta t + {}^{G}_{I_k}\mathbf{R} \, \Delta\mathbf{p} + \tfrac{1}{2} {}^{G}\mathbf{g} \, \delta t^2, & \Delta\mathbf{p} &= \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R} \, {}^{I_s}\mathbf{a} \, ds \, d\tau, \\
{}^{G}\mathbf{v}_{I_{k+1}} &= {}^{G}\mathbf{v}_{I_k} + {}^{G}_{I_k}\mathbf{R} \, \Delta\mathbf{v} + {}^{G}\mathbf{g} \, \delta t, & \Delta\mathbf{v} &= \int_{t_k}^{t_{k+1}} {}^{I_k}_{I_\tau}\mathbf{R} \, {}^{I_\tau}\mathbf{a} \, d\tau, \\
\mathbf{b}_{g_{k+1}} &= \mathbf{b}_{g_k} + \textstyle\int_{t_k}^{t_{k+1}} \mathbf{n}_{wg} \, d\tau, & \Delta\mathbf{b}_g &= \int_{t_k}^{t_{k+1}} \mathbf{n}_{wg} \, d\tau, \\
\mathbf{b}_{a_{k+1}} &= \mathbf{b}_{a_k} + \textstyle\int_{t_k}^{t_{k+1}} \mathbf{n}_{wa} \, d\tau, & \Delta\mathbf{b}_a &= \int_{t_k}^{t_{k+1}} \mathbf{n}_{wa} \, d\tau.
\end{aligned}} \tag{394}$$

This is exactly the integrated function $\mathbf{x}_{k+1} = \mathbf{f}_{\text{int}}(\mathbf{x}_k, {}^{I}\boldsymbol{\omega}_m, {}^{I}\mathbf{a}_m, \mathbf{n}_k)$.

> **To this step, no assumption or approximation is made — every step is accurate.** The accuracy of the integration is decided by how $\Delta\mathbf{R}$, $\Delta\mathbf{p}$, $\Delta\mathbf{v}$ are computed from ${}^{I}\boldsymbol{\omega}$ and ${}^{I}\mathbf{a}$.

5.6 Computing $\Delta\mathbf{R}, \Delta\mathbf{p}, \Delta\mathbf{v}$: preintegration and three methods

The way to compute $\Delta\mathbf{R}$, $\Delta\mathbf{p}$, $\Delta\mathbf{v}$ given $\boldsymbol{\omega}_m$ and $\mathbf{a}_m$ defines the algorithms. A few discussions follow.

(1) Connection to preintegration. $\Delta\mathbf{R}$, $\Delta\mathbf{p}$, $\Delta\mathbf{v}$ are also called the *pre-integrated measurements*: they are only affected by ${}^{I}\boldsymbol{\omega}_m$, ${}^{I}\mathbf{a}_m$ and $\mathbf{b}_g$, $\mathbf{b}_a$ (not by the global state). Inverting (394),

$$\Delta\mathbf{R} \triangleq {}^{I_k}_{I_{k+1}}\mathbf{R} \qquad = {}^{G}_{I_k}\mathbf{R}^{\top} \, {}^{G}_{I_{k+1}}\mathbf{R}, \tag{395}$$

$$\Delta\mathbf{p} \triangleq \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R} \, {}^{I_s}\mathbf{a} \, ds \, d\tau \qquad = {}^{G}_{I_k}\mathbf{R}^{\top} \big({}^{G}\mathbf{p}_{I_{k+1}} - {}^{G}\mathbf{p}_{I_k} - {}^{G}\mathbf{v}_{I_k} \, \delta t - \tfrac{1}{2} {}^{G}\mathbf{g} \, \delta t^2 \big), \tag{396}$$

$$\Delta\mathbf{v} \triangleq \int_{t_k}^{t_{k+1}} {}^{I_k}_{I_\tau}\mathbf{R} \, {}^{I_\tau}\mathbf{a} \, d\tau \qquad = {}^{G}_{I_k}\mathbf{R}^{\top} \big({}^{G}\mathbf{v}_{I_{k+1}} - {}^{G}\mathbf{v}_{I_k} - {}^{G}\mathbf{g} \, \delta t \big), \tag{397}$$

$$\Delta\mathbf{b}_g \triangleq \int_{t_k}^{t_{k+1}} \mathbf{n}_{wg} \, d\tau \qquad = \mathbf{b}_{g_{k+1}} - \mathbf{b}_{g_k}, \tag{398}$$

$$\Delta\mathbf{b}_a \triangleq \int_{t_k}^{t_{k+1}} \mathbf{n}_{wa} \, d\tau \qquad = \mathbf{b}_{a_{k+1}} - \mathbf{b}_{a_k}. \tag{399}$$

So the IMU integration/propagation and pre-integration are connected: they compute the same quantities.

(2) Discrete preintegration on the manifold. For example, IMU preintegration on manifold (Forster, Carlone, et al.) [9] and VINS-Mono (Tong Qin, Shaojie Shen) [13]. Assume ${}^{I_k}\boldsymbol{\omega}$ and ${}^{I_k}\mathbf{a}$ are constant in $[t_k, t_{k+1}]$ *and* $\delta t = t_{k+1} - t_k$ is small, so ${}^{I_k}_{I_\tau}\mathbf{R} \approx \mathbf{I}$ as $\delta t \to 0$:

$$\Delta\mathbf{R} \triangleq {}^{I_k}_{I_{k+1}}\mathbf{R} \approx \operatorname{Exp}\left({}^{I_k}\boldsymbol{\omega} \cdot \delta t\right), \tag{400}$$

$$\Delta\mathbf{p} = \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R}\, {}^{I_s}\mathbf{a}\, ds\, d\tau \approx \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \mathbf{I}_3 \cdot {}^{I_k}\mathbf{a}\, ds\, d\tau = \frac{1}{2}\, {}^{I_k}\mathbf{a}\, \delta t^2, \tag{401}$$

$$\Delta\mathbf{v} = \int_{t_k}^{t_{k+1}} {}^{I_k}_{I_\tau}\mathbf{R}\, {}^{I_\tau}\mathbf{a}\, d\tau \approx {}^{I_k}\mathbf{a}\, \delta t. \tag{402}$$

Then the propagation becomes

$$\begin{cases} {}^{G}_{I_{k+1}}\mathbf{R} = {}^{G}_{I_k}\mathbf{R} \cdot \operatorname{Exp}\left({}^{I_k}\boldsymbol{\omega}\, \delta t\right), \\ {}^{G}\mathbf{p}_{I_{k+1}} = {}^{G}\mathbf{p}_{I_k} + {}^{G}\mathbf{v}_{I_k}\, \delta t + \frac{1}{2}\, {}^{G}_{I_k}\mathbf{R}\, {}^{I_k}\mathbf{a}\, \delta t^2 + \frac{1}{2}\, {}^{G}\mathbf{g}\, \delta t^2, \\ {}^{G}\mathbf{v}_{I_{k+1}} = {}^{G}\mathbf{v}_{I_k} + {}^{G}_{I_k}\mathbf{R}\, {}^{I_k}\mathbf{a}\, \delta t + {}^{G}\mathbf{g}\, \delta t, \\ \mathbf{b}_{g_{k+1}} = \mathbf{b}_{g_k} + \int_{t_k}^{t_{k+1}} \mathbf{n}_{wg}\, d\tau, \qquad \mathbf{b}_{a_{k+1}} = \mathbf{b}_{a_k} + \int_{t_k}^{t_{k+1}} \mathbf{n}_{wa}\, d\tau. \end{cases} \tag{403}$$

This is easy and simple, but it might lose some accuracy for fast rotation and low sampling rate (δt larger).

(3) Analytic integration: ACI^2 and CPI. ACI^2 — analytic combined IMU integration (Yulin/Chuchu) [14]; CPI — continuous preintegration (Kevin/Patrick) [15, 10]. If we assume only that ${}^{I_k}\boldsymbol{\omega}$ and ${}^{I_k}\mathbf{a}$ are constant in $[t_k, t_{k+1}]$ (without the small-δt approximation), then ${}^{I_k}_{I_\tau}\mathbf{R} = \operatorname{Exp}({}^{I_k}\boldsymbol{\omega}\, \delta\tau)$ and

$$\Delta\mathbf{R} \triangleq {}^{I_k}_{I_{k+1}}\mathbf{R} = \operatorname{Exp}\left({}^{I_k}\boldsymbol{\omega} \cdot \delta t\right), \tag{404}$$

$$\Delta\mathbf{p} = \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} {}^{I_k}_{I_s}\mathbf{R}\, {}^{I_s}\mathbf{a}\, ds\, d\tau = \int_{t_k}^{t_{k+1}} \int_{t_k}^{\tau} \operatorname{Exp}\left({}^{I_k}\boldsymbol{\omega}\, \delta s\right) {}^{I_k}\mathbf{a}\, ds\, d\tau, \tag{405}$$

$$\Delta\mathbf{v} = \int_{t_k}^{t_{k+1}} {}^{I_k}_{I_\tau}\mathbf{R}\, {}^{I_\tau}\mathbf{a}\, d\tau = \int_{t_k}^{t_{k+1}} \operatorname{Exp}\left({}^{I_k}\boldsymbol{\omega}\, \delta\tau\right) {}^{I_k}\mathbf{a}\, d\tau, \tag{406}$$

which admit closed-form (analytic) solutions.

(4) Two ways to solve for $\hat{\mathbf{x}}_{k+1}$ and $\mathbf{P}_{k+1}$. We can use different ways to solve for $\hat{\mathbf{x}}_{k+1}$ and $\mathbf{P}_{k+1}$.

Way 1 — discretize, then linearize. Integrate the nonlinear ODE first, then linearize the integrated function (the route taken in this lecture):

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{n}_I) \quad \to \quad \mathbf{x}_{k+1} = \mathbf{f}_{\text{int}}(\mathbf{x}_k, \mathbf{n}_k) \quad \to \quad \begin{cases} \hat{\mathbf{x}}_{k+1} = \mathbf{f}_{\text{int}}(\hat{\mathbf{x}}_k, \mathbf{0}), \\ \tilde{\mathbf{x}}_{k+1} = \boldsymbol{\Phi}_{(k+1,k)}\, \tilde{\mathbf{x}}_k + \mathbf{G}_k\, \mathbf{n}_{kd}. \end{cases} \tag{407}$$

Way 2 — linearize, then integrate. Linearize the continuous ODE first to get the continuous error-state equation, then integrate it:

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{n}_I) \quad \Rightarrow \quad \dot{\tilde{\mathbf{x}}} = \mathbf{F}\, \tilde{\mathbf{x}} + \mathbf{J}\, \mathbf{n}_I, \tag{408}$$

whose solution over $[t_k, t_{k+1}]$ is

$$\tilde{\mathbf{x}}_{k+1} = \mathbf{\Phi}_{(k+1,k)}\,\tilde{\mathbf{x}}_k + \int_{t_k}^{t_{k+1}} \mathbf{\Phi}_{(k+1,\tau)}\,\mathbf{J}(\tau)\,\mathbf{n}_I\,d\tau, \qquad \mathbf{\Phi}_{(k+1,k)} = \exp\left(\int_{t_k}^{t_{k+1}} \mathbf{F}\,dt\right). \tag{409}$$

Different systems treat this state-transition matrix differently:

- **VINS-Mono** [13]: take $\mathbf{F}$ constant and $\delta t \to 0$, so $\mathbf{\Phi}_{(k+1,k)} = \exp\left(\int_{t_k}^{t_{k+1}} \mathbf{F}\,dt\right) \approx \mathbf{I} + \mathbf{F}\,\delta t$;
- **OC-VINS** (Joel Hesch) [16] / **CPI** (Kevin/Patrick) [15] / **ACI**^{2} (Yulin/Chuchu) [14]: keep $\mathbf{\Phi}_{(k+1,k)} = \exp\left(\int_{t_k}^{t_{k+1}} \mathbf{F}\,dt\right)$, evaluated with ${}^I\boldsymbol{\omega}_m$, ${}^I\mathbf{a}_m$ constant in $[t_k, t_{k+1}]$.

How to address the noise term of (409),

$$\mathbf{n}_{\text{int}} \triangleq \int_{t_k}^{t_{k+1}} \mathbf{\Phi}_{(k+1,\tau)}\,\mathbf{J}(\tau)\,\mathbf{n}_I\,d\tau\,? \tag{410}$$

(1) Pull the *discrete noise* $\mathbf{n}_{Id}$ out of the integral:

$$\mathbf{n}_{\text{int}} = \int_{t_k}^{t_{k+1}} \mathbf{\Phi}_{(k+1,\tau)}\,\mathbf{J}(\tau)\,d\tau \cdot \mathbf{n}_{Id}, \tag{411}$$

$$\Rightarrow\ \text{Cov}(\mathbf{n}_{\text{int}}) = \int_{t_k}^{t_{k+1}} \mathbf{\Phi}_{(k+1,\tau)}\mathbf{J}(\tau)\,d\tau \cdot \mathbf{Q}_{Id} \cdot \left(\int_{t_k}^{t_{k+1}} \mathbf{\Phi}_{(k+1,\tau)}\mathbf{J}(\tau)\,d\tau\right)^{\top}. \tag{412}$$

(2) Keep the noise inside the integral:

$$\text{Cov}(\mathbf{n}_{\text{int}}) = \mathbb{E}\left(\int_{t_k}^{t_{k+1}} \mathbf{\Phi}_{(k+1,\tau)}\,\mathbf{J}(\tau)\,\mathbf{n}_{Id}\,\mathbf{n}_{Id}^{\top}\,\mathbf{J}^{\top}(\tau)\,\mathbf{\Phi}_{(k+1,\tau)}^{\top}\,d\tau\right) \tag{413}$$

$$= \int_{t_k}^{t_{k+1}} \mathbf{\Phi}_{(k+1,\tau)}\,\mathbf{J}(\tau)\,\mathbf{Q}_{Id}\,\mathbf{J}^{\top}(\tau)\,\mathbf{\Phi}_{(k+1,\tau)}^{\top}\,d\tau. \tag{414}$$

Either way, the *linearization* itself can be carried out with different error parametrizations, all 15 DoF for $\mathbf{x}_I$:

$$\boxed{\begin{cases} SO(3) + \mathbb{R}^3 + \mathbb{R}^3 + \mathbb{R}^3 + \mathbb{R}^3 & \Rightarrow\ 15 \text{ DoF for } \mathbf{x}_I, \\ SE_2(3) + \mathbb{R}^3 + \mathbb{R}^3 & \Rightarrow\ 15 \text{ DoF for } \mathbf{x}_I, \\ SE(3) + \mathbb{R}^3 + \mathbb{R}^3 + \mathbb{R}^3 & \Rightarrow\ 15 \text{ DoF for } \mathbf{x}_I. \end{cases}} \tag{415}$$

We now use the analytic form from (3) for the detailed mean and covariance propagation, working out the first two parametrizations: $SO(3) + \mathbb{R}^{12}$ in §5.7 and $SE_2(3) + \mathbb{R}^6$ in the final subsection.

5.7 Mean and covariance propagation with $SO(3) + \mathbb{R}^3$ errors

The input, discretized. As in §5.5, but discretizing the noises over the interval ($\mathbf{n}_g \to \mathbf{n}_{gd}$, $\mathbf{n}_a \to \mathbf{n}_{ad}$):

$${}^I\boldsymbol{\omega} = {}^I\hat{\boldsymbol{\omega}} + {}^I\tilde{\boldsymbol{\omega}}, \quad \begin{cases} {}^I\hat{\boldsymbol{\omega}} = {}^I\boldsymbol{\omega}_m - \hat{\mathbf{b}}_g \\ {}^I\tilde{\boldsymbol{\omega}} = -\tilde{\mathbf{b}}_g - \mathbf{n}_{gd} \end{cases} \qquad {}^I\mathbf{a} = {}^I\hat{\mathbf{a}} + {}^I\tilde{\mathbf{a}}, \quad \begin{cases} {}^I\hat{\mathbf{a}} = {}^I\mathbf{a}_m - \hat{\mathbf{b}}_a \\ {}^I\tilde{\mathbf{a}} = -\tilde{\mathbf{b}}_a - \mathbf{n}_{ad} \end{cases} \tag{416}$$

Let's integrate $\Delta\mathbf{R}$, $\Delta\mathbf{p}$, $\Delta\mathbf{v}$, $\Delta\mathbf{b}_g$, $\Delta\mathbf{b}_a$ with these perturbed inputs.

Perturbing $\Delta\mathbf{R}$. With $\hat{\boldsymbol{\theta}} \triangleq {}^{I_k}\hat{\boldsymbol{\omega}}\,\delta t$ and the right Jacobian $\mathbf{J}_r$ from Lecture 2,

$$\begin{aligned}\Delta\mathbf{R} = \operatorname{Exp}\left({}^{I_k}\boldsymbol{\omega}\,\delta t\right) &= \operatorname{Exp}\left(({}^{I_k}\hat{\boldsymbol{\omega}} + {}^{I_k}\tilde{\boldsymbol{\omega}})\,\delta t\right) &(417)\\ &= \operatorname{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta t\right)\,\operatorname{Exp}\left(\mathbf{J}_r(\hat{\boldsymbol{\theta}})\,{}^{I_k}\tilde{\boldsymbol{\omega}}\,\delta t\right) &(418)\\ &\approx \operatorname{Exp}\left(\hat{\boldsymbol{\theta}}\right)\left(\mathbf{I} + \left\lfloor \mathbf{J}_r(\hat{\boldsymbol{\theta}})\,\delta t\,{}^{I_k}\tilde{\boldsymbol{\omega}}\right\rfloor_\times\right). &(419)\end{aligned}$$

Perturbing $\Delta\mathbf{p}$. Substituting the split inputs and expanding to first order (using $\lfloor\mathbf{x}\rfloor_\times\mathbf{y} = -\lfloor\mathbf{y}\rfloor_\times\mathbf{x}$ and dropping second-order error terms),

$$\begin{aligned}\Delta\mathbf{p} &= \int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau}\operatorname{Exp}\left(({}^{I_k}\hat{\boldsymbol{\omega}} + {}^{I_k}\tilde{\boldsymbol{\omega}})\,\delta s\right)ds\,d\tau\,\left({}^{I_k}\hat{\mathbf{a}} + {}^{I_k}\tilde{\mathbf{a}}\right) &(420)\\ &= \int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau}\operatorname{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta s\right)\left(\mathbf{I} + \left\lfloor\mathbf{J}_r({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta s)\,\delta s\,{}^{I_k}\tilde{\boldsymbol{\omega}}\right\rfloor_\times\right)ds\,d\tau\,\left({}^{I_k}\hat{\mathbf{a}} + {}^{I_k}\tilde{\mathbf{a}}\right) &(421)\\ &= \underbrace{\int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau}\operatorname{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta s\right)ds\,d\tau\;\;{}^{I_k}\hat{\mathbf{a}}}_{\Delta\hat{\mathbf{p}}} + \underbrace{\int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau}\operatorname{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta s\right)ds\,d\tau}_{\boldsymbol{\Xi}_2}\;{}^{I_k}\tilde{\mathbf{a}}\\ &\quad - \underbrace{\int_{t_k}^{t_{k+1}}\int_{t_k}^{\tau}\operatorname{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta s\right)\left\lfloor{}^{I_k}\hat{\mathbf{a}}\right\rfloor_\times\,\mathbf{J}_r\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta s\right)\delta s\,ds\,d\tau}_{\boldsymbol{\Xi}_4}\;{}^{I_k}\tilde{\boldsymbol{\omega}} &(422)\\ &= \Delta\hat{\mathbf{p}} + \boldsymbol{\Xi}_2\,{}^{I_k}\tilde{\mathbf{a}} - \boldsymbol{\Xi}_4\,{}^{I_k}\tilde{\boldsymbol{\omega}}. &(423)\end{aligned}$$

Perturbing $\Delta\mathbf{v}$. By the same expansion,

$$\begin{aligned}\Delta\mathbf{v} &= \int_{t_k}^{t_{k+1}}\operatorname{Exp}\left(({}^{I_k}\hat{\boldsymbol{\omega}} + {}^{I_k}\tilde{\boldsymbol{\omega}})\,\delta\tau\right)d\tau\,\left({}^{I_k}\hat{\mathbf{a}} + {}^{I_k}\tilde{\mathbf{a}}\right) &(424)\\ &= \int_{t_k}^{t_{k+1}}\operatorname{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta\tau\right)\left(\mathbf{I} + \left\lfloor\mathbf{J}_r({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta\tau)\,\delta\tau\,{}^{I_k}\tilde{\boldsymbol{\omega}}\right\rfloor_\times\right)d\tau\,\left({}^{I_k}\hat{\mathbf{a}} + {}^{I_k}\tilde{\mathbf{a}}\right) &(425)\\ &= \underbrace{\int_{t_k}^{t_{k+1}}\operatorname{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta\tau\right)d\tau\;\;{}^{I_k}\hat{\mathbf{a}}}_{\Delta\hat{\mathbf{v}}} + \underbrace{\int_{t_k}^{t_{k+1}}\operatorname{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta\tau\right)d\tau}_{\boldsymbol{\Xi}_1}\;{}^{I_k}\tilde{\mathbf{a}}\\ &\quad - \underbrace{\int_{t_k}^{t_{k+1}}\operatorname{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta\tau\right)\left\lfloor{}^{I_k}\hat{\mathbf{a}}\right\rfloor_\times\,\mathbf{J}_r\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta\tau\right)\delta\tau\,d\tau}_{\boldsymbol{\Xi}_3}\;{}^{I_k}\tilde{\boldsymbol{\omega}} &(426)\\ &= \Delta\hat{\mathbf{v}} + \boldsymbol{\Xi}_1\,{}^{I_k}\tilde{\mathbf{a}} - \boldsymbol{\Xi}_3\,{}^{I_k}\tilde{\boldsymbol{\omega}}. &(427)\end{aligned}$$

The perturbed propagation. Then we have

$$\begin{cases}{}^{G}_{I_{k+1}}\mathbf{R} = {}^{G}_{I_k}\mathbf{R}\cdot\operatorname{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta t\right)\cdot\operatorname{Exp}\left(\mathbf{J}_r(\hat{\boldsymbol{\theta}})\,\delta t\,{}^{I_k}\tilde{\boldsymbol{\omega}}\right),\\ {}^{G}\mathbf{p}_{I_{k+1}} = {}^{G}\mathbf{p}_{I_k} + {}^{G}\mathbf{v}_{I_k}\,\delta t + {}^{G}_{I_k}\mathbf{R}\left(\Delta\hat{\mathbf{p}} + \boldsymbol{\Xi}_2\,{}^{I_k}\tilde{\mathbf{a}} - \boldsymbol{\Xi}_4\,{}^{I_k}\tilde{\boldsymbol{\omega}}\right) + \frac{1}{2}{}^{G}\mathbf{g}\,\delta t^2,\\ {}^{G}\mathbf{v}_{I_{k+1}} = {}^{G}\mathbf{v}_{I_k} + {}^{G}_{I_k}\mathbf{R}\left(\Delta\hat{\mathbf{v}} + \boldsymbol{\Xi}_1\,{}^{I_k}\tilde{\mathbf{a}} - \boldsymbol{\Xi}_3\,{}^{I_k}\tilde{\boldsymbol{\omega}}\right) + {}^{G}\mathbf{g}\,\delta t,\\ \mathbf{b}_{g_{k+1}} = \mathbf{b}_{g_k} + \mathbf{n}_{wgd}\,\delta t,\\ \mathbf{b}_{a_{k+1}} = \mathbf{b}_{a_k} + \mathbf{n}_{wad}\,\delta t.\end{cases} \tag{428}$$

First: the mean propagation. Setting all errors and noises to zero,

$$\boxed{\begin{cases} {}^{G}_{I_{k+1}}\hat{\mathbf{R}} = {}^{G}_{I_k}\hat{\mathbf{R}} \cdot \operatorname{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\, \delta t\right), \\ {}^{G}\hat{\mathbf{p}}_{I_{k+1}} = {}^{G}\hat{\mathbf{p}}_{I_k} + {}^{G}\hat{\mathbf{v}}_{I_k}\, \delta t + {}^{G}_{I_k}\hat{\mathbf{R}}\, \Delta\hat{\mathbf{p}} + \frac{1}{2}{}^{G}\mathbf{g}\, \delta t^2, \\ {}^{G}\hat{\mathbf{v}}_{I_{k+1}} = {}^{G}\hat{\mathbf{v}}_{I_k} + {}^{G}_{I_k}\hat{\mathbf{R}}\, \Delta\hat{\mathbf{v}} + {}^{G}\mathbf{g}\, \delta t, \\ \hat{\mathbf{b}}_{g_{k+1}} = \hat{\mathbf{b}}_{g_k}, \qquad \hat{\mathbf{b}}_{a_{k+1}} = \hat{\mathbf{b}}_{a_k}. \end{cases}} \tag{429}$$

Hence $\hat{\mathbf{x}}_k \to \hat{\mathbf{x}}_{k+1}$: solved.

Second: the linearization for covariance propagation. Use $SO(3) + \mathbb{R}^3 + \mathbb{R}^3 + \mathbb{R}^3 + \mathbb{R}^3$ for example, i.e., a right rotation perturbation ${}^{G}_{I}\mathbf{R} = {}^{G}_{I}\hat{\mathbf{R}}\, \operatorname{Exp}(\delta\boldsymbol{\theta})$ and additive errors elsewhere. For the rotation,

$${}^{G}_{I_{k+1}}\hat{\mathbf{R}}\, \operatorname{Exp}(\delta\boldsymbol{\theta}_{k+1}) = {}^{G}_{I_k}\hat{\mathbf{R}}\, \operatorname{Exp}(\delta\boldsymbol{\theta}_k)\, \operatorname{Exp}\left({}^{I_k}\hat{\boldsymbol{\omega}}\, \delta t\right)\, \operatorname{Exp}\left(\mathbf{J}_r({}^{I_k}\hat{\boldsymbol{\omega}}\, \delta t)\, \delta t\, {}^{I_k}\tilde{\boldsymbol{\omega}}\right), \tag{430}$$

so, moving $\operatorname{Exp}(\delta\boldsymbol{\theta}_k)$ across $\operatorname{Exp}({}^{I_k}\hat{\boldsymbol{\omega}}\, \delta t) = \Delta\hat{\mathbf{R}}$ with the adjoint identity,

$$\delta\boldsymbol{\theta}_{k+1} = {}^{I_k}_{I_{k+1}}\hat{\mathbf{R}}^{\top}\, \delta\boldsymbol{\theta}_k + \mathbf{J}_r\left({}^{I_k}\hat{\boldsymbol{\omega}}\, \delta t\right) \delta t\, {}^{I_k}\tilde{\boldsymbol{\omega}}. \tag{431}$$

For the position and velocity (substituting ${}^{G}_{I_k}\mathbf{R} = {}^{G}_{I_k}\hat{\mathbf{R}}\, \operatorname{Exp}(\delta\boldsymbol{\theta}_k) \approx {}^{G}_{I_k}\hat{\mathbf{R}}(\mathbf{I} + \lfloor\delta\boldsymbol{\theta}_k\rfloor_\times)$ and keeping first-order terms),

$${}^{G}\tilde{\mathbf{p}}_{I_{k+1}} = {}^{G}\tilde{\mathbf{p}}_{I_k} + {}^{G}\tilde{\mathbf{v}}_{I_k}\, \delta t - {}^{G}_{I_k}\hat{\mathbf{R}}\, \lfloor\Delta\hat{\mathbf{p}}\rfloor_\times\, \delta\boldsymbol{\theta}_k + {}^{G}_{I_k}\hat{\mathbf{R}}\left(\boldsymbol{\Xi}_2\, {}^{I_k}\tilde{\mathbf{a}} - \boldsymbol{\Xi}_4\, {}^{I_k}\tilde{\boldsymbol{\omega}}\right), \tag{432}$$

$${}^{G}\tilde{\mathbf{v}}_{I_{k+1}} = {}^{G}\tilde{\mathbf{v}}_{I_k} - {}^{G}_{I_k}\hat{\mathbf{R}}\, \lfloor\Delta\hat{\mathbf{v}}\rfloor_\times\, \delta\boldsymbol{\theta}_k + {}^{G}_{I_k}\hat{\mathbf{R}}\left(\boldsymbol{\Xi}_1\, {}^{I_k}\tilde{\mathbf{a}} - \boldsymbol{\Xi}_3\, {}^{I_k}\tilde{\boldsymbol{\omega}}\right), \tag{433}$$

$$\tilde{\mathbf{b}}_{g_{k+1}} = \tilde{\mathbf{b}}_{g_k} + \mathbf{n}_{wgd}\, \delta t, \qquad \tilde{\mathbf{b}}_{a_{k+1}} = \tilde{\mathbf{b}}_{a_k} + \mathbf{n}_{wad}\, \delta t, \tag{434}$$

where the input errors are

$${}^{I_k}\tilde{\mathbf{a}} = -\tilde{\mathbf{b}}_a - \mathbf{n}_{ad}, \qquad {}^{I_k}\tilde{\boldsymbol{\omega}} = -\tilde{\mathbf{b}}_g - \mathbf{n}_{gd}. \tag{435}$$

Hence, stacking the error state $\tilde{\mathbf{x}} = [\delta\boldsymbol{\theta}^\top, {}^{G}\tilde{\mathbf{p}}_I^\top, {}^{G}\tilde{\mathbf{v}}_I^\top, \tilde{\mathbf{b}}_g^\top, \tilde{\mathbf{b}}_a^\top]^\top$,

$$\tilde{\mathbf{x}}_{k+1} = \boldsymbol{\Phi}_{(k+1,k)}\, \tilde{\mathbf{x}}_k + \mathbf{G}_k\, \mathbf{n}_{Id}, \qquad \mathbf{n}_{Id} = \begin{bmatrix} \mathbf{n}_{gd} \\ \mathbf{n}_{ad} \\ \mathbf{n}_{wgd} \\ \mathbf{n}_{wad} \end{bmatrix}, \tag{436}$$

with the state-transition matrix

$$\boldsymbol{\Phi}_{(k+1,k)} = \begin{bmatrix} {}^{I_k}_{I_{k+1}}\hat{\mathbf{R}}^{\top} & \mathbf{0}_3 & \mathbf{0}_3 & -\mathbf{J}_r\left({}^{I_k}\hat{\boldsymbol{\omega}}\, \delta t\right) \delta t & \mathbf{0}_3 \\ -{}^{G}_{I_k}\hat{\mathbf{R}}\, \lfloor\Delta\hat{\mathbf{p}}\rfloor_\times & \mathbf{I}_3 & \mathbf{I}_3\, \delta t & {}^{G}_{I_k}\hat{\mathbf{R}}\, \boldsymbol{\Xi}_4 & -{}^{G}_{I_k}\hat{\mathbf{R}}\, \boldsymbol{\Xi}_2 \\ -{}^{G}_{I_k}\hat{\mathbf{R}}\, \lfloor\Delta\hat{\mathbf{v}}\rfloor_\times & \mathbf{0}_3 & \mathbf{I}_3 & {}^{G}_{I_k}\hat{\mathbf{R}}\, \boldsymbol{\Xi}_3 & -{}^{G}_{I_k}\hat{\mathbf{R}}\, \boldsymbol{\Xi}_1 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix}, \tag{437}$$

and the noise Jacobian

$$\mathbf{G}_k = \begin{bmatrix} -\mathbf{J}_r\left({}^{I_k}\hat{\boldsymbol{\omega}}\,\delta t\right)\delta t & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ {}^{G}_{I_k}\hat{\mathbf{R}}\,\boldsymbol{\Xi}_4 & -{}^{G}_{I_k}\hat{\mathbf{R}}\,\boldsymbol{\Xi}_2 & \mathbf{0}_3 & \mathbf{0}_3 \\ {}^{G}_{I_k}\hat{\mathbf{R}}\,\boldsymbol{\Xi}_3 & -{}^{G}_{I_k}\hat{\mathbf{R}}\,\boldsymbol{\Xi}_1 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3\,\delta t & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3\,\delta t \end{bmatrix}. \tag{438}$$

Hence the covariance propagation

$$\boxed{\mathbf{P}_{k+1} = \boldsymbol{\Phi}_{(k+1,k)}\,\mathbf{P}_k\,\boldsymbol{\Phi}_{(k+1,k)}^{\top} + \mathbf{G}_k\,\mathbf{Q}_d\,\mathbf{G}_k^{\top} \quad \rightarrow \quad \text{solved.}} \tag{439}$$

5.8 The linearization using $SE_2(3)$

Now the linearization using $SE_2(3)$ (Lecture 2). The states live on $SE_2(3) + \mathbb{R}^3 + \mathbb{R}^3$:

$$\mathbf{x}_I = (\mathbf{X}_n, \mathbf{x}_b), \qquad \mathbf{X}_n = \left({}^{G}_{I}\mathbf{R}, {}^{G}\mathbf{p}_I, {}^{G}\mathbf{v}_I\right) = \begin{bmatrix} {}^{G}_{I}\mathbf{R} & {}^{G}\mathbf{p}_I & {}^{G}\mathbf{v}_I \\ \mathbf{0}_{1\times 3} & 1 & 0 \\ \mathbf{0}_{1\times 3} & 0 & 1 \end{bmatrix}, \qquad \mathbf{x}_b = (\mathbf{b}_g, \mathbf{b}_a). \tag{440}$$

The error states. Define a left (group) error on $\mathbf{X}_n$ and additive errors on the biases,

$$\mathbf{x}_I = \hat{\mathbf{x}}_I \boxplus \delta\mathbf{x}_I = \big(\operatorname{Exp}(\delta\mathbf{x}_n)\,\hat{\mathbf{X}}_n,\ \hat{\mathbf{x}}_b + \tilde{\mathbf{x}}_b\big), \tag{441}$$

which, writing $\delta\mathbf{x}_n = [\delta\boldsymbol{\theta}^\top, \delta\mathbf{p}^\top, \delta\mathbf{v}^\top]^\top$ and using the $SE_2(3)$ exponential (with the left Jacobian $\mathbf{J}_l$),

$$\begin{cases} {}^{G}_{I}\mathbf{R} = \operatorname{Exp}(\delta\boldsymbol{\theta})\,{}^{G}_{I}\hat{\mathbf{R}}, \\ {}^{G}\mathbf{p}_I = \operatorname{Exp}(\delta\boldsymbol{\theta})\,{}^{G}\hat{\mathbf{p}}_I + \mathbf{J}_l(\delta\boldsymbol{\theta})\,\delta\mathbf{p}, \\ {}^{G}\mathbf{v}_I = \operatorname{Exp}(\delta\boldsymbol{\theta})\,{}^{G}\hat{\mathbf{v}}_I + \mathbf{J}_l(\delta\boldsymbol{\theta})\,\delta\mathbf{v}, \\ \mathbf{b}_g = \hat{\mathbf{b}}_g + \delta\mathbf{b}_g, \qquad \mathbf{b}_a = \hat{\mathbf{b}}_a + \delta\mathbf{b}_a. \end{cases} \tag{442}$$

The propagation as a group factorization. The propagation equations

$$\begin{cases} {}^{G}_{I_{k+1}}\mathbf{R} = {}^{G}_{I_k}\mathbf{R}\cdot\Delta\mathbf{R}, \\ {}^{G}\mathbf{p}_{I_{k+1}} = {}^{G}\mathbf{p}_{I_k} + {}^{G}\mathbf{v}_{I_k}\,\delta t + {}^{G}_{I_k}\mathbf{R}\,\Delta\mathbf{p} + \frac{1}{2}{}^{G}\mathbf{g}\,\delta t^2, \\ {}^{G}\mathbf{v}_{I_{k+1}} = {}^{G}\mathbf{v}_{I_k} + {}^{G}_{I_k}\mathbf{R}\,\Delta\mathbf{v} + {}^{G}\mathbf{g}\,\delta t, \\ \mathbf{b}_{g_{k+1}} = \mathbf{b}_{g_k} + \mathbf{n}_{wgd}\,\delta t, \qquad \mathbf{b}_{a_{k+1}} = \mathbf{b}_{a_k} + \mathbf{n}_{wad}\,\delta t \end{cases} \tag{443}$$

factor into a product of three $SE_2(3)$ elements,

$$\boxed{\mathbf{X}_{n_{k+1}} = \boldsymbol{\Gamma}_k\cdot\boldsymbol{\Psi}(\mathbf{X}_{n_k})\cdot\mathbf{Y}_k,} \tag{444}$$

with

$$\boldsymbol{\Gamma}_k = \begin{bmatrix} \mathbf{I}_3 & \frac{1}{2}{}^{G}\mathbf{g}\,\delta t^2 & {}^{G}\mathbf{g}\,\delta t \\ \mathbf{0}_{1\times 3} & 1 & 0 \\ \mathbf{0}_{1\times 3} & 0 & 1 \end{bmatrix}, \qquad \mathbf{Y}_k = \begin{bmatrix} \Delta\mathbf{R} & \Delta\mathbf{p} & \Delta\mathbf{v} \\ \mathbf{0}_{1\times 3} & 1 & 0 \\ \mathbf{0}_{1\times 3} & 0 & 1 \end{bmatrix}, \tag{445}$$

$$\boldsymbol{\Psi}(\mathbf{X}_{n_k}) = \begin{bmatrix} {}^{G}_{I_k}\mathbf{R} & {}^{G}\mathbf{p}_{I_k} + {}^{G}\mathbf{v}_{I_k}\,\delta t & {}^{G}\mathbf{v}_{I_k} \\ \mathbf{0}_{1\times 3} & 1 & 0 \\ \mathbf{0}_{1\times 3} & 0 & 1 \end{bmatrix}. \tag{446}$$

The properties of $\mathbf{\Psi}(\cdot)$. The map $\mathbf{\Psi}$ satisfies

$$\begin{cases} \mathbf{\Psi}(\mathbf{X}_{n_i}\cdot\mathbf{X}_{n_j}) = \mathbf{\Psi}(\mathbf{X}_{n_i})\cdot\mathbf{\Psi}(\mathbf{X}_{n_j}), \\ \mathbf{\Psi}\big(\operatorname{Exp}(\delta\mathbf{x}_n)\big) \approx \operatorname{Exp}\left(\mathbf{F}\,\delta\mathbf{x}_n\right), \\ \mathbf{\Psi}\big(\operatorname{Exp}(\delta\mathbf{x}_n)\,\hat{\mathbf{X}}_n\big) = \mathbf{\Psi}\big(\operatorname{Exp}(\delta\mathbf{x}_n)\big)\cdot\mathbf{\Psi}(\hat{\mathbf{X}}_n) = \operatorname{Exp}\left(\mathbf{F}\,\delta\mathbf{x}_n\right)\mathbf{\Psi}(\hat{\mathbf{X}}_n), \end{cases} \qquad \mathbf{F} \triangleq \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{I}_3\,\delta t \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix}. \tag{447}$$

The linearization. Hence we can do the linearization as

$$\begin{aligned} \mathbf{X}_{n_{k+1}} = \operatorname{Exp}\left(\delta\mathbf{x}_{n_{k+1}}\right)\hat{\mathbf{X}}_{n_{k+1}} &= \mathbf{\Gamma}_k\,\mathbf{\Psi}(\mathbf{X}_{n_k})\,\mathbf{Y}_k && (448)\\ &= \mathbf{\Gamma}_k\,\mathbf{\Psi}\big(\operatorname{Exp}(\delta\mathbf{x}_{n_k})\,\hat{\mathbf{X}}_{n_k}\big)\,\hat{\mathbf{Y}}_k\cdot\tilde{\mathbf{Y}}_k && (449)\\ &\approx \mathbf{\Gamma}_k\,\operatorname{Exp}\left(\mathbf{F}\,\delta\mathbf{x}_{n_k}\right)\mathbf{\Psi}(\hat{\mathbf{X}}_{n_k})\,\hat{\mathbf{Y}}_k\,\tilde{\mathbf{Y}}_k && (450)\\ &= \operatorname{Exp}\left(\operatorname{Ad}_{\mathbf{\Gamma}_k}\mathbf{F}\,\delta\mathbf{x}_{n_k}\right)\underbrace{\mathbf{\Gamma}_k\,\mathbf{\Psi}(\hat{\mathbf{X}}_{n_k})\,\hat{\mathbf{Y}}_k}_{\hat{\mathbf{X}}_{n_{k+1}}}\tilde{\mathbf{Y}}_k && (451)\\ &= \operatorname{Exp}\left(\operatorname{Ad}_{\mathbf{\Gamma}_k}\mathbf{F}\,\delta\mathbf{x}_{n_k}\right)\hat{\mathbf{X}}_{n_{k+1}}\tilde{\mathbf{Y}}_k && (452)\\ &= \operatorname{Exp}\left(\mathbf{\Phi}_{nn}\,\delta\mathbf{x}_{n_k}\right)\hat{\mathbf{X}}_{n_{k+1}}\tilde{\mathbf{Y}}_k, \qquad \boxed{\mathbf{\Phi}_{nn}\triangleq\operatorname{Ad}_{\mathbf{\Gamma}_k}\mathbf{F}} && (453) \end{aligned}$$

where the preintegration factor splits as $\mathbf{Y}_k = \hat{\mathbf{Y}}_k\,\tilde{\mathbf{Y}}_k$,

$$\mathbf{Y}_k = \hat{\mathbf{Y}}_k\,\tilde{\mathbf{Y}}_k = \begin{bmatrix} \Delta\hat{\mathbf{R}} & \Delta\hat{\mathbf{p}} & \Delta\hat{\mathbf{v}} \\ \mathbf{0}_{1\times3} & 1 & 0 \\ \mathbf{0}_{1\times3} & 0 & 1 \end{bmatrix}\begin{bmatrix} \Delta\tilde{\mathbf{R}} & \Delta\hat{\mathbf{R}}^\top\Delta\tilde{\mathbf{p}} & \Delta\hat{\mathbf{R}}^\top\Delta\tilde{\mathbf{v}} \\ \mathbf{0}_{1\times3} & 1 & 0 \\ \mathbf{0}_{1\times3} & 0 & 1 \end{bmatrix} = \hat{\mathbf{Y}}_k\cdot\operatorname{Exp}\left(\begin{bmatrix} \delta\Delta\boldsymbol{\theta} \\ \Delta\hat{\mathbf{R}}^\top\Delta\tilde{\mathbf{p}} \\ \Delta\hat{\mathbf{R}}^\top\Delta\tilde{\mathbf{v}} \end{bmatrix}\right), \tag{454}$$

with $\delta\Delta\boldsymbol{\theta} = \operatorname{Log}(\Delta\tilde{\mathbf{R}})$ (to first order). From the perturbations of §5.7,

$$\begin{aligned} \begin{bmatrix} \delta\Delta\boldsymbol{\theta} \\ \Delta\hat{\mathbf{R}}^\top\Delta\tilde{\mathbf{p}} \\ \Delta\hat{\mathbf{R}}^\top\Delta\tilde{\mathbf{v}} \end{bmatrix} &= \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \Delta\hat{\mathbf{R}}^\top & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \Delta\hat{\mathbf{R}}^\top \end{bmatrix}\begin{bmatrix} \delta\Delta\boldsymbol{\theta} \\ \Delta\tilde{\mathbf{p}} \\ \Delta\tilde{\mathbf{v}} \end{bmatrix} && (455)\\ &= \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \Delta\hat{\mathbf{R}}^\top & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \Delta\hat{\mathbf{R}}^\top \end{bmatrix}\begin{bmatrix} -\mathbf{J}_r(\Delta\hat{\boldsymbol{\theta}})\,\delta t & \mathbf{0}_3 \\ \mathbf{\Xi}_4 & -\mathbf{\Xi}_2 \\ \mathbf{\Xi}_3 & -\mathbf{\Xi}_1 \end{bmatrix}\begin{bmatrix} \tilde{\mathbf{b}}_g + \mathbf{n}_{gd} \\ \tilde{\mathbf{b}}_a + \mathbf{n}_{ad} \end{bmatrix} && (456)\\ &= \begin{bmatrix} -\mathbf{J}_r(\Delta\hat{\boldsymbol{\theta}})\,\delta t & \mathbf{0}_3 \\ \Delta\hat{\mathbf{R}}^\top\mathbf{\Xi}_4 & -\Delta\hat{\mathbf{R}}^\top\mathbf{\Xi}_2 \\ \Delta\hat{\mathbf{R}}^\top\mathbf{\Xi}_3 & -\Delta\hat{\mathbf{R}}^\top\mathbf{\Xi}_1 \end{bmatrix}\begin{bmatrix} \tilde{\mathbf{b}}_g + \mathbf{n}_{gd} \\ \tilde{\mathbf{b}}_a + \mathbf{n}_{ad} \end{bmatrix}, && (457) \end{aligned}$$

with $\Delta\hat{\boldsymbol{\theta}} = {}^{I_k}\hat{\boldsymbol{\omega}}\,\delta t$. Finally, move $\tilde{\mathbf{Y}}_k$ across $\hat{\mathbf{X}}_{n_{k+1}}$ with the adjoint:

$$\begin{aligned} \operatorname{Exp}\left(\delta\mathbf{x}_{n_{k+1}}\right) &= \operatorname{Exp}\left(\mathbf{\Phi}_{nn}\,\delta\mathbf{x}_{n_k}\right)\hat{\mathbf{X}}_{n_{k+1}}\tilde{\mathbf{Y}}_k\,\hat{\mathbf{X}}_{n_{k+1}}^{-1} && (458)\\ &= \operatorname{Exp}\left(\mathbf{\Phi}_{nn}\,\delta\mathbf{x}_{n_k}\right)\operatorname{Exp}\left(\operatorname{Ad}_{\hat{\mathbf{X}}_{n_{k+1}}}\begin{bmatrix} -\mathbf{J}_r(\Delta\hat{\boldsymbol{\theta}})\,\delta t & \mathbf{0}_3 \\ \Delta\hat{\mathbf{R}}^\top\mathbf{\Xi}_4 & -\Delta\hat{\mathbf{R}}^\top\mathbf{\Xi}_2 \\ \Delta\hat{\mathbf{R}}^\top\mathbf{\Xi}_3 & -\Delta\hat{\mathbf{R}}^\top\mathbf{\Xi}_1 \end{bmatrix}\begin{bmatrix} \tilde{\mathbf{b}}_g + \mathbf{n}_{gd} \\ \tilde{\mathbf{b}}_a + \mathbf{n}_{ad} \end{bmatrix}\right), && (459) \end{aligned}$$

so that

$$\delta\mathbf{x}_{n_{k+1}} = \boldsymbol{\Phi}_{nn}\,\delta\mathbf{x}_{n_k} + \boldsymbol{\Phi}_{nb}\begin{bmatrix}\tilde{\mathbf{b}}_g + \mathbf{n}_{gd}\\ \tilde{\mathbf{b}}_a + \mathbf{n}_{ad}\end{bmatrix}, \qquad \boxed{\boldsymbol{\Phi}_{nb} \triangleq \mathrm{Ad}_{\hat{\mathbf{X}}_{n_{k+1}}}\begin{bmatrix}-\mathbf{J}_r(\Delta\hat{\boldsymbol{\theta}})\,\delta t & \mathbf{0}_3\\ \Delta\hat{\mathbf{R}}^\top\,\boldsymbol{\Xi}_4 & -\Delta\hat{\mathbf{R}}^\top\,\boldsymbol{\Xi}_2\\ \Delta\hat{\mathbf{R}}^\top\,\boldsymbol{\Xi}_3 & -\Delta\hat{\mathbf{R}}^\top\,\boldsymbol{\Xi}_1\end{bmatrix}} \tag{460}$$

The stacked system. Then

$$\begin{cases}\delta\mathbf{x}_{n_{k+1}} = \boldsymbol{\Phi}_{nn}\,\delta\mathbf{x}_{n_k} + \boldsymbol{\Phi}_{nb}\begin{bmatrix}\tilde{\mathbf{b}}_g\\ \tilde{\mathbf{b}}_a\end{bmatrix} + \boldsymbol{\Phi}_{nb}\begin{bmatrix}\mathbf{n}_{gd}\\ \mathbf{n}_{ad}\end{bmatrix},\\ \tilde{\mathbf{b}}_{g_{k+1}} = \tilde{\mathbf{b}}_{g_k} + \mathbf{n}_{wgd}\,\delta t, \qquad \tilde{\mathbf{b}}_{a_{k+1}} = \tilde{\mathbf{b}}_{a_k} + \mathbf{n}_{wad}\,\delta t,\end{cases} \tag{461}$$

which stacks into $\tilde{\mathbf{x}}_{k+1} = \boldsymbol{\Phi}_{(k+1,k)}\tilde{\mathbf{x}}_k + \mathbf{G}_k\,\mathbf{n}_{Id}$ with

$$\boldsymbol{\Phi}_{(k+1,k)} = \begin{bmatrix}\boldsymbol{\Phi}_{nn} & \boldsymbol{\Phi}_{nb}\\ \mathbf{0}_{6\times 9} & \mathbf{I}_6\end{bmatrix} = \begin{bmatrix}\mathbf{I}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \boldsymbol{\Phi}_{14} & \mathbf{0}_3\\ \frac{1}{2}\left\lfloor {}^G\mathbf{g}\right\rfloor_\times \delta t^2 & \mathbf{I}_3 & \mathbf{I}_3\,\delta t & \boldsymbol{\Phi}_{24} & \boldsymbol{\Phi}_{25}\\ \left\lfloor {}^G\mathbf{g}\right\rfloor_\times \delta t & \mathbf{0}_3 & \mathbf{I}_3 & \boldsymbol{\Phi}_{34} & \boldsymbol{\Phi}_{35}\\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3 & \mathbf{0}_3\\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3\end{bmatrix}, \tag{462}$$

where the first three block-columns are exactly $\boldsymbol{\Phi}_{nn} = \mathrm{Ad}_{\boldsymbol{\Gamma}_k}\,\mathbf{F}$ and the $\boldsymbol{\Phi}_{i4}, \boldsymbol{\Phi}_{i5}$ blocks are the columns of $\boldsymbol{\Phi}_{nb}$ ($\boldsymbol{\Phi}_{15} = \mathbf{0}_3$ since the accel error does not enter the rotation), and

$$\mathbf{G}_k = \begin{bmatrix}\boldsymbol{\Phi}_{nb} & \mathbf{0}_{9\times 6}\\ \mathbf{0}_{6\times 6} & \mathbf{I}_6\,\delta t\end{bmatrix} = \begin{bmatrix}\mathbf{G}_{11} & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3\\ \mathbf{G}_{21} & \mathbf{G}_{22} & \mathbf{0}_3 & \mathbf{0}_3\\ \mathbf{G}_{31} & \mathbf{G}_{32} & \mathbf{0}_3 & \mathbf{0}_3\\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3\,\delta t & \mathbf{0}_3\\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{I}_3\,\delta t\end{bmatrix}, \tag{463}$$

with $[\mathbf{G}_{i1}, \mathbf{G}_{i2}]$ the blocks of $\boldsymbol{\Phi}_{nb}$. Hence, as before,

$$\boxed{\tilde{\mathbf{x}}_{k+1} = \boldsymbol{\Phi}_{(k+1,k)}\,\tilde{\mathbf{x}}_k + \mathbf{G}_k\,\mathbf{n}_{Id} \quad \Rightarrow \quad \mathbf{P}_{k+1} = \boldsymbol{\Phi}_{(k+1,k)}\,\mathbf{P}_k\,\boldsymbol{\Phi}_{(k+1,k)}^\top + \mathbf{G}_k\,\mathbf{Q}_d\,\mathbf{G}_k^\top \quad \rightarrow \quad \text{solved.}} \tag{464}$$

Note that $\boldsymbol{\Phi}_{nn}$ in (462) depends only on gravity and δt — not on the state estimate — which is the hallmark of the (right-)invariant error design mentioned in Lecture 4.

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